BY ORDER OF THE
SECRETARY OF THE AIR FORCE

AIR FORCE MANUAL 11-2C-17V3

19 AUGUST 2024
Corrective Action, 28 AUGUST 2024
Flying Operations

C-17 OPERATIONS PROCEDURES

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available for downloading or ordering on the e-Publishing website at **www.e-Publishing.af.mil**.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: AMC/A3V

Certified by: AF/A3T
(Mr Howard D. Hall)

Supersedes: AFMAN11-2C-17V3, 30 July 2019

Pages: 146

This manual implements Air Force Policy Directive (AFPD) 11-2, *Aircrew Operations* and references Air Force Manual (AFMAN) 11-202, Volume 3, *Flight Operations*, as well as Air Force Tactics Techniques and Procedures (AFTTP) 3-3.C-17, *Combat Fundamentals C-17.* This is a specialized publication intended for use by Airmen who have graduated from technical training related to this publication. It provides guidance for the operation of the C-17 aircraft to safely and successfully accomplish worldwide mobility missions. It applies to individuals at all levels operating C-17 aircraft, including the Air Force Reserve Command (AFRC) and National Guard Bureau except where noted otherwise. This publication does not apply to the United States Space Force. This manual may be supplemented at any level, but all supplements that directly implement this publication must be routed to the office of primary responsibility (OPR) for coordination prior to certification and approval. Refer recommended changes and questions about this publication to the OPR using Department of the Air Force (DAF) Form 847, *Recommendation for Change of Publication*; route DAF Forms 847 from the field through the appropriate functional chain of command. The authorities to waive wing/unit level requirements in this publication are identified with a Tier ("T-0, T-1, T-2, T-3") number following the compliance statement. See DAFMAN 90-161, *Publishing Processes and Procedures,* for a description of the authorities associated with the Tier numbers. Submit requests for waivers through the chain of command to the appropriate Tier waiver approval authority, or to the publication OPR for non-tiered compliance items. Report all deviations in accordance with (IAW) the current **paragraph 1.4**. Ensure all records generated as a result of processes prescribed in this publication adhere to Air Force Instruction 33-322, *Records Management and Information Governance Program*, and are disposed in accordance with the Air Force Records Disposition Schedule, which is located in the Air Force Records Information

Management System. The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include the removal of sections and paragraph renumbering due to the revision of AFMAN 11-202V3, AMC Supplement, *Flight Operations*. It standardizes format, paragraphs, and information with other Air Mobility Command (AMC) Mission Design Series (MDS)-specific volumes by relocating and renumbering paragraphs and chapters throughout. The Minimum Equipment List has been realigned with the Minimum Essential Subsystem List (MESL) utilized by maintenance personnel.

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Chapter 1

GENERAL INFORMATION

1.1. General. This manual provides guidance for operating the C-17A aircraft. When guidance in this manual conflicts with another source document, that document takes precedence. For matters where this manual is the source document, reference **paragraph 1.4**.

1.2. Applicability. This AFMAN applies to aircrew members, maintenance/support personnel, and managers involved with employing C-17 aircraft. Unit commanders and agency directors involved with or supporting C-17A operations shall make current copies of this manual available to appropriate personnel. **(T-2)**

1.3. Key Words Explained.

1.3.1. "Will" and "Must" indicate a mandatory requirement.

1.3.2. "Should" indicates a preferred, but not mandatory, method of accomplishment.

1.3.3. "May" indicates an acceptable or suggested means of accomplishment.

1.3.4. **"Note"** indicates operating procedures, techniques, etc., considered essential to emphasize.

1.3.5. **"Caution"** indicates operating procedures, techniques, etc., which could result in damage to equipment if not carefully followed.

1.3.6. **"WARNING"** indicates operating procedures, techniques, etc., which could result in personal injury or loss of life if not carefully followed.

1.4. Deviations and Waivers. Do not deviate from policies in this manual except when the situation demands immediate action to ensure safety.

1.4.1. Deviations. The pilot in command (PIC) will report deviations or exceptions taken without a waiver through command channels to their Chief, major command (MAJCOM) Standardization/Evaluation (Stan/Eval), who will notify Chief, AMC Stan/Eval (lead command) as appropriate for follow-on action. **(T-2)** **Note:** For Air Education and Training Command (AETC), MAJCOM Stan/Eval or A3V refers to 19AF Stan/Eval (19AF/A3V).

1.4.2. Waivers. Unless otherwise directed, waiver authority for the contents of this manual is the MAJCOM/A3 with mission execution authority. **Note:** For AETC, MAJCOM/A3 refers to 19 AF Director of Operations (19AF/A3) or appropriate delegated authority. For United States Transportation Command/AMC operational missions, reference the most current AMC tiered waiver delegation memo on the Air Mobility Command Standardization/Evaluations (AMC/A3V) SharePoint® when flying 618th Air Operations Center (AOC) controlled missions. For aircrews that change operational control to a different Combatant Commander, the Commander, Air Force Forces is the waiver authority.

1.4.2.1. Waivers may affect multiple aircraft and/or missions but are not permanent in nature (expire at a specific date/time). MAJCOM Stan/Eval will send AMC Stan/Eval (lead command) copies of MAJCOM/A3-approved waivers. **(T-2)**

1.4.2.2. When coordinating a waiver for a mission in execution, the PIC will provide the Command and Control (C2) agent the nature of the request, individual crewmember

qualifications, mission leg(s) requiring the waiver, weather or other adverse conditions, and the governing directive of waiver request, to include volume, chapter, and paragraph. **(T-2)** Initiate waiver requests as soon as possible; plan at least 1 hour for waiver processing time. If beyond C2 communication capability, or when it is necessary to protect the crew or aircraft, the PIC may deviate from policies in this manual when the situation demands immediate action to ensure safety.

1.5. Supplemental Procedures. This AFMAN is a basic directive. Each MAJCOM or operational theater may supplement this AFMAN according to AFPD 11-2 and DAFMAN 90-161. Stipulate unique MAJCOM procedures (cannot be less restrictive than this basic document) in the MAJCOM supplement.

1.5.1. Combined Command Operations. Plan and conduct all operations that include forces from multiple MAJCOMs using provisions in this manual. MAJCOM/A3s planning multiple command operations are responsible for coordinating MAJCOM/theater unique procedures to include special instructions (SPINS) and air operations directive review with supporting MAJCOM/A3.

1.5.2. Forward MAJCOM-approved supplements (attach DAF Form 673, *Department of the Air Force Publication/Form Action Request*) to Air Mobility Command/A3V, 510 POW/MIA Dr., Suite W101, Scott AFB IL, 62225-5302.

1.6. Local Supplement Coordination Process. Operations Group commanders (OG/CCs) may define local operating procedures to this instruction in a unit supplement. OG/CCs must obtain approval from MAJCOM prior to releasing their supplement. **(T-2)** Send an electronic copy of the approved version through Stan/Eval channels to MAJCOM/A3V. MAJCOM/A3V will send approved copies to AMC/A3V. **(T-2)**

1.7. Improvement Recommendations. Send comments and suggested improvements to this manual on a DAF Form 847, *Recommendation for Change of Publication*, through Stan/Eval channels to AMC/A3V, IAW procedures in DAFMAN 90-161.

1.8. Definitions. Find explanations or definitions of terms and abbreviations commonly used in the aviation community in Title 14, Code of Federal Regulations, Part 1, *Definitions and Abbreviations*; *DoD FLIP General Planning*, Chapter 2; and The DoD Dictionary of Military and Associated Terms. See **Attachment 1** for common terms used in this manual.

1.9. Aircrew Operational Reports. The reporting requirements in this manual are exempt from licensing IAW AFI 33-324, *The Air Force Information Collections and Reports Management Program.*

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Major Command (MAJCOM). MAJCOMs will provide guidance and approve waivers (as required), where specified throughout this manual.

2.2. Pilot in Command (PIC). The PIC is the aircrew member designated by competent authority, regardless of rank, as being responsible for, and is the final authority for the operation of the aircraft. The PIC will ensure the aircraft is not operated in a careless, reckless, or irresponsible manner that could endanger life or property. The PIC will ensure compliance with this publication and the following:

2.2.1. Headquarters Air Force, MAJCOM, and Mission Design Series (MDS)-specific guidance.

2.2.2. Flight Information Publication (FLIP) and Foreign Clearance Guide (FCG).

2.2.3. Air Traffic Control (ATC) clearances.

2.2.4. Notices to Airmen (NOTAM).

2.2.5. Aircraft Technical Orders (T.O.'s).

2.2.6. Combatant Commander's instructions and other associated directives.

2.3. Aircrew. Individuals designated on the flight authorization are responsible to fulfill specific aeronautical tasks regarding operating USAF aircraft as specified in this manual or by other competent, supplemental authority.

Chapter 3

AIRCREW COMPLEMENT/MANAGEMENT

3.1. General. This chapter provides guiding principles to form and manage mobility aircrews to enable work/rest schedules that optimize the efficiency of mobility forces engaged in worldwide operations.

3.2. Aircrew Complement.

3.2.1. Squadron commanders (SQ/CCs) will form aircrews based on fragmentation order/mission directive, Crew Duty Time (CDT) and Flight Duty Period (FDP) requirements, aircrew member qualification, and other constraints to safely accomplish the mission tasking. **(T-3)** Table 3.1 below summarizes the minimum crewmember complement for different crew types. When a mission requires more than one aircrew member at a position, the SQ/CC will determine whether an instructor and a Non-Mission Ready crewmember meet mission requirements. **(T-3)**

3.2.2. SQ/CCs will form augmented aircrews for missions planned to take longer than a basic CDT. **(T-3)** Augmenting aircrew members must be Mission Ready in accordance with AFMAN 11-2C-17V1, *C-17 Aircrew Training*, unless an instructor is on the Flight Authorization for that crew position. **(T-3)** SQ/CCs will augment an aircrew for the full Flight Duty Period (FDP). **(T-3)** The MAJCOM/A3 may identify aircrews as augmented during mission execution (see **paragraph 3.4** for more on CDT/FDP).

Table 3.1. Aircrew Complement.

Crew Position	Crew Complement (1, 2, 3, 4)		
	Basic	**Augmented**	**Basic + 1 (5)**
Aircraft Commander (AC)	1	1	1
First Pilot	1	2 (6)	1
Loadmaster (LM)	1	2 (7)	1

Notes:
1. Two loadmasters or one loadmaster and one pilot are required in the cargo compartment if more than 40 passengers are scheduled to be carried. Both crewmembers must remain in the cargo compartment, one forward and one aft, for takeoffs and landings. **(T-2) Exception:** On personnel airdrop sorties, only one loadmaster is required with more than 40 jumpers planning to jump. This exception does not apply to airland segments of an airdrop mission.
2. One loadmaster and another qualified crewmember/physiological technician are required in the cargo compartment on any mission segment where cabin altitude exceeds 13,000 feet Mean Sea level (MSL).
3. Two loadmasters are required for dual-row airdrop and Joint Precision Airdrop System (JPADS)/Improved Container Delivery System (ICDS) missions. Both loadmasters must be qualified in the specific mission. **(T-2) Exception:** For dropsonde and ICDS airdrops, a single loadmaster may concurrently perform LM1 and LM2 duties.
4. For formation flight/airdrop, the following lead pilot requirements apply:

a. Single-element formations. An airdrop instructor or lead pilot is required in the formation (any position) when the element consists of three aircraft. Two-ship elements do not require a lead certified pilot.
b. Multiple-element formation. In a formation greater than two elements, a lead certified pilot is required within the formation, and should be in the lead element unless required for training. An airdrop instructor or lead certified pilot is required in each element of this type of formation. In a formation of two elements, an airdrop instructor pilot is required within the formation, and will be in the lead element unless required for training. Any crew can fly the last ship of a formation even if it is an element lead position. **(T-2)** The lead or airdrop instructor pilot should be in the first aircraft of the element they are in, unless required for training purposes.

5. Basic + 1 is 2 pilots, 1 loadmaster, and 1 additional mission-qualified C-17 pilot or loadmaster (additional crewmember will be dictated by mission requirements).
6. Pilots who have graduated from the Aircraft Commander Initial Qualification course may augment the crew. Pilots who have graduated from the Pilot Initial Qualification course, when approved by the SQ/CC, may act as the augmenting crewmember.
7. For augmented airdrop missions, one loadmaster must be airdrop qualified. The other loadmaster may be airland qualified.

3.3. Loadmasters.

3.3.1. A non-current or unqualified loadmaster may serve as a primary aircrew member on any mission when supervised by a qualified instructor or flight examiner loadmaster (direct supervision for critical phases of flight).

3.3.2. Multiple Loadmaster Crew Resource Management (CRM). To ensure good CRM when there is a multiple loadmaster requirement, the primary loadmaster assumes overall responsibility for the completion of all checklists. They coordinate all loadmaster mission responsibilities and ensures there is no confusion as to which duties have been accomplished.

3.4. Aircrew Management. SQ/CCs and command and control (C2) agencies ensure work/rest cycles permit an aircrew adequate time to safely accomplish mission duties and personal time for rest.

3.4.1. Flight Duty Period (FDP). Maximum FDP for basic and augmented aircrews is IAW AFMAN 11-202V3_AMCSUP.

3.4.1.1. Reduce FDP for autopilot failure after departure IAW AFMAN 11-202V3_AMCSUP. If the autopilot fails after departure, consider mission requirements and determine the best course of action to preclude further mission delays due to reduced FDP. The best course of action may include diverting to an airfield with maintenance capability. Contact C2, coordinate intentions, and comply with limitations.

3.4.1.2. For augmented FDPs, two aircraft commanders are required if Air-to-Air Refueling (AAR) is accomplished after 14 hours in the FDP. **(T-3)** The intent of having two ACs is to manage rest cycles in a way that allows one AC to accomplish any AAR prior to the 14-hour point, and for the second AC to accomplish any AAR past the 14-hour point. The second AC fulfills the requirement for an additional pilot.

3.4.2. Crew Duty Time (CDT). CDT is that period of time an aircrew may perform combined ground/flight duties. An aircrew member may perform mission-related duties for other

missions when approved by member's home station SQ/CC or equivalent. Assistant Vice Chief of Staff of the Air Force, Special Air Missions (CVAM)-directed special assignment airlift missions (SAAM) aircrew require 89th Air Wing Commander (89 AW/CC) approval for CDT/FDP extensions.

3.5. Standby Forces. J-Alert and P-Alert are standing C-17 alert taskings. Scheduling, duties, and length of alert duty may be adapted and managed at the wing level to optimize alert force availability and tasking fulfillment. Length of alert duty will not exceed 7 days (168 hours). **(T-2)**

3.6. Crew Rest/Enroute Ground Time. MAJCOMs will follow AFMAN 11-202V3 and appropriate supplement.

3.6.1. Off-station/Enroute Ground Time. Mobility planners will provide aircrews at least 16+30 hours ground time between engine shutdown and subsequent takeoff. **(T-2)**

3.6.2. Mission planners, PICs, or C2 agents may modify ground time as follows:

3.6.2.1. In the interest of safety.

3.6.2.2. To alert the aircrew no earlier than 12 hours from the time the aircrew entered crew rest. Before reducing ground time, PICs should consider time to complete mission planning, cargo on-/off-load, and non-standard mission related duties. C2 agents will not ask PICs to accept less than 16+30 hours ground time. **(T-2)**

3.7. Alerting Procedures. Aircrew alert time is normally 3+45 hours (4+15 for airdrop missions) before scheduled takeoff time. This allows 1 hour for reporting and 2+45 hours for mission preparation (3+15 for airdrop missions). The OG/CC may increase or decrease this time depending on specific capabilities at the aircrew's home station. OG/CCs may establish self-alert procedures for local training missions.

3.8. Flight Attendants on Distinguished Visitor Missions. Flight attendants may fly as primary crewmembers on designated C-17 missions. They fall under the authority of the PIC, or mission commander (if assigned), throughout the mission. An egress briefing will be given to the flight attendants prior to the first mission leg.

Chapter 4

AIRCRAFT OPERATING RESTRICTIONS

4.1. Objective. Redundant systems may allow crews to safely perform some missions when a component/system is degraded. The PIC is the final authority in determining the overall suitability of an aircraft for the mission. The PIC will ensure a detailed explanation of any discrepancy is entered in the AFTO Form 781A, *Maintenance Discrepancy and Work Document*. **(T-2)** The following maintenance identifiers should be used to effectively communicate aircraft status:

4.1.1. Mission Essential (ME). An item, system, or subsystem component essential for safe aircraft operation.

4.1.2. Mission Contributing (MC). An item, system, or subsystem component which is not currently essential for safe aircraft operation. These discrepancies should be cleared at the earliest opportunity. If circumstances change or mission safety would be compromised, re-designate as ME. Do not delay a mission to clear a MC discrepancy.

4.1.3. Open Item (OI). Discrepancies not expected to adversely impact the current mission or any subsequent mission. These items are normally cleared at home station.

4.2. Minimum Equipment List (MEL) Guidance. The MEL is a pre-launch document that lists the minimum equipment/systems to operate the aircraft. It is impractical to prepare a list that would anticipate all possible combinations of equipment malfunctions and contingent circumstances.

4.2.1. Consider equipment and systems not listed but obviously required for airworthiness (wings, primary flight controls, flaps/slats, etc.) as grounding items. Also not included are items that do not affect airworthiness such as galley equipment, passenger convenience items, speakers, etc. PICs should carefully consider mission requirements when accepting degraded items not listed on the MEL.

4.2.2. The PIC retains the authority to request a waiver to operate with degraded items. A PIC who accepts an aircraft with degraded equipment/systems is not committed to subsequent operations with the same degraded equipment. PICs are not committed to operations with degraded equipment accepted by another PIC.

4.2.3. The PIC must account for the possibility of additional failures during continued operation with inoperative systems or components. **(T-2)** The MEL is not intended for continued operation over an indefinite period with systems/subsystems inoperative.

4.2.4. Units will install all emergency equipment unless there is a mission requirement/directive exemption. **(T-2)**

4.3. Waiver Request Guidance. Waivers to operate with degraded equipment are granted on a case-by-case basis. The PIC determines the need for a waiver after coordinating with the lowest practical level of command.

4.3.1. The wing commander (WG/CC) or equivalent, delegable no lower than the OG/CC, is the waiver authority for locally tasked missions (e.g., training, exercises).

4.3.2. Mission execution authority (i.e., 618 AOC for TRANSCOM missions) is the waiver authority for AOC and MAJCOM tasked missions. **(T-2)**

4.3.3. A PIC prepared to operate with a degraded MEL item will request a waiver through C2 channels. **(T-3)** A PIC operating with waiver(s) for degraded equipment will coordinate mission requirements (e.g., revised departure times, fuel requirements, maintenance requirements, etc.) with the controlling C2 agency and/or flight manager (FM). **(T-3)**

4.3.4. When coordinating a waiver, the PIC must provide the C2 agent the nature of the request, individual crewmember qualifications, mission leg(s) requiring the waiver, weather or other adverse conditions, and the governing directive of waiver request to include volume, chapter, and paragraph. **(T-2)** Initiate waiver requests as soon as possible; plan at least 1 hour for waiver processing time. If beyond C2 communication capability, or when it is necessary to protect the crew or aircraft from a situation not covered by this chapter, the PIC may deviate from policies in this manual when the situation demands immediate action to ensure safety.

4.3.5. Report deviations (without waiver) through appropriate channels to MAJCOM/A3 within 48 hours. OG/CCs will collect background information and submit a follow-up written report to MAJCOM/A3 upon request. **(T-2)**

4.4. Technical Assistance. The PIC may request technical support and additional assistance from their home unit or MAJCOM C2 agency.

4.4.1. Engineering Disposition (ED). Dispositions are requested when aircraft are damaged and/or established maintenance technical order procedures cannot be followed or do not exist. The on-site maintenance authority is responsible for determining the need for and requesting an ED. Most EDs allow maintenance to repair the aircraft and return it to unrestricted status; dispositions of this nature do not concern aircrews. However, EDs affecting aircrew operations require MEL waiver authority approval.

4.4.1.1. PICs will coordinate dispositions containing flight restrictions, prohibitions, additional operating limits, or modified/nonstandard operating procedures with the appropriate MEL waiver authority. **(T-2)**

4.4.1.2. PICs will not accept dispositions appearing incomplete, in error, or unsafe. **(T-3)** Prior to rejecting a disposition, the PIC will contact the appropriate MEL waiver authority. **(T-3)** The waiver authority will attempt to resolve the issue. **(T-3)** **Note:** Requests for deviations or waivers to specific flight manuals are submitted through the operating MAJCOM Stan/Eval to the MAJCOM/A3 for approval IAW AFI 11-215, *Flight Manuals Program*. **(T-2)**

4.4.2. CONFERENCE HOTEL Procedures. Contact the nearest or controlling C2 agency to initiate a CONFERENCE HOTEL when technical order guidance is unable to resolve an issue and/or further guidance is required. Reference AFMAN 11-202V3_AMCSUP for more detailed information.

4.5. Gear Down Flight Operations. Limit gear down flight operations to sorties required to move the aircraft to a suitable repair facility. **(T-2)** Consider gear down flight only after the PIC exhausts all avenues to repair the aircraft in place.

4.5.1. PICs will not takeoff until there is reasonable assurance that they will achieve/maintain adequate obstacle clearance (to include enroute stops and alternates). **(T-1)** The appropriate drag index is required to correctly calculate fuel requirements and to determine terrain clearance using T.O. 1C-17A-1-1, *Performance Data*, “Climbout Flight Path – 3-ENG” charts.

4.5.2. Time and communications capability permitting, validate takeoff data with Operations Group Stan/Eval (OG/OGV) or MAJCOM Stan/Eval.

4.6. MEL Table Definitions/Column Identifiers. MEL tables are arranged by aircraft system to provide the PIC a mechanism to determine minimum system requirements. Components are listed by number installed and minimum required for flight. Requirements are defined as Home Station Departure/Main Operating Base (MOB) (Column A) and non-MOB locations (Column B). An asterisk (*) in the required column indicates the number required is situation dependent; refer to the Remarks/Limitations/Exception column for clarification.

4.6.1. Home Station Departures utilize Column A. Local training missions, to include off-station trainers, utilize Column B.

4.6.2. MOBs for all AMC, Pacific Air Forces (PACAF), and Air Reserve Component (ARC) aircraft (includes all AMC-gained aircraft) are: Joint Base Charleston, SC; Dover AFB, DE; Joint Base Lewis-McChord, WA; Joint Base McGuire-Dix-Lakehurst, NJ; and Travis AFB, CA.

4.6.3. Any aircraft transiting Joint Base Elmendorf-Richardson, AK and Joint Base Pearl Harbor-Hickam, HI will utilize Column B. **Exception:** PACAF may coordinate agreements to provide Column A support for PACAF aircraft.

4.6.4. MOBs for ARC aircraft (dependent on their component) are March ARB, CA (AFRC); Wright-Patterson AFB, OH (AFRC); Pittsburgh IAP, PA (AFRC); Jackson, MS (ANG); Martinsburg, WV (ANG); Memphis, TN (ANG); Stewart, NY (ANG); and Charlotte, NC (ANG).

4.6.5. AETC will consider Altus AFB, OK a MOB.

4.6.6. When transiting a MOB on a de-positioning leg use Column B. Example: A McChord aircraft transiting Charleston enroute to Ramstein Air Base (active) will use Column A. However, when transiting Charleston enroute to McChord (de-positioning) use Column B. **Note:** Column B requirements will not normally be waived when transiting a MOB on a de-positioning leg. Aircraft transiting other component MOBs will use column B. Example: An AMC aircraft transiting Stewart ANGB will use Column B for both active and de-positioning legs.

4.6.7. Remarks/Limitations/Exceptions. Some technical information and procedures are contained in this column. This is not all-inclusive; refer to the flight manual and other directives for procedures, techniques, limitations, etc.

4.6.7.1. One-time Flight Clarification. Although a one-time flight may be authorized IAW the MEL, maintenance release (“clearing the Red X”) will still be required. A Red X discrepancy must be downgraded through maintenance channels prior to flight. **(T-3)** Contact C2 channels to coordinate the Red X discrepancy downgrade from the Maintenance Group Commander that owns the aircraft. The priority is to move the airplane to a repair capable facility. PICs must coordinate with C2 to ensure repair capability exists at the destination. **(T-3)**

4.6.7.1.1. One-time flight to nearest repair capable facility: Flight is limited to the nearest (shortest enroute time) repair capable base.

4.6.7.1.2. One-time flight to a repair capable facility: Flight is not restricted to the nearest repair capable facility.

4.6.7.1.3. One-time flights may include enroute stops only when necessary to recover the airplane. Example: An airplane departs on a gear-down flight from Djibouti IAP and requires an enroute fuel stop (Cairo) before landing at the nearest repair capable facility, Sigonella Naval Air Station.

4.6.7.1.4. An approved one-time flight does not prohibit carrying cargo and passengers unless stipulated otherwise by the waiver.

4.6.7.2. Other Mission and Repair Clarifications:

4.6.7.2.1. Will be repaired at next repair capable facility: Mission may continue as scheduled, and the item will be repaired upon reaching a repair capable facility. **(T-3)** Designate item missions essential (ME) upon reaching repair facility. Once maintenance action is initiated, and it is determined repairs are not possible, the PIC will discuss possible courses of action with C2 agency to return aircraft to service. **(T-3)**

4.6.7.2.2. Per mission requirements: PIC considers the entire mission profile, not just the next leg. Example: An airplane is departing an enroute station with repair capability. After engine start it is discovered that the #1 engine anti-ice is inoperative. Icing conditions are not forecasted for the next leg. However, because the mission spans several days and repair capability does not exist at the scheduled enroute stops, the PIC elects to have the item repaired prior to departing.

4.7. C-17 MEL. The MEL listed below in **Table 4.1** applies to all C-17 models and lists the minimum equipment and systems to launch the aircraft under routine operations.

4.7.1. The MEL is not applicable for troubleshooting component failure(s) in flight. However, it may be used inflight to determine aircraft status after recovery. Additional guidance specific to Special Operations Low Level II (SOLL II) operations is listed in AFMAN 11-2C-17V3, Addenda B, *C-17 Special Operations*. Additional guidance for Operation DEEP FREEZE operations is listed in AFMAN 11-2C-17V3, Addenda-C, *C-17A Antarctic Operations*.

4.7.2. Navigation systems and equipment listed in Flight Information Publication for permitting compliance with Minimum Navigation Performance Specifications (MNPS) is mandatory. Loss of any component before airspace entry requires return to a station with maintenance capability or re-filing via routes permitting equipment degradation. **(T-2)**

Table 4.1. C-17 Minimum Equipment List (MEL).

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
AIR CONDITIONING / PRESSURIZATION					
10,000 Foot Pressure Warning Aneroid Switch		1	1	1*	A one-time flight below 10,000 feet MSL to nearest repair capable facility is authorized IAW **paragraph 4.6.7.1.1**.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Air Conditioning Outlet Air Valves		13	0	0	
Air Conditioning Pack	L/R Pack DISAG	2	2	1*	1 complete air conditioning pack system must be fully operational for flight. Both packs required for takeoff into known icing conditions, or any Aeromedical Evacuation (AE) mission.
Air Conditioning Supply Check Valve, Cargo Compartment		4	2*	2*	1 per side required, inop valve(s) will be closed.
Avionics Cooling Differential Pressure Sensor		2	1	1	
Avionics Cooling Equipment Air Shutoff Valve		10	0*	0*	Valve required to be closed if corresponding avionics equipment is not installed.
Avionics Cooling Fan		3	2*	2*	3 required for airdrop above 25,000 feet MSL.
Avionics Cooling Fan Check Valves		3	3	3	
Avionics Cooling Skin Heat Exchanger		1	0*	0*	Will have air conditioning pack operating for ground operations. Valve will be manually locked open if heat exchanger is inop.
Avionics Ground Cooling Inlet Filter		1	1	1	Will have filter installed for ground operation of avionics equipment.
Cabin Pressure Controller	AUTO PRESS SINGLE	2	1*	1*	These requirements include Cabin Pressure Sensor and Air Conditioning Outflow Valve Motor. 1 complete cabin pressure system will be operational.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Cabin Pressure Outflow Valve		1	1	1*	Outflow valve must be operational in manual mode.
Cabin Pressurization Panel Indicator Unit		1	1*	1*	2 of 3 indicators on the panel will be operational. The Cabin Altitude indicator must be operational.
Cargo Compartment Exhaust Fan		2	0*	0*	Required during prolonged operation of engine-powered equipment inside the aircraft.
Cargo Compartment Recirculation Fan		1	1	0*	Will be operable if 1 air conditioning pack is inop.
Cargo Compartment Redistribution Fan		1	0*	0*	Required for Aerial Bulk Fuel Delivery System and Dispensing System operations.
Engine Anti-Ice Systems (includes valves, cockpit switches, temperature sensors)		4	4	0*	Will be operational for flights into known or forecast icing. If icing is anticipated, manually open Shutoff Valve after associated engine has been started.
Environmental Control Panel, Auxiliary Power Unit (APU) Air		1	1	0*	Required if air cart is unavailable for engine start.
Environmental Control Panel, Pack Discharge Temperature/ Supply/ Compartment Temperature Indicator		8	0	0	
Environmental System Controller	ECS X	2	2	1*	If 1 inop, continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1**.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Environmental System Fire Detection Panel	ENV PANEL SINGLE/ INOP	1	1	1*	If ENV PANEL SINGLE, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Ground Inlet Shutoff Valve	GND COOLING INLET	1	1	0*	Will be manually closed for flight if inop.
HI-FLOW On Switch		1	0*	0*	Avionics Cooling Override Switch will be operational if HI-FLOW Switch is inop.
Ice Detector Probe		1	1	0*	Crew will monitor for ice if inop.
Inlet Air Temperature Sensor		6	3*	3*	1 sensor per zone will be operational.
L/R Pack DISAG Switch		2	1*	1*	Operable switch will correspond to operating pack.
Ram Air Inlet/Exhaust Doors		2	1*	1*	Inoperative Ram Air Inlet Door will be wired open. Cargo compartment heat may be degraded. Both doors are required for AE missions.
Ram Air Ventilation Valve		1	1	1	Left Ram Air ducts must be operational.
Ramp Temperature Sensor		1	0	0	
Remote Temp Control Switch		1	0*	0*	Loadmaster Temp Control Selector will be operational if Remote Temp Controller Switch is inop.
TAT Heater	TAT HEATER L/R DISAGREE	2	2	2	
Trim Air Check Valve		3	2*	2*	Center check valve may be inop.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Trim Air Regulator		2	1*	1*	These requirements cover the Trim Air Differential Pressure Sensor. Inop valve will be locked closed. All associated equipment will be operational on same side as operational trim air regulator.
Trim Air Switch /Flight Deck Overhead Panel		1	0	0	
Window Defog		1	1*	1*	Defog must be operational for windshields (front) and sliding windows.
Windshield Ice Protection		2	2	0*	Will be operational for flights into known or forecast icing.
Windshield Wipers		2	2	0*	Required for operations in precipitation.
Wing Ice Protection System (includes valves, cockpit switch, temperature sensor)		2	2	0*	Will be operational for flights into known or forecast icing. Failed valve will be locked closed.
Zone Temperature Control		3	3*	3*	Auto or manual control. Must have control of all 3 zones.
Zone Temperature Sensor		6	3*	3*	1 sensor per zone will be operational.
BLEED AIR					
Cowl Ice Protection Burst Duct Differential Switch		8	8	4*	1 per engine will be operable.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Engine Shut-Off Valve (SOV)		4	4	2*	1 SOV per wing may be failed in the closed position if flight is not conducted into known or forecast icing conditions. Any aircraft with an SOV failed in the open position will be considered Non Mission Capable. All components will be functioning on the operational bleed system.
Manifold Failure Detection System	MANF FAIL DET/ MANIFOLD FAIL	*	*	*	MANF FAIL DET indication warns of a loss of detection capability for region(s). Ensure all regions have monitoring capability prior to flight.
Manifold Failure Detector Controller	MANF FAIL DET	1	1	1	
Pneumatic Ground Service System		2	1	0*	APU required if ground service system inop.
Wing Ice Protection Burst Duct Differential		4	4	2*	1 per wing will be operable.
Wing Isolation Valve		1	1	1*	May be manually closed after engine start. If manually closed, all engine bleed air sources will be operational for each operating pack.
CARGO MISSION SYSTEMS					
Aerial Delivery System Controller (ADSC)		1	1	1*	Manual operation permissible to continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** unless AE or airdrop. Consideration will be given to cargo jettison capability. All associated components for airdrop are required.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Aircrew Data Transfer Device		1	1	0*	Will be operational if upload of Worldwide Navigation Database (WWNDB) and Terrain Avoidance Warning System (TAWS) is required.
AOC Hub		2	2	0*	Continue mission if WWNDB and TAWS databases will remain current for the duration of the mission.
Cargo Loading Stabilizer Struts		2	2	0*	Required to integrally jack the aircraft. Mission may continue to a station with repair capability if struts are not needed for mission accomplishment IAW **paragraph 4.6.7.2.1**.
Cargo Rails and Locks (ADS and Logistic)			*	*	**Home Station Departure:** All rails, locks, lips, and rollers will be fully operational. **Enroute:** Rails, locks, lips, and/or rollers may be inop if supplemental restraint is possible. Cargo will not be placed over missing/inop rollers.
Cargo Winch		1	1	0*	Mission may continue if winch is not needed for mission accomplishment.
Cockpit Printer		1	0	0	
Ramp Toes		4	4	0*	At least 1 ramp toe requires an operational proximity sensor. May have less than 4 operational toes if not needed for mission accomplishment.
Troop Doors		2	2	0*	As required for personnel airdrop or egress.
CARGO MISSION SYSTEMS (AIRDROP)					
Buffer Stop Assembly		1	1*	0*	As required for Container Delivery System (CDS) airdrop.
Drogue Parachute Video System		1	1*	0*	Required for heavy equipment airdrop unless a clear, unobstructed view of the drogue chute is available.
Gate Release Mechanism		6	6*	0*	As required for CDS airdrop.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Left Rail Bridge Assembly		1	1*	0*	Left rail bridge assembly required for heavy equipment airdrop.
Paratroop Retrieval Systems		2	2*	0*	As required for personnel airdrop.
Ramp Edge Cover		1	1*	0*	Required for heavy equipment airdrop.
Retrieval Winches		2	2*	0*	Required for airdrop using non-breakaway static lines.
Roller Conveyor Release Latches			*	*	Airdrop prohibited if release latch in load path is missing, damaged, safety wired, and/or taped to the cargo floor.
Station Keeping Equipment (SKE)/ Formation Flight System (FFS)		1	0*	0*	Per mission requirements.
Tow Release Mechanism		1	1*	0*	Required for heavy equipment airdrop.
Troop Door Air Deflector		2	0*	0*	As required for personnel airdrop.
COMMUNICATIONS					
AERO-I, Airline Operational Control (AOC), Controller-Pilot Direct Link Communication (CPDLC)	AERO-I INOP CMU 1, 2	1	1	0*	As required for ATC airspace restrictions.
Comm 1		1	1*	1*	ARC-210 Very High Frequency (VHF) and Ultra High Frequency (UHF) capability required. ARC-210 Secure Voice and SATCOM will be operational IAW mission requirements.
Comm 2/UHF/ VHF		3	3	2	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Communication Control Unit (CCU)	CCU FAULT X	2	2	2*	With 1 inop, a one-time flight to a repair capability facility is authorized IAW **paragraph 4.6.7.1.2**.
Communication Navigation Control Panel (CNC)		2	2	1*	Pilot CNC must be operational.
Control, Intercom Set (ICS)		7	4*	4*	Pilot, copilot, and forward and aft loadmaster intercom control sets will be operational.
High Frequency (HF) Radios		2	0*	0*	Per mission requirements. 1 required for flights over water.
INMARSAT AERO-H Antenna		1	0*	0*	Per mission requirements.
Public Address (PA) System		1	1	0*	When carrying passengers, PA will be operational unless other suitable means of communication are available.
UHF SATCOM/ GPS Antennas		3	0*	0*	Per mission requirements.
DEFENSIVE SYSTEMS					
CMDS		1	0*	0*	Per mission requirements.
IRCM		1	0*	0*	Per mission requirements.
MWS		1	0*	0*	Per mission requirements.
DOORS					
Cargo Door Ditching Locks		4	4	4*	Manual operation permissible to continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** unless AE or airdrop. Consideration will be given to cargo jettison capability.
Cargo Door Downlock Assemblies		2	2	2	
Cargo Door Uplocks		2	2	2	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Cargo Door/Ramp Proximity Indicating Systems			*	*	All proximity sensors and indicating systems associated with the cargo door and ramp system will be operational. May be inop on unpressurized flights if it can be determined that the locks are positively locked. All proximity sensors and indicating systems affecting the ADSC, Loadmaster's Forward Control Panel (LFCP), Loadmaster's Aft Control Panel (LACP), and Precision Airdrop System (PADS) will be operational for airdrop.
Cargo Ramp Latches		22	22	22*	All cargo ramp latches will be operational. Manual operation permissible to continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** unless AE or airdrop. Consideration will be given to cargo jettison capability.
Crew Entrance Door	ENTRY DOOR	1	1	1	Indicating systems will be operational.
Door Warning Indication Systems (Crew, Troop, Emergency Exit, Air Deflector)	X DOOR		*	*	Indicating systems will be operational.
Electro-Mechanical Ramp Locks		2	2	2*	All cargo ramp electrical safety locks will be operational. Manual operation permissible to continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** unless AE or airdrop. Consideration will be given to cargo jettison capability.
Emergency Exit Door	EMERG EXIT	1	1	1	Indicating systems will be operational.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Horizontal / Vertical Stab Access Doors, Maintenance Ditching Hatch, Oxygen Access Doors, Service Hatch (Belly), Proximity Indicating Systems	HATCH/ ACCESS		*	*	May be inop if the door is visually verified closed and locked.
Sidewall Jamb Spindles		34	34	34	
ELECTRICAL					
60hz Power Supply System		1	1	0*	Will be operational for AE missions.
AC BUS TIE Relays		4	4	4*	With 1 inop and all Integrated Drive Generators (IDGs) operational, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. On missions with Air Transportable Galley Lavatory (ATGL) or Distinguished Visitor (DV) Module installed and powered, all 4 IDGs must be operational.
AC X TIE		1	1	1*	If inop, continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** if 4 IDGs and all AC bus ties are operational. If operating with 3 IDGs or any AC bus tie is failed, a one-time flight to nearest repair capable facility is authorized IAW **paragraph 4.6.7.1.1**.
Batteries and Battery Chargers	BATT (X) NOT CHARGING	2	2	2	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
DC BUS TIE Relays		2	2	2*	With 1 inop and transformer rectifiers operational, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
DC X TIE		1	1	1	
Emergency Power Relay		1	1	1	EMER PWR IN USE illuminates when relay is tied.
External Power		1	1	0*	APU Generator will be operational. External power is required for AE missions.
Integrated Drive Generators (IDG)	GEN/OFF Switchlight Illuminated	4	3*	3*	With 2 inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. On missions with ATGL or DV Module installed and powered, all 4 IDGs must be operational.
Loadmaster Bus 1	LM 1 BUSES	1	1	1	
Loadmaster Bus 4	LM 4 BUSES	1	1	1	
Static Inverter		1	1	1	
Transfer Buses	AC/DC XFER BUS OFF	2	2	2	
Transformer Rectifiers		4	3*	3*	DC X TIE and both DC BUS TIEs will be operational.
ENGINES/APU					
APU		1	1	0*	APU will be operational for any mission departure into a field without alternate electric & air sources when engine shutdown is planned.
EEC	EEC FAULT X	4	4	4*	1 channel (A or B) may be inop. If channel A is inop, engine will operate in N1 mode. Continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1**.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Exhaust Gas Temperature (EGT) Thermocouple Probes		24	20*	20*	6 probes per engine, 5 required. 2 channels per engine, 1 required.
Engines		4	4	4	
Ignition System		8	8	4*	Channel B may be inop. If only 1 engine has channel A inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Low Oil Pressure Indication	ENG OIL PRESS (X)	4	4	4	
Oil Quantity Transmitter	ENG OIL ABNORMAL	4	4	0*	Oil pressure and temp indications will be operational. Verify oil quantity prior to flight.
Oil Temperature Indication		4	4	4	
Standby Engine Display (SED)		1	1	1	
Starter Control Valve		4	4	4*	Starter control valve will be manually operable. For manual operation, starter position indicator (amber engine start button light) must be operable.
Thrust Reverser (TR)		4	2*	0*	Inop TRs will be locked out symmetrically per job guide.
Turbine Cooling Air Valve (TCA)	ENG STABILITY X	4	0*	0*	Inop Valve must be safed open for flight.
EMERGENCY EQUIPMENT					
Cockpit Voice Recorder (CVR)		1	1	1*	If inop, continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** if the Signal Data Recorder-Reproducer is operational.
Crash Axes		2	2	2	
Emergency Locator Transmitter (ELT)		1	1	1	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Flotation Equipment Deployment System (FEDS)			*	*	All FEDS components will be operational. All 4 ladders must be installed.
FEDS Life Rafts		3	0*	0*	Raft quantity will be adequate to accommodate total persons onboard when flight exceeds power off gliding distance from land.
Portable Fire Extinguishers		9	9	7*	Required placement areas: flight deck/crew rest area (3), forward loadmaster station (1), crew entrance door (1). The remaining fire extinguishers will be placed in the cargo compartment at the loadmaster's discretion.
Ramp Blow Down System		1	1	1	
Underwater Beacon		1	1	0*	Required for overwater flight.
Warning Horn		4	2*	2*	1 cargo bay and 1 underfloor warning horn must be operational.
EQUIPMENT AND FURNISHINGS					
Lavatory/Toilet		1	1	1*	Continue the mission (if practical) to a repair capable facility IAW **paragraph 4.6.7.2.1**. Can be inop with comfort pallet onboard. See AFMAN 11-2C-17V3, Addenda A, *C-17 Configuration and Mission Planning,* for passenger restrictions.
Potable Water System		1	0	0	If inop, ensure adequate supply of water.
Refrigerator		1	0*	0*	Mission may continue if meal refrigeration is not required, or alternate cold storage is available.
FIRE PROTECTION					
APU Fire Bottle	AGENT LOW	1	1	0*	If inop, the APU will not be used.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
APU Fire Detection System		1	1*	0*	Column A: Either loop A or B will be operational. Column B: If both loops are inop, the APU will not be used.
Avionics Smoke Detector		2	1	1	
Cargo Compartment Smoke Detector		14	6*	6*	Sensors 9, 10, 13 & 14 plus 2 others will be operable.
Crew Rest Smoke Detector		1	0	0	
Engine Fire Bottle	AGENT X LOW	4	4	4*	A one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Engine Fire Detection System		4	4*	4*	Either loop A or B for each engine will be operational.
Inertial Reference Unit (IRU) Smoke Detector		4	4	3*	Corresponding IRU must be deactivated. Reference IRU guidance (this chapter) and T.O. 1C-17A-1, Sect III IRU INOP procedure.
Lavatory Smoke Detector		1	0	0	
FLIGHT CONTROLS (AUTO-FLIGHT)					
Automatic Flight Control System (AFCS) Panel		1	1	1	
Aileron Trim Actuator		1	1	1	
Air Refueling (A/R) Mode	A/R MODE INOP		*	*	Required for A/R missions.
Alpha Limiter System	ALPHA LIMIT INOP	1	1	1	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Angle of Attack (AOA) Vanes	FCC_AOAx on Avionic Fault List	6	6	5*	1 AOA may be failed (single Cat 2 fault) if both Aircraft/Propulsion Data Management Computers (A/PDMCs) are operational. Verify only 1 AOA vane is INOP by comparing Avionic, Electronic Flight Control System (EFCS), A/PDMC Fault List and Warning Annunciator Panel (WAP).
Autopilot and Autothrottles		1	0*	0*	Per mission requirements.
Autopilot/Air Refueling Disconnect Switch		2	0*	0*	Required for A/R missions.
Autothrottle/ Air Refueling Disengage Button		1	0*	0*	Required for A/R missions.
Axis Fail Operation Modes	Roll, Yaw, or Pitch FAIL OP		Req	Req	Do not takeoff with any axis FAIL OP conditions.
Direct Lift Control (DLC)/Speed Brake Switch		2	1	1	
EFCS Pre-flight BLIN Codes	EFCS RESET		*	*	No more than 1 CAT 2 fault for Home Station and no more than 2 CAT 2 faults for Enroute departures.
Electronic Flight Control Axis	FCS cue with any EFCS Axis “FAIL” Light	5	5	5	
Flap Position Indicator		1	0*	0*	Multi-function display (MFD) indication will be operational.
Flap Position Transducer		4	4	4	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Flight Control Computer (FCC)	FCC X	4	4	4*	With 1 inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. For one-time flights, both Spoiler Control/Electronic Flap Computer (SC/EFC) will be operational, no (Pitch, Yaw, Roll, Pitch Trim Fail) FAIL OP messages will be illuminated, and the FCC pre-flight built-in test (PFBIT) must have been accomplished in the previous 72 hours. Air Refueling may be restricted IAW T.O. 1C-17A-1. If FCC 1 or FCC 4 is the failed FCC, flight is limited to FL200.
Ground Proximity Warning System (GPWS)	GPWS FAIL	1	1	1*	If inop, TAWS is required. A one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Integrated Flight Control Module (IFCM), Rudder		2	2	2*	With the upper IFCM inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. Lower rudder must be fully operational.
Pitch Autopilot Actuator	PITCH AP ACT FAIL	1	1	1*	Continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1**. if the FCC PFBIT has been accomplished within the previous 72 hours.
Pitch Ratio Change Actuator	PITCH RATIO DISAG	1	1	1	
Roll Autopilot Actuator	ROLL AP ACT FAIL	1	1	1*	Continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1** if the FCC PFBIT has been accomplished within the previous 72 hours.
Roll Ratio Change Actuator	ROLL RATIO DISAG	1	1	1	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
SC/EFC Pre-flight BLIN Codes		*	*	*	No more than 1 CAT 2 fault for Home Station and no more than 2 CAT 2 faults for Enroute departures.
Slat Actuator		16	14*	14*	1 actuator per wing may be inop.
Speed Brake Indicator		1	0*	0*	MFD indication will be operational.
Spoiler Control/ Electronic Flap Computer	SC/EFC X	2	2	2*	With 1 inop and 4 FCCs operational, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**, provided SC/EFC PFBIT was accomplished in the previous 72 hours and flight control or hydraulic system integrity has not been compromised. Air refueling may be restricted IAW T.O. 1C-17A-1.
Stall Warning System	STALL WARN INOP	1	1	1*	One-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**, if Alpha Limiting System is operational. STALL WARN INOP will be annunciated when both stall system channels are inop.
Stick Shaker		2	2	2*	With 1 inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Surface Fail Operation Modes	Aileron, Flap, Elevator, Rudder, or Slat FAIL OP		Req	*	With any single surface fail op condition, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
System Fail Operation Modes	ALS, Pitch Trim, ADC FAIL OP		Req	*	A one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
TOGA Button	TOGA BUTTON FAIL	1	1	1	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Trim Indicators: Aileron, Rudder, Horizontal Stabilizer		1	0*	0*	MFD indication will be operational.
FUEL					
A/R Amp Override Switch, Overhead Panel		1	0*	0*	Required if Emergency Boom Latching authorized by mission directive.
A/R Annunciator Dimming Unit		1	0*	0*	Required for night A/R missions.
A/R Annunciator Lights (READY, DISC, & LATCHED)		1	0*	0*	For A/R missions READY light may be inop if overhead panel READY light is operational.
A/R L/R Master DISAG Switch, Overhead Panel		2	1*	0*	Separation valve will be operable. Inop valves will be closed prior to takeoff. Required for A/R missions.
A/R Master Valves		2	0*	0*	Inop valve will be manually closed prior to takeoff. With any inop valve, the center separation valve will be operable. 1 required for A/R missions.
A/R Rheostat Annunciator/ Slipway, Overhead Panel		1	0*	0*	For night missions, A/R may only be accomplished if at least 1 tanker boom nozzle light is operable.
Crossfeed Valves		4	4	4	
Dump Valves		2	2	1*	Left and Right Master, or separation valve, will be operational.
Fill Valve		4	4	2*	4 required for A/R missions. Fill valves 1 and 4 will be operational. Over wing refueling is required for affected tanks.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Fuel Manifold Drain & Check Valves & Pump		1	1	0*	May be inop but manifold must stay dry and have manifold drain capability.
Fuel Quantity Computer	FUEL QTY SYSTEM	1	1	1*	With 1 channel inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Fuel Quantity Display, Overhead Panel		4	4	3*	Total fuel quantity indication will be operational with any single overhead fuel quantity display inop. Erratic or fluctuating indications will be considered inop.
Fuel Quantity Probes – Extended Range (ER) aircraft			*	*	1 probe in main tanks 1, 2, 3, & 4 may be inop and 1 additional probe in ER tanks 2 & 3 may be inop. No more than 6 probes per aircraft may be inop.
Fuel Quantity Probes (Non-ER aircraft)			*	*	1 probe per tank may be inop. No more than 4 probes per aircraft may be inop.
Ground Refuel L/R Valve Switch, Overhead Panel		2	0*	0*	Ground Refueling Panel will be Operational.
Ground Refueling Control Panel		1	1	0*	Left or Right ground refueling and separation valve switchlights on overhead refuel panel must be operational.
Ground Refueling Isolation Valve		2	0*	0*	Inop valve will be closed manually prior to takeoff.
Ground Refueling Manual Drain Valve		1	1	1	
Ground Refueling Receptacle		2	1*	1*	Separation valve must be operational.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Hi-Level Shutoff Test Valve		4	4	0*	4 required for A/R missions. Quantity Select method required for ground refueling.
Low-Level Fuel Dump Shutoff		4	4	4	
Main Tank Boost Pumps		8	6*	6*	1 per wing may be inop if inboard transfer pumps and crossfeed valves are operational on affected side.
Overhead Fuel Panel	FUEL PANEL SINGLE/ INOP	1	1	1*	If FUEL PANEL SINGLE, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Separation Valve		1	0*	0*	If failed closed, both A/R isolation valves will be operable and both ground refueling receptacles must be operational.
Total Fuel Quantity Indicator		1	1	0*	Required if any fuel quantity display inop.
Transfer Pump, Extended Range (ER) Tanks		4	0*	0*	For extended range missions, 1 pump required per side.
Transfer Pump, Wing Tanks		4	2*	2*	If fuel quantity in tank 2 or 3 is greater than 36K lbs., respective XFER pump will be operational. 1 transfer pump/switch per wing may be inop; tank with inop pump will have both boost pumps and crossfeed valve operational.
UARRSI Door Assembly & Handle		1	0*	0*	Door will be verified open before flight for A/R missions.
UARRSI System		1	0*	0*	Required for A/R missions.
Valve Assembly, Primary Climb/Dive		2	0*	0*	Secondary climb/dive valve, override solenoid valve, and fuel tank pressure transducer will be fully operational.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Valve Assembly, Secondary Climb/Dive		2	0*	0*	Primary climb/dive valve, override solenoid valve, and fuel tank pressure transducer will be fully operational.
Valve Assembly, Solenoid, Fuel Vent Override, Fuel Tank Pressure Transducer		2	0*	0*	Primary and secondary climb/dive valve will be operational.
HYDRAULICS					
AUX Pumps		4	4	3*	If the failed Aux pump is on #2 or #3 system, the transfer pump will be operational. If failed Aux pump is on #1 or #4, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. Open the circuit breaker and select the pump to AUTO (prevents associated "HYDRAULIC" takeoff warning).
Engine Driven Hydraulic Pumps		8	8	5*	System #1 and #4 must have one of the two engine driven pumps operational. System #2 and #3 may have one engine driven pump inop provided both system #2 and #3 have Aux and transfer pumps operational. If an engine driven pump fails to depressurize, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. **Note**: "HYDRAULIC" will annunciate on takeoff roll. Crew should brief takeoff considerations.
Hydraulic Manifold Press Transducer		4	3*	3*	Associated pump LOW pressure light and temp indicator required.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Hydraulic Quantity Transducer		4	0*	0*	Associated system reservoir low quantity proximity sensor required.
Hydraulic Reservoir Low Quantity Proximity Sensor		4	4	3*	May be inop on system #1 or #4 only. Associated system reservoir hydraulic quantity transducer required.
Hydraulic System Control Panel	HYD PANEL SINGLE/ INOP	1	1	1*	If HYD PANEL SINGLE, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Hydraulic System Controllers	HSC SINGLE/ INOP	2	2	2*	If HSC SINGLE, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Ram Air Turbine		1	1	1	Will be stowed prior to departure.
Transfer Pump		1	1	0*	If inop, all system 2 and 3 (engine driven and AUX) pumps will be operational.
INDICATING SYSTEMS					
Annunciator (Light), WACS Fail		2	1	1	
Central Aural Warning Computer	AURAL WARNING INOP	1	1	1	
Loudspeaker, CAWS		2	1	1	
Proximity Sensor Interface Unit (PSDAU, PIU)	PROX UNIT (X)	2	2	2	
Quick Access Recorder (QAR)		1	1	1*	If inop, continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1**.
Signal Data Recorder-Reproducer (SDRR)	FLIGHT RECORDER	1	1	1*	If inop and the CVR is operational, continue the mission to a station with repair capability IAW **paragraph 4.6.7.2.1**.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Switch, Master Caution & Reset		2	1	1	
Switch, Master Warning & Reset		2	1	1	
Warning and Caution Computer (WCC)	WAC COMPUTER (X)	2	2	2	
WAP		1	1	1	
LANDING GEAR AND BRAKES					
Anti-Skid Brake Temperature Monitor Control Unit		1	1	1	
Anti-Skid Braking	ANTI-SKID INOP	1	1	1	
Brake Accumulator		2	2	2	
Brake Pressure Indicator, Cockpit		1	1	1	
Landing Gear Indicators		2	2	2*	Accurate gear indication will be available on either the configuration page or the landing gear control indication panel. With 1 inop, continue the mission to a station with repair capability IAW **paragraph 4.6.7.2.1**.
Multiple Disc Brakes		12	12	10*	A brake worn beyond limits does not automatically require deactivation. If deactivation is required, then 1 brake/hose per side may be deactivated and no more than 2 brakes may be deactivated on the aircraft at a time.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Nose Wheel Steering Control (Tiller)		2	2	2*	With 1 inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. The Pilot Flying will have the operable tiller.
Parking Brake	PARK BRAKE INOP	2	2	1*	With PARK BRAKE INOP, the parking brake levers will engage, allowing the IRUs to align. A one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Sensor, Temperature, Brake Monitor		12	8*	8*	1 sensor per bogie may be inop. Erroneous indications verified by qualified maintenance personnel and documented in AFTO 781A may be considered inop.
Steering Cylinder Assembly		2	2	2	
Transducer, Motional Pickup, Wheel Speed, Main Landing Gear		12	12	10*	Brakes (7 or 8) and (11 or 12) transducers will be operational. Brake on affected wheel will be deactivated.
Wheel & Tire Assembly, Main Landing Gear		12	12	12	See Tire Limits in T.O. 1C-17A-1 Chapter 5. With any tire exceeding the Maximum Wear Limit (3 cords showing), a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Wheel & Tire Assembly, Nose Landing Gear		2	2	2	See Tire Limits in T.O. 1C-17A-1 Chapter 5. With any tire exceeding the Maximum Wear Limit (3 cords showing), a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
LIGHTING					

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
A/R Flood Light		1	0*	0*	Required for night A/R.
Emergency Exit Lighting Systems		3	3	3	
Emergency Exit Signs		13	13	13	
Emergency Lighting, Battery Power Supply		3	3	3	
Flight Compartment Lighting		1	1	1*	Main instrument panel floodlight, cockpit dome, and thunderstorm lights will be operational for night flight.
Fuselage In-Trail Light		2	0*	0*	2 required for night formation.
Nose Landing Light, Covert		2	0*	0*	A Column: Both Infrared (IR) nose landing lights will be operational for night vision goggle (NVG)-required missions. B Column: One IR nose landing light will be operational for NVG-required missions.
Nose Landing Light, Overt		2	2	1*	1 wingtip or nose landing light on each side will be operational.
Nose Taxi Light		2	0*	0*	Nose taxi light may be inop provided the nose landing light on the same side is operational. Not required if mission will be conducted in day/Visual Meteorological Conditions (VMC).
Nurse Call Light & Buttons		2	0*	0*	Will be operable if required by Medical Crew Director.
Rudder-Based Tailcone In-Trail Light		2	0*	0*	1 required for night formation.
Runway Turnoff Light, Overt/Covert		2/2	0	0	

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
UARRSI Perimeter Light Panel		3	0*	0*	For night missions, A/R may only be accomplished if at least 1 tanker Tail Mounted Floodlight is operable.
UARRSI Slipway Light		4	0*	0*	Required for night A/R.
Upper & Lower Anti-Collision Light		2	2*	2*	See AFMAN 11-202V3 requirements. If either the upper or lower light is inop, a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**. For Semi-Prepared Runway Operations (SPRO), lower light may be removed, upper light must be operational.
Wing In-Trail Light		2	0*	0*	2 required for night formation. Wing tip position lights can be used as an alternate for training only.
Winglet Retractable Landing Lights, Covert		2	0*	0*	Both IR wingtip lights will be operational for NVG-required missions.
Wingtip Landing Lights, Overt		2	2	1*	1 wingtip or nose landing light on each side will be operational.
Wingtip Navigation Lamp, Aft Position		4	4	2*	1 position light assembly per wing will be operational. **Note:** There are 2 light assemblies with single white bulbs on each wing.
Wingtip Navigation Lamp, Forward Position		4	4	2*	1 position light assembly per wing will be operational. **Note:** There are 2 bulbs in each lighting assembly.
Wingtip Recognition Lights (Strobe)		4	0*	0*	See AFMAN 11-202V3 requirements.
NAVIGATION SYSTEMS					

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
1553 Databus	1553 Degraded (Scratchpad)	1	1	0*	Continue the mission to repair capable facility IAW **paragraph 4.6.7.2.1**.
Air Data Computer (ADC)	ADC 1A, 1B, 2A, 2B	2	2	2*	Each ADC has 2 channels. Must have 3 operational channels, to include both 1A and 2B.
A/PDMC	APDMC X	2	2	2	
Bearing Distance Heading Indicator (BDHI)	Blank display or OFF flags	2	2	1*	Pilot will have a full set of standby indicators.
Commercial Global Positioning System (CGPS)		2	1	0*	Both may be inoperative if MGPS is operational. Refer to Precision Landing System Receiver (PLSR) 1/2 (this chapter) for ADS-B restrictions.
Core Integrated Processor (CIP)	MC X	2	2	2	
Distance Measuring Equipment (DME) 1/2		2	1*	1*	Per mission requirements.
Heads Up Display (HUD)		2	2	1*	5 of 6 displays (HUD/MFDs) will be operational. 2 HUDs required for assault landings and NVG.
Identification Friend or Foe (IFF)		1	1	1*	Mode S required for ADS-B operations. Mode 1, 2, & 5 may be inop based on mission/airspace requirements.
IRU & Battery	IRU INOP X	4	4	3*	Positions 1 and 4 must be operational.
LF/ADF		1	0*	0*	Per mission requirements.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Military Global Positioning System (MGPS)		2	1	0*	As required for ATC airspace restrictions. Both may be inoperative if CGPS is operational. If both MGPS are inop: -- Weather Radar & Flight Path Vector (FPV) will revert to degraded mode. -- FFS will not function; airdrop not authorized.
Mission Computer/ Communication Display (MCD)		4	4	3	
Mission Computer/ Communication Keyboard (MCK)		2	2	2	
Multi-Function Control (MFC)		2	2	2	
Multi-Function Display (MFD)	MFD (X)	4	4	3*	5 of 6 displays (HUD/MFDs) will be operational.
Pitot Static Probes	P/S XX MAST/HEAD HTR	4	4	4*	Verify heating elements on all 4 probes are operational. If fault is an indication only, continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1**.
PLSR 1/2	NAV X	2	2	1*	PLSR #1 must be operational for ADS-B operations. Both are required when CAT II ILS operations are expected. Aircrew will comply with single FM Immunity receiver procedures in the applicable Area Planning series.
Radar Altimeter	RADAR ALT (X)	2	1	1	
Standby Altimeter Airspeed Indicator		2	2	1*	Pilot will have a full set of standby indicators. Altimeter set function will be operational on both.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Standby Attitude Indicator		2	2	1*	Pilot will have a full set of standby indicators.
Tactical Aircraft Control and Navigation (TACAN)		1	1	0*	Per mission requirements.
Traffic Alerting and Collision Avoidance System (TCAS)	TCAS INOP	1	1	1*	Required if ATC airspace mandates equipment; otherwise a one-time flight to a repair capable facility is authorized IAW **paragraph 4.6.7.1.2**.
Terrain Avoidance Warning System	TAWS FAIL	1	1	1*	If inop, continue the mission to a repair capable facility IAW **paragraph 4.6.7.2.1**.
Weather Radar		1	1	0*	Required for air refueling when TCAS is inop or when thunderstorms are forecast for the planned route of flight.
OBIGGS					
OBIGGS I (ER aircraft)		2	1*	1*	Single system operation allowed if supply crossfeed valve is operational.
OBIGGS I (Non-ER aircraft)		2	0	0	
OBIGGS II		2	1*	1*	Single system operation allowed if crossfeed valve is operational. Reference T.O. 1C-17A-1 Section II, Operation With OBIGGS Shutoff Valve(s) Locked Closed.
OXYGEN					
25-liter Crew LOX Converter	CREW OXY LOW	1	1	1*	May be inop if passenger and/or auxiliary system and crossfeed are operational.
75-liter AUX Converter	OXY LOW	1	1	1*	May be inop if passenger system is operational.
75-liter PAX LOX Converter	OXY LOW	1	1	1*	May be inop if auxiliary system is operational.

Item/System	Message/Cue /Alert	Installed	Required		Remarks/Limitation/Exceptions
			A	B	
Portable Oxygen Bottles		10	10	6*	2 bottles (minimum) required for each primary crewmember. Ensure requirements of AFMAN 11-202V3 are met. Crew rest and forward loadmaster station will be equipped with Unmodified or Modified2 bottles.
Quick-Don Mask		15	15	*	Required for each occupied crew seat. When oxygen sources with attached mask are not prepositioned throughout the cargo compartment, crewmembers mobile in the cargo compartment will have a portable oxygen walk-around bottle with quick-don mask/helmet available for use in the event of an emergency.
Regulators		10	3*	3*	Pilot, copilot, and loadmaster regulators will be operational. Other regulator(s) required for each occupied crew position.
Note: An asterisk (*) in the required column indicates the number required is situation dependent; refer to the Remarks/Limitations/Exception column for clarification.					

Chapter 5

OPERATIONAL PROCEDURES

5.1. Duty Station. Both pilots will be in their seats during flight. **(T-2)** With both pilots in their seats, PICs may authorize rest periods for one pilot occupying a primary duty station during non-critical phases of flight (the other pilot will be awake and alert).

5.1.1. All crewmembers will be at their duty stations during all critical phases of flight, unless crew duties dictate otherwise. **(T-2)** Headsets or helmets will be worn at all times while crewmembers are operating at their primary duty station. **(T-2)**

5.1.2. During non-critical phases of flight, crewmembers may leave their duty stations, for brief periods, to meet physiological needs and perform normal crew duties provided one pilot remains at a primary duty station during flight. Crewmembers will coordinate with the pilot flying (PF) before departing their assigned primary duty stations. **(T-3)**

5.2. Takeoff and Landing Policy. An aircraft commander, or above, will occupy either the left or the right seat during all takeoffs and landings. **(T-2)** The designated PIC (A-code) is not required to occupy a primary position, but still retains overall authority for conduct of the mission.

5.2.1. An Aircraft Commander (AC) or Instructor Pilot (IP) will make all takeoffs and landings during:

5.2.1.1. Aircraft emergencies unless conditions prevent compliance. **(T-3)**

5.2.1.2. Emergency Nuclear Airlift Operations. **(T-3)** Additional information is in AFMAN 13-526, *Nuclear Airlift Operations.*

5.2.1.3. Assault or substandard airfield operations. **(T-3) Exception:** Pilots receiving upgrade training or receiving an evaluation.

5.2.1.4. Category II Instrument Landing System (ILS) approaches and landings when the weather is below Category I minimums. **(T-3)**

5.2.2. Unless the other pilot in the seat is a certified AC or higher, the PIC with less than 100 primary assigned aircraft hours since AC certification will make all takeoffs and landings under any of the following conditions:

5.2.2.1. Ceiling/visibility less than 300 feet and/or Runway Visual Range (RVR) 4000 (3/4 SM visibility). **(T-3)**

5.2.2.2. Runway Condition Reading (RCR) less than 12. **(T-3)**

5.2.2.3. Crosswind component greater than 15 knots. **(T-3)**

5.3. Landing Gear and Slat/Flap Operating Policy. The PF commands configuration changes. The pilot monitoring (PM) verifies appropriate airspeed and configuration prior to echoing the gear or slat/flap actuation command. The pilot in the right seat operates the landing gear. The slats/flaps are operated by the PM.

5.4. Observer/Jump Seat Duties. Unless crew duties dictate otherwise, an extra pilot or loadmaster should occupy the augmented crew station to the maximum extent possible, especially in times of increased crew workload (e.g., critical phases of flight) to assist in scanning, promote

effective Crew Resource Management/Threat and Error Management (CRM/TEM), and perform other tasks as determined by the PIC.

5.5. Seat Belts.

5.5.1. All occupants will have a designated seat with a seat belt. **(T-2)** A crew bunk does not meet the requirement of a designated seat.

5.5.2. Equipment will be properly secured, and all crewmembers and passengers will be seated with seat belts and shoulder harnesses fastened during critical phases of flight unless crew duties dictate otherwise. **(T-1).** The PIC may authorize passengers to observe AAR, but seat changes will be completed no closer than 100ft in trail. The PIC may authorize passenger movement via the aircraft seat belt signs. Crewmembers may remove the shoulder harness during non-critical phases of flight. **(T-2)**

5.6. Aircraft Lighting. Configure lighting IAW AFMAN 11-202V3, AFMAN 11-218, *Aircraft Operations and Movement on the Ground*, and applicable T.Os. See mission directives, SPINS, or Operations Order (OPORD) for NVG lighting configuration.

5.7. Aircraft Armor. If a flight control check determines interference or binding is occurring PICs are approved to remove and securely stow armor plates. **(T-3)** The PIC will then notify maintenance, if available, and write the removal up in the aircraft AFTO 781A. **(T-3)**

5.7.1. Aircrew are not permitted to replace the armor themselves. **(T-3)** Only maintenance can replace armor once it is removed.

5.7.2. If a mission requires aircraft armor, notify maintenance and have the armor replaced and secured properly prior to takeoff, if possible. If this is not possible, the PIC will assess the risk and make the decision whether or not to continue the mission to the location requiring aircraft armor.

5.8. Advisory Calls. Advisory calls are IAW AFMAN 11-202V3_AMCSUP. **Note:** Automated aircraft advisories satisfy this requirement if the crew acknowledges the advisory. On instrument approaches, if the PF can maintain visual contact with the landing runway, and states “Visual,” subsequent instrument advisory calls are not required.

5.9. Communications Policy.

5.9.1. Remain cognizant of critical phases of flight and do not allow non-critical communications to cause distractions. Communications with C2 should be avoided during critical phases of flight and while the aircraft is taxiing in congested areas. If a C2 interruption occurs during these times, answering AOC or AERO-I should be postponed or the crewmember monitoring the C2 radio should simply acknowledge with “Standby.”

5.9.2. Aircraft Interphone (ICS). Primary crewmembers will actively monitor ICS during critical phases of flight and remain connected to ICS during all phases of flight, unless crew duties dictate otherwise. **(T-2)** Crewmembers will advise the PF before checking off interphone or operating in ISOLATE. **(T-2)** Crewmembers will ensure personnel on headset, or within listening distance, are cleared prior to discussing classified information over interphone. **(T-2)**

5.9.3. Use secure and jam-resistant communications to the maximum extent possible. In a threat environment, limit radio transmissions with the objective area to those required for safety of flight or factors affecting force employment.

5.10. Transportation of Animals. Pets are defined in DoDI 4515.13, *Air Transportation Eligibility*. Transporting pets in conjunction with the sponsor's permanent change of station is authorized. Other pets or animals are normally prohibited but may be moved according to DoDI 4515.13. Service animals are working animals, not pets. Follow the guidance prescribed in Section 10 of DoDI 4515.13, when transporting a qualified individual with a disability who is accompanied by a service animal. **(T-0)** The service animal must be properly harnessed and leashed. **(T-3)** Proper sanitation is the responsibility of the passenger and must be maintained at all times. **(T-3)** Government-owned animals (military working dogs) will be transported in pet crates and manifested as cargo. **(T-3)**

5.11. Runway, Taxiway, and Airfield Requirements.

5.11.1. Minimum Runway and Taxiway Requirements. Minimum runway width is 90 feet/27.5 meters. Minimum runway length is 3,500 feet. Minimum taxiway width is 50 feet/15.5 meters. Minimum width for star turns is 90 feet/27.5 meters. **(T-2)**

5.11.2. Runway Length for Takeoff and Landing. Do not takeoff if computed critical field length exceeds runway available. Minimum runway for a normal landing (3/4 or full flap) is computed landing distance with IDLE reverse. **(T-2)**

5.11.3. Minimum runway length for a full flap assault landing, to include SPRO, is computed ground roll, with MAX reverse including applicable NEAT corrections, plus marked landing zone distance (marked landing zone distance is normally 500 feet). Touchdown is within the marked landing zone. Weather minimums will be circling minimums, no lower than 600-2. **(T-2)** Prior to landing, the PIC reviews takeoff performance to ensure they can take off after the planned offload/onload.

5.11.4. Pilots may accomplish intersection takeoffs provided the operating environment (i.e., gross weight, obstructions, climb criteria, weather, etc.) allows a safe takeoff and departure. Calculate takeoff performance based on the runway remaining from the point at which the takeoff is initiated.

5.11.5. During operations on runways partially covered with snow or ice, base takeoff computations on the reported runway surface condition (RSC) and/or RCR for the cleared portion of the runway. A minimum of 45 feet either side of centerline should be cleared. If 45 feet either side of centerline is not cleared, compute takeoff data based on the un-cleared portion up to 45 feet either side of centerline.

5.11.6. Use of Overruns. If approach end overruns are available and stressed or authorized for normal operations, they may be used to increase the runway available for takeoff. Departure end overruns (if stressed and authorized) may also be used for landing if needed.

5.11.7. Arresting Cables.

5.11.7.1. Do not land prior to or touchdown on approach end arresting cables (does not include recessed cables). If the aircraft lands before the cable, the crew should contact the tower to have the cable inspected. Contact the airfield manager if landing distance calculations require landing prior to the arresting gear to have the cables derigged.

5.11.7.2. Do not takeoff over an approach end cable that has been reported as slack, loose, or improperly rigged by NOTAM, automated terminal information service (ATIS), or ATC.

5.11.7.3. When conditions permit (aircraft gross weight, runway length, weather, winds, TOLD, etc.) and the pilot in command has considered the potential for damaging the aircraft, make takeoffs and landings beyond raised cable barriers. Use the entire length of runway only if necessary. Be aware that operations over arresting gear barriers at speeds in excess of taxi speed may result in damage to the aircraft.

5.11.8. Other Airfield Requirements.

5.11.8.1. Consult with AMC Airfield Suitability Branch (AMC/A3AS) for suitability guidance, COMM: 618-792-7114. Airfield certification requirements are detailed in the Airfield Suitability Restriction Report (ASRR). Aircrews and planning agencies will contact AMC/A3AS for all questions pertaining to airfield weight bearing capacity and will review the Global Decision Support System (GDSS) or ASRR. **(T-2)**

5.11.8.2. A current landing zone (LZ) survey (within the past five-years as specified in DAFMAN 13-217, *Drop Zone, Landing Zone, and Helicopter Landing Zone Operations*) is needed before using other than hard-surfaced runways or taxiways.

5.11.9. RCR Considerations. Runway Condition Assessment Matrix (RCAM) should be used in conjunction with reported RCR at some airfields. When no RCR is available, refer to the RCAM in T.O. 1C-17A-1-1 or T.O. 1C-17A-1CL-1, *Pilot's Abbreviated Flight Crew Checklist*. During periods of reported low RCR, the taxiways and ramps may have an even lower RCR than reported for the runway. The runway surface should be considered wet when water on the runway causes a reflective glare.

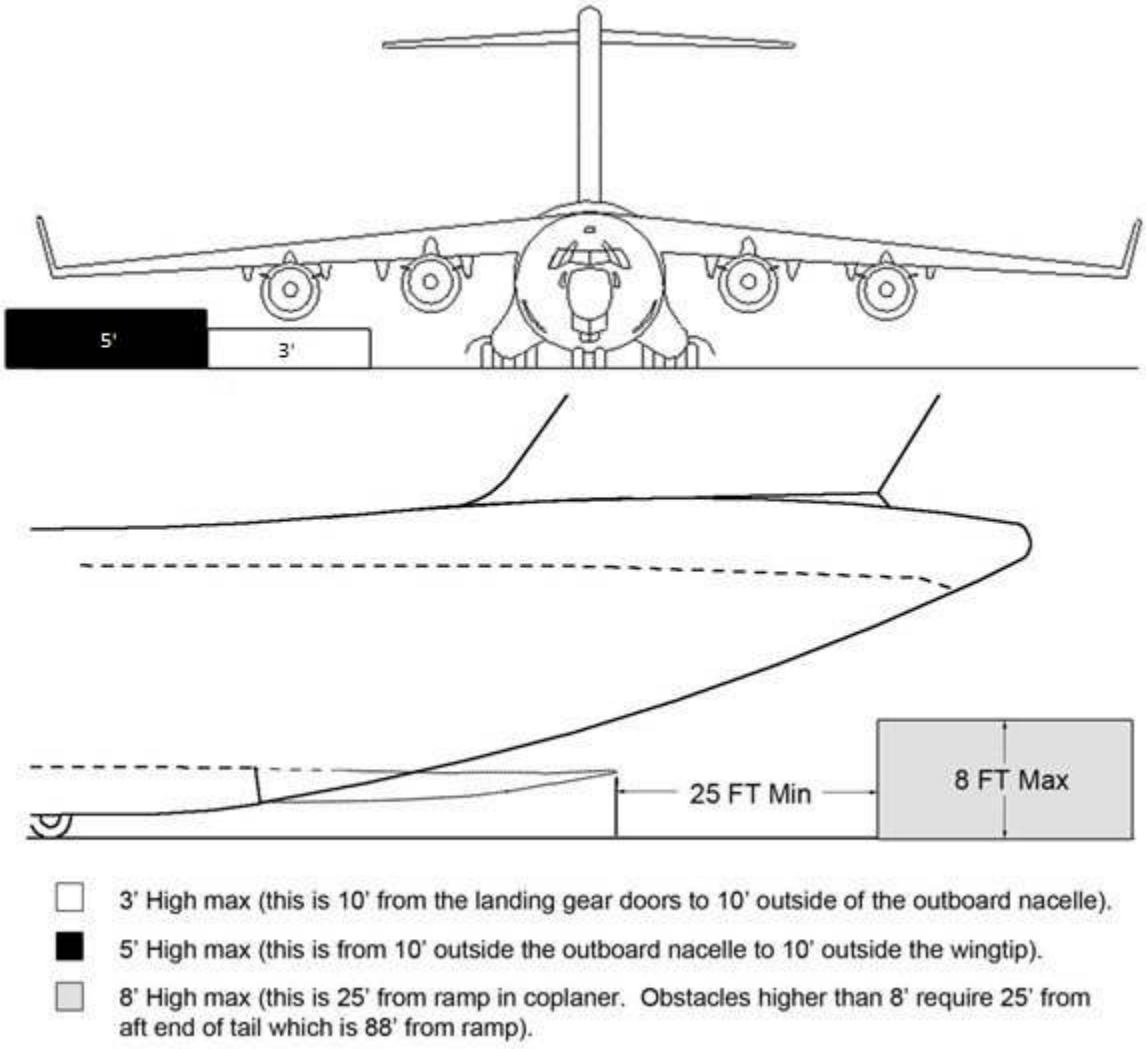
5.11.10. Semi-Prepared Runway Operations. See T.O. 1C-17A-1-1, *Performance Data,* Appendix B. OG/CC approval is required for semi-prepared runway operations within the Contiguous United States (CONUS), Alaska, and Hawaii. MAJCOM/A3 approval is required in all other instances. For semi-prepared LZs other than matted surfaces, the PIC will ensure proper engineering evaluations (i.e., Dynamic Cone Penetrometer (DCP) or equivalent) are completed by qualified personnel within one week of the first landing to verify the LZ meets C-17 requirements. **(T-2)**

5.12. Aircraft Taxi and Taxi Obstruction Clearance Criteria and Foreign Object Damage (FOD) Avoidance.

5.12.1. The pilot will coordinate taxi directions and signals to be used with the loadmaster and marshaller (when available). **(T-3)** Do not taxi an aircraft within 25 feet of obstructions without wing walkers monitoring the clearance between aircraft and obstruction. With wing walkers, avoid taxi obstructions by at least 10 feet. Wing walkers do not absolve the crew of their responsibility for obstruction clearance. For small permanent obstructions underneath the wing and tail, follow clearance criteria in **Figure 5.1**. During aircraft backing operations, do not taxi within 25 feet of an obstruction with or without a marshaller/wing walker. **Exception:** Stations where the WG/CC has established local procedures for operating under a waiver to the 25-foot requirement along taxi lines IAW AFMAN 11-218.

5.12.2. Adjacent aircraft are considered a permanent obstruction, provided the aircraft is parked properly in its designated spot and is not moving. Aerospace Ground Equipment (AGE) and vehicles are considered permanent obstructions, provided they are parked entirely within a designated area. Areas will be designated by permanent markings such as painted boxes or lines on the ramp or another suitable means. **(T-3)**

Figure 5.1. Taxi Obstruction Diagram.



5.12.3. Permanent obstacles (e.g., fuel hydrants), 3 feet high or shorter, are not considered obstructions as long as they remain a minimum of 10 feet away from the landing gear doors. A marshaller or crewmember verifies the height and location of 3 feet, 5 feet, and 8 feet high obstacles to ensure specified clearances from the wing, engines, landing gear doors, and ramp toes or ramp.

5.12.4. Consider positioning loadmaster(s) at paratroop doors to relay taxi obstruction clearances.

5.12.5. Stop the aircraft any time clearance is in question and deplane a crewmember to verify clearance and walk the wing, if necessary. If wing walkers are unavailable, deplane one or

more crewmembers to maintain obstruction clearance and provide marshaling signals IAW AFMAN 11-218, *Aircraft Operations and Movement on the Ground*. Use wing walkers, deplaned crewmembers, or a crewmember on interphone positioned at the paratroop door(s) or ramp to act as an observer while maneuvering on narrow taxiways. During night taxi operations, marshallers will have an illuminated wand in each hand. **(T-3)** Wing walkers are only required to have one illuminated wand. Observers should be in a position to see wing walkers at all times (through door or windows) and communicate with the pilot.

5.12.6. FOD Avoidance. Make every effort to minimize the potential for engine FOD. Crews should:

5.12.6.1. Carefully review airfield layout paying particular attention to taxi routes, turn requirements, and areas for potential FOD.

5.12.6.2. Minimize power settings during all taxi operations.

5.12.6.3. Where possible, avoid 180° turns. If required to accomplish a 180° turn on a narrow runway, the turn should be accomplished at an intersection of a link taxiway or at a designated turn around pad.

5.12.6.4. Where possible, avoid taxi operations that position an engine over an unprepared or un-swept surface. If unavoidable, leave the engine in idle (to the maximum extent possible) until the engine is over an improved surface.

5.12.7. Aircraft Backing:

5.12.7.1. The pilot will coordinate backing directions and signals to be used with the loadmaster and marshaller (when available). **(T-3)** During the entire backing operation, if the loadmaster states the word "Stop", the pilot will immediately stop the aircraft. **(T-2)** Failure to comply may result in damage to aircraft or injury to personnel.

5.12.7.2. Confirm all passengers are seated with seatbelts fastened and cargo is secure. **(T-2)**

5.12.7.3. The loadmaster must have enough maneuverability to observe and direct backing operations and report any hazards. **(T-3)** The loadmaster provides the pilot with continuous interphone instructions on conditions in the maneuvering area. This includes turns, distance remaining (for wingtips, empennage, and main gear), and stopping point. During aircraft backing, if the pilot and loadmaster lose interphone contact the pilot will stop the aircraft. **(T-3)**

5.12.7.4. When backing the aircraft at night, the pilots and loadmaster will ensure the taxi area is sufficiently illuminated. **(T-3)** Use the staging lights, retracted landing lights, or any other source that provides adequate lighting of the taxi area.

5.12.7.5. During backing operations, stop at least 25 feet from an obstruction with or without a marshaller/wing walker. The aircraft tail cannot be seen by the loadmaster from inside the aircraft and could possibly strike an obstruction well before the main gear tires are near the ramp edge. If any doubt exists as to sufficient clearance, stop the aircraft**.**

5.13. Functional Check Flights (FCFs), Acceptance Check Flights (ACFs) and Operational Check Flights (OCFs). Check flights are accomplished IAW T.O. 1C-17A-6*, Inspection Requirements Manual*, Department of the Air Force Instruction (DAFI) 21-101, *Aircraft and*

Equipment Maintenance Management, T.O. 1-1-300, *Maintenance Operational Checks and Check Flights*, and T.O. 1C-17A-6CF-1, *Acceptance and Functional Check Flight Procedures Manual.* Periodic Ram Air Turbine (RAT) checks are not considered part of an FCF check. RAT checks are conducted IAW T.O. 1C-17A-1, Ram Air Turbine Functional Deployment Procedure.

5.14. Traffic Alerting and Collision Avoidance System (TCAS).

5.14.1. For TCAS Traffic Alert and Warning procedures refer to T.O. 1C-17A-1*, Flight Manual*.

5.14.1.1. PM confirms the action being performed by the PF. Advise the PF of any deviation from the vertical path indicated by flight director guidance, the TCAS Plan Position Indicator (PPI) display, and/or other annunciations.

5.14.1.2. If a Ground Proximity Warning System (GPWS) or stall warning occurs, terminate the resolution advisory (RA) maneuver.

5.14.2. Multi-ship formation.

5.14.2.1. Lead aircraft (or designated alternates) will operate TCAS in the "TA only" mode. **(T-3)**

5.14.2.2. Consideration should be given to having the last aircraft in multi-element formations operating TCAS in "TA only" mode.

5.14.3. Low-level operations. This system was not designed for use in the low-level environment but could provide valuable awareness of light aircraft or other military aircraft using military airspace.

5.14.4. North Atlantic High Level Airspace (NAT HLA). IAW the NAT Doc 007, excessive climb and descent rates in excess of 1500 feet per minute within about 1500 feet of level-off altitude could lead to inadvertent TCAS warnings.

5.14.5. Resolution Advisory Documentation. The PIC will document all pertinent information surrounding an RA event on the AF Form 651, *Hazardous Air Traffic Report (HATR)*, and submit the report to the nearest USAF Safety office. **(T-3)**

5.15. Radar Altimeter. Any crewmember detecting the "TOO LOW" annunciation on the Primary Flight Display (PFD)/HUD) will immediately notify the PF. **(T-3)** Terrain clearance and aircraft position must be verified. **(T-3)**

5.16. Engine Starts.

5.16.1. Crossbleed Starts. If using an engine to provide bleed air for engine start, the PIC will ensure a FOD check is accomplished under and around the engines. **(T-3)** A higher chance of ingesting FOD exists due to the inlet vortices created when above idle power on the donor engine**.**

5.16.2. Buddy Starts. Buddy starts must be approved by MAJCOM/A3 or equivalent (OG/CC for local training missions). **(T-3)** Repeated buddy starts are not authorized for multiple scheduled enroute stops.

5.17. Ground Proximity Warning System (GPWS) / Terrain Alert Warning System (TAWS). During GPWS/TAWS WARNINGs, apply the Terrain Warning Maneuver IAW T.O. 1C-17A-1, unless during VMC operations with terrain/obstacle clearly in sight. Do not delay pull-

up for diagnosis of the low altitude WARNINGs. Failure to roll wings level during the maneuver described above decreases stall margin.

5.17.1. Exceptions to this policy are addressed in the GDSS Giant Report and ASRR for specific airfields. The PIC may only disregard GPWS/TAWS warnings at night or in Instrument Meteorological Condition (IMC) if all the following conditions are met: **(T-1)**

5.17.1.1. The airfield has a known GPWS/TAWS anomaly as listed in the GDSS Giant Report and ASRR.

5.17.1.2. The warning occurs inside the Final Approach Fix (FAF) and within any parameters listed in the Giant Report remark.

5.17.1.3. A plan of action is briefed to the crew prior to commencing the approach.

5.17.1.4. Either the aircraft is at or above the ILS glideslope or Derived Decision Altitude (DDA) on a non-precision approach with both PLSRs being monitored, or the runway is in sight with the aircraft at or above the glideslope as determined by the Visual Approach Slope Indicator (VASI), Precision Approach Path Indicator (PAPI), or similar device. If any of these conditions are not met execute the escape maneuver.

5.17.2. Ensure the mode of the GPWS/TAWS is commensurate with the aircraft's phase of flight.

5.17.3. Aircrews will annotate all TAWS and GPWS alerts deemed to be nuisance warnings in the aircraft forms (AFTO 781A). Write-ups must include type of alert (TAWS/GPWS), aural warning received, location, and time of incident. Maintenance, in turn, will forward the event data IAW the deficiency report procedures through appropriate command channels. **(T-2)** In addition, crews will annotate nuisance warning information on the TAWS/GPWS Nuisance Event Worksheet and forward to their unit OGV office for processing. **(T-2)**

5.18. Standard Automatic Flight Control System (AFCS) Terminology. Under certain conditions, complete commands may be required. This would require an action, the proper axis, then the setting (e.g., "SELECT HEADING 060").

5.18.1. SELECT or ENGAGE directs the selection of a value on the AFCS panel which results in the value being placed in the "Engaged" (Top) portion of the Flight Mode Annunciation (FMA). SELECT is normally used with rotary knobs labeled "SEL," however, ENGAGE is acceptable. The Flight Director (FD), Takeoff Go-Around (TOGA), Approach (APPR), Attitude (ATT), Airdrop (AD), Autopilot (AP), Autothrottle (AT) are normally engaged. For added clarification, selecting an Engine Pressure Ratio (EPR) rating means pressing a thrust rating button on the SED panel; engaging EPR means pressing the AFCS "EPR" pushbutton.

5.18.2. For ATC assigned altitudes, one pilot selects and arms the assigned altitude in the ALT window. Both pilots confirm the altitude in the ALT window and FMA is correct.

5.18.3. ARM directs the selection of a value on the AFCS panel which results in the value being placed in the "Armed" (Bottom) portion of an FMA.

5.18.4. For LNAV and VNAV selections, command the actual value intended, not "ARM LNAV" or "ENGAGE VNAV." For example, "ARM VOR" or "ENGAGE VPROF" are proper commands.

5.18.5. Standard terminology is precise but can unnecessarily congest cockpit communication. In-flight context generally allows simplified direction and execution. For example, “ARM 10,000” is obviously an altitude command and does not require the noun “altitude” to clearly communicate the PF’s desires. Airspeeds appended with “PITCH” or “THRUST” are sufficiently clear and do not require the use of the noun “speed.” VERT SPDS appended with “UP” or “DN” are sufficiently clear. If requested by the PF, ATC vectors (headings) may be automatically selected, and altitudes automatically armed by the PM. At any time the PM is unsure of the command, they will ask for clarification. **(T-2)** If the PF sees an uncommanded FMA, they clearly restate the command and ensure the actual value is made.

5.19. C-17 HUD/MFD Endorsement. The C-17 MFD and Replacement Heads Up Display (RHUD) are certified as a single medium display and may be used as a primary flight reference (PFR). Due to the lack of a full-time attitude reference, the Legacy Heads Up Display (LHUD) is endorsed as a PFR as long as one primary flight display is present on an MFD. To the maximum extent possible, keep a primary flight display (PFD) on one of the MFDs at all times(including RHUD equipped aircraft). When mission requirements dictate, the PFD may be replaced with another display for short periods of time.

5.20. C-17 Engine Start Policy. Maintenance personnel or a crewmember should be used as “ground” during engine starts. The engines may be started without someone outside the aircraft. If this option is used, ensure all crewmembers are thoroughly briefed.

5.21. Aircrew Data Transfer Device (ADTD). Issued devices from mission planning cells are approved. All other devices to be used on the ADTD require approval from AMC/A3V.

5.21.1. Do not make changes to any of the operating properties of the ADTD. Installing or running programs not specifically authorized in the flight manual, is prohibited.

5.21.2. The ADTD has not been granted approval for viewing/accessing electronic pubs, regardless of whether the program or file is saved to the hard drive or read and executed from a removable media source.

5.21.3. The ADTD requires a separate DVD/CD drive. This drive is not part of the aircraft equipment; therefore, they are included in the Mission Kit. If the drives are missing or damaged, do not make a write-up in the AFTO 781A.

Chapter 6

AIRCREW PROCEDURES

Section 6A—Pre-Mission

6.1. Large Aircraft Infrared Counter-Measure (LAIRCM).

6.1.1. Aircrew should utilize the LAIRCM to the maximum extent possible and IAW SPINS.

6.1.2. Removable User Data Module (UDM) cards will be stored in the secure stowage facility (gun box) on board the aircraft anytime the LAIRCM system is not in use. **(T-3)**

6.1.3. A Standard Form (SF) 702*, Security Container Check Sheet,* will be annotated by the individual opening and closing the gun box. **(T-3)**

6.1.4. The PIC is responsible for ensuring the UDM card is in the gun box prior to the flight, and before leaving the aircraft at the end of the duty day. If the UDM card is missing, notify appropriate C2 agency.

6.2. Aircrew Publications Requirements. All crewmembers will be issued an Electronic Flight Bag (EFB) and will maintain as a minimum, the publications specified in AFMAN 11-202V3, applicable MAJCOM supplement, and **Table 6.1** **(T-2)** Operating crewmembers will carry an EFB meeting these requirements on all sorties (N/A for flight surgeons). **(T-3)** EFBs are the primary reference source for abbreviated aircrew checklists. In the event of device failure(s), a minimum of 3 EFBs with a current set of ePubs and FLIP dataset are required for continued flight. **(T-2)** Reference AFI 11-215, *Flight Manuals Program*, for guidance on electronic publications.

6.2.1. OGVs will ensure units maintain and provide a current paper copy of T.O. 1C-17A-1CL-1, *Pilot's Abbreviated Flight Crew Checklist,* T.O. 1C-17A-1CL-2, *Loadmaster's Abbreviated Flight Crew Checklist,* and T.O. 1C-17A-1-6CL-1 in the aircraft mission kits, and T.O. 1C-17A-1CL-1-2, *Quick Reaction Card*, in the flight deck's center console paper holders. **(T-2)** Paper copies are intended as a backup medium in the event of EFB failures and must be accessible in flight. **(T-3) Exception:** The Quick Reaction Card (QRC) will be referenced in accordance with MAJCOM guidance.

6.2.2. Units will ensure a process is in place to confirm crews have synced their EFBs prior to departure from home station for missions, or prior to stepping to the aircraft for locals. **(T-3)**

Table 6.1. Minimum Aircrew ePubs.

Item	Medium
T.O. 1C-17A-1, *Flight Manual*	Digital
T.O. 1C-17A-1-1, *Performance Data*	Digital
T.O. 1C-17A-1-2, *Mission Computer*	Digital
T.O. 1C-17A-1-4, *Airdrop Mission Crew Manual*	Digital (Airdrop only)
T.O. 1C-17A-1-5, *Special Operations Crew Manual*	Digital (SOLL II only)
T.O. 1C-17A-9, *Loading Instructions*	Digital

Abbreviated Checklists	Digital: All abbreviated checklists for required parent manuals. Paper: One paper copy of T.O. 1C-17A-1CL-1, T.O. 1C-17A-1CL-2, T.O. 1C-17A-1-6CL-1, and T.O. 1C-17A-1CL-1-2 will be carried on all aircraft until removal directed by command guidance.
Fanfold Checklists	Digital and Paper
Checklist Inserts/Briefing Guides	Digital
AFMAN 11-202V3 (Parent and HHQ supplements)	Digital
AFMAN 11-2C-17V3	Digital
NATO Standard ATP-3.3.4.2, *Air-to-Air Refueling*	Digital
AFMAN 13-526 (CUI), *Nuclear Airlift Operations*	Digital or Paper (PNAF only)
AFI 91-115, *Safety Rules for Nuclear Airlift Operations*	Digital or Paper (PNAF only)
T.O. 1C-17A-16-1, *Loading and Air Transport of Nuclear Weapons (Non-palletized)*	Digital or Paper (PNAF only)

6.3. Airfield Review. As a minimum, airfield review will include:

6.3.1. Airspace/Airfield Review. FLIP, Flight Information Region (FIR)/Upper Information Region (UIR)/Area Defense Identification Zone (ADIZ) procedures.

6.3.2. Airspace classifications, GDSS/ASRR, and Giant Report.

6.3.3. Special Pilot in Command (SPIC) airport and Airfield Qualification Program (AQP) reviews.

Section 6B—Pre-Departure

6.4. Global Decision Scheduling System (GDSS) Account. Pilots will obtain a GDSS account prior to operating on Integrated Flight Management (IFM)-planned sorties. **(T-3)**

6.5. Mission Kits. Carry mission kits on all operational missions. Forms may be maintained and carried electronically provided a printing capability exists (e.g., a portable printer.) Suggested items are listed below. **Note:** An asterisk (*) indicates mandatory for all 618 AOC or AMC missions away from home station and directed by C2 authority.

6.5.1. Forms:

6.5.1.1. *CBP Form 6059B, *Customs and Border Protection Customs Declaration*.

6.5.1.2. *DD Form 1748-2, *Joint Airdrop Malfunction Report (Personnel-Cargo)* (required on airdrop missions only).

6.5.1.3. *DD Form 2131, *Passenger Manifest*.

6.5.1.4. *DD Form 1385, *Cargo Manifest*.

6.5.1.5. *CBP Form 7507, *General Declaration (Outward/Inward).*

6.5.1.6. AF Form 457, *Hazard Report*.

6.5.1.7. *AF Form 651, *Hazardous Air Traffic Report (HATR)*.

6.5.1.8. *AFTO Form 781, *ARMS Aircrew/Mission Flight Data Document*.

6.5.1.9. *DAF Form 1297, *Temporary Issue Receipt*.

6.5.1.10. AF Form 711B, *USAF Mishap Report*.

6.5.1.11. AMC Form 54, *Aircraft Commander's Report on Services/Facilities*.

6.5.1.12. *AMC Form 97, *AMC In-Flight Emergency and Unusual Occurrence Worksheet*.

6.5.1.13. *SF 44, *Purchase Order-Invoice-Voucher (Storage Safeguard Form)*.

6.5.1.14. *SF 702, *Security Container Check Sheet*.

6.5.1.15. *Japanese Customs Service Forms.

6.5.2. Orders:

6.5.2.1. *DD Form 1610, *Request and Authorization for TDY Travel of DoD Personnel*.

6.5.2.2. DAF Form 1631, *NATO Travel Order* (when required).

6.5.2.3. *AF Form 4327a, *Crew Flight (FA) Authorization*.

6.5.3. Miscellaneous:

6.5.3.1. *Box car seals.

6.5.3.2. *Masking tape.

6.5.3.3. *Pad lock (compatible with crew entry door).

6.5.3.4. CD/DVD drive.

6.6. Briefing Requirements. The AMC Briefing Guide addresses all mission phases and should be used for each brief. Reference AFTTP 3-3.C-17 for additional briefing information.

6.7. Departure Planning. Use AFMAN 11-202V3, appropriate MAJCOM supplements, and the C-17A Departure Planning Checklist. Regardless of the type of departure flown, Instrument Flight Rules (IFR) or Visual Flight Rules (VFR), review the following (as appropriate): IFR Departure Procedure, instrument approach plate, NOTAMS, GDSS Giant Report, and suitable terrain charts.

6.7.1. In the event the aircraft is unable to meet the published ALL ENGINE climb gradient, download cargo/fuel or delay until more favorable conditions exist.

6.7.2. Minimum climb gradient.

6.7.2.1. If a Special Departure Procedure (SDP) is not available, the crew must ensure compliance with any obstacle-based minimum climb gradients for the selected departure, with one-engine inoperative. **(T-1)** Contact AMC Airspace Ranges/Airfield Operations Division (AMC/A3A) for ad hoc SDP requests.

6.7.2.2. Onboard Performance Tool (OPT). To facilitate easier reference to Jeppesen SDPs, the OPT application is configured with C-17A data and is authorized for use only if current SDP data is uploaded to the application. OPT SDP Data is updated on a 28 day cycle and should be synced every month. OPT can be referenced in lieu of the Jeppesen Runway Analysis (JRA) MilPlanner website.

6.7.3. If operational requirements justify increased risk, the mission execution authority may authorize the PIC to subtract up to 48 feet per nautical mile (NM) from the published (or standard) climb gradient for One-Engine Inoperative departure planning. For all 618 AOC, 613 AOC, CVAM, and training missions with external users, the PIC is authorized to subtract up to 48 feet/NM.

6.7.4. Preflight Predictive Receiver Autonomous Integrity Monitoring (RAIM) Check. Pilots or mission planners are required to accomplish a predictive RAIM check during planning when terminal area navigation (RNAV) operations using Standard Instrument Departure (SIDs), Standard Terminal Arrival Routes (STARs), or other approaches are planned and no other backup navigation aids/procedures are available to identify planned outages or lack of adequate coverage. **(T-1)**

6.7.4.1. If RAIM will not be available at the time(s) and location(s) when GPS updating is required, the mission must be altered to a time when RAIM will be available. **(T-3)** If the predictive RAIM check cannot be completed, do not file terminal RNAV procedures. **(T-3)**

6.7.4.2. The following websites are available to meet this requirement and contain KGPS NOTAM coverage.

6.7.4.2.1. Eurocontrol. Using an EFB or a non-government computer, visit **http://augur.eurocontrol.int/**. Select the "Terminal/Approach Tool". Select the box below "Airports", input desired International Civil Aviation Organization (ICAO), then click "Save Changes". Under Settings, select the following from the dropdown options: "FDE", "SA on", "Baro aiding", "5 deg mask angle", and "Approach (RNP 0.3)". Select "Check Outages" to calculate results (depending on your browser, you may need to scroll down the page to see the results). Two sets of results must be generated, one for approach and one for terminal. **(T-3)** The "Export in TXT" option provides RAIM outage results at the selected ICAOs in a printable format.

6.7.4.2.2. National Airspace System (NAS). Visit **https://sapt.faa.gov** and scroll down to "Grid Display Tool" under the RAIM Prediction Tool section. Select the desired airspace, Terminal or non-precision approach (NPA). Ensure Baro-Aiding is selected "On". Select "Click to View" for each Terminal and NPA airspace selection. **(T-3)** Alternatively, users can also click the applicable images in the "RAIM Summary Pages" section below "Grid Display Tool".

6.7.4.2.3. Other Outside Contiguous United States (OCONUS) consult General Planning (GP) /Aeronautical Information Publication (AIP) for RAIM prediction tools.

6.8. Weather Minimums for Takeoff. Reference AFMAN 11-202V3_AMCSUP for departures with weather below landing minimums.

6.9. Adverse Weather.

6.9.1. The C-17A is a category III aircraft for turbulence. Crews should confirm the type of aircraft the forecast turbulence applies to, or what type of aircraft reported the encounter, to gain a more accurate picture for their route of flight. The Aircraft Turbulence Category table is found in Air Force Handbook (AFH) 11-203V2/ATC 3-04.14-2, *Weather for Aircrews – Products and Services*.

6.9.2. The 618 AOC issues charts for CAT II aircraft. C-17 crews must interpret the conditions for their category using the Aircraft Turbulence Intensity Conversion Chart found in AFH 11-203V2/ATC 3-04.14-2 (e.g., turbulence reported as MOD/OCL SEV [M/(S)] for CAT II is actually MOD [M] for CAT III.)

6.9.3. FMs will not plan flights into freezing levels below -60 degrees Celsius. **(T-1)**

Section 6C—Preflight

6.10. Aircraft Servicing and Ground Operations. Reference **paragraph 10.5** for additional information.

6.10.1. Aircrew T.O. 1C-17A-1, Preflight Inspection Requirements.

6.10.1.1. The aircrew T.O. 1C-17A-1 preflight inspection remains valid until either aircraft ground time exceeds 12 hours (72 hours provided the aircraft is sealed, not flown, and documented entry control is maintained) or another maintenance T.O. 1C-17A-6 preflight is performed.

6.10.1.2. Crews should consider adding an info note in the AFTO 781A indicating the date and time the FCC and SC/EFC Hydraulic PFBIT was accomplished.

6.10.1.3. When an aircrew assumes a preflighted spare or quick turn, perform a thorough visual inspection. **(T-3)**

6.11. Fire Protection and Crash Rescue Requirements.

6.11.1. The aircraft engine/APU fire extinguisher system fulfills the minimum requirements for fire protection during engine/APU start.

6.11.2. A fireguard should be used for all engine starts. A crewmember or ground controller may act as fireguard.

Section 6D—Departure

6.12. NVG Departures.

6.12.1. NVG Departure Weather Minimums. Weather minimums for NVG departures for crewmembers who are non-current and/or unqualified is 1500-3. **(T-1)** Current and qualified NVG aircrews may fly NVG departures weather down to 600-2 (OG/CC or equivalent may approve down to 300-1). Give careful consideration to potential hazards during the critical phase of flight. Other weather limitations are IAW this manual and AFMAN 11-202V3 and MAJCOM supplements. NVGs have inherent limitations which can further be reduced by poor weather conditions. Consider weather conditions, moon illumination and position, sky

glow at dawn and dusk, cultural lighting, and weapon/expendable effects when planning NVG operations.

6.12.2. NVG Crosswind Limits for Departure.

6.12.2.1. Runways 90 to 120 feet wide. Maximum crosswind component is 15 knots. **(T-3)**

6.12.2.2. Runways wider than 120 feet. Maximum crosswind component is 20 knots. **(T-3)**

6.12.3. NVG Malfunctions During Takeoff. During an NVG takeoff, if the PF experiences NVG failure, the takeoff may be continued at the discretion of the PIC. The PIC will brief NVG failure on takeoff procedures. **(T-2)** The PM should be ready to immediately assume aircraft control if the PF experiences spatial disorientation or an NVG malfunction.

Section 6E—Enroute

6.13. Flight Progress. In-flight, monitor mission computer navigation performance. Immediately report malfunctions or any loss of navigation capability that degrades centerline accuracy to the controlling air route traffic control center (ARTCC). **(T-2)**

6.14. Oceanic Procedures. Pilots will use the following procedures prior to entering oceanic airspace to comply with coast out/in (gross nav) navigation accuracy checks:

6.14.1. Select a Navigation Aid (NAVAID) that provides a DME signal (within its standard service volume range) as close to the beam position from the aircraft as possible. DME distance should be no closer than the first two digits of the flight level value (e.g., FL310 equals a minimum distance of 31 miles).

6.14.2. Display progress page on a mission computer display (MCD) and enter the NAVAID identifier at BRG DST TO Line Select Key (LSK) 5R.

6.14.3. Change one MFD to the Navigational Display (ND) Compass display and associated MFC HDG REF SEL switch to TRUE.

6.14.4. Change the NAVAID Course Deviation Indicator (CDI) course selector to center the CDI for a course TO the NAVAID, note the DME distance then immediately record the progress page course and distance information displayed onto the Master Document.

6.14.5. Record the following navigation accuracy check information on the Master Document: NAVAID identifier, universal coordinated time (UTC), Mission Computer (MC) Actual Navigation Performance (ANP) value, and courses and distances from progress page display and ND Compass display.

6.14.6. If the noted DME distance and course in the CNC window are not within 4 miles and 4 degrees attempt another navigation accuracy check with another NAVAID, check MC position during over flight of a VOR/Non-Directional Beacon (NDB), or use ATC radar position information referencing a NAVAID or airfield compared with the progress page 1 BRG DST TO LSK 5R.

6.14.7. Ten-minute plotting and position verification can be performed either manually on an approved oceanic plotting chart or via software program. If a software program is used, crews

will record the information from the mission computer, not the GPS information from the software program (e.g., ForeFlight). Aircrews will annotate the following:

6.14.7.1. Full Lat/Long position.

6.14.7.2. UTC time at that position.

6.14.7.3. Flight Level/Altitude.

6.14.7.4. MC position update source with MC ANP.

6.14.7.5. Pilot ID switch position.

6.14.7.6. Hourly altimeter checks on the Master Document.

6.14.8. When approaching each waypoint, recheck coordinates for the next waypoint.

6.14.9. Aircrews operating missions requiring unique or specially developed routing will normally be briefed at home station, onload station, and/or by the last C2 facility transited before performing the critical portion of the mission.

6.15. Communications. Controller-Pilot Direct Link Communication/Automatic Dependent Surveillance – Contract (CPDLC/ADS-C) data link communications are only used on AERO-I. AOC communications may utilize HF DATA if AERO-I is unavailable.

Section 6F—Arrival

6.16. Instrument Approach Procedures.

6.16.1. Aircraft approach category. The C-17 is a category "D" aircraft. If maneuvering speed exceeds 165 Knots Indicated Airspeed (KIAS), the minimums for category "E" will be used. **(T-1)**

6.16.2. Derived Decision Altitude (DDA). Use 50 feet above the Minimum Descent Altitude (MDA) as the DDA when executing a Continuous Descent Final Approach (CDFA).

6.16.3. Prior to starting an instrument approach, pilots will confirm their aircraft can comply with the missed approach climb gradient requirements established in AFMAN 11-202V3. **(T-0)** If unable to meet required climb gradients, pilots must coordinate alternate missed approach procedures with ATC, which will ensure terrain clearance, prior to commencing the approach. **(T-1)** If this is not possible, do not attempt the approach.

6.16.4. Weather minimums. Before starting an instrument approach or beginning an enroute descent, pilots will confirm the existing weather is reported to be IAW AFMAN 11-202V3 requirements (i.e., straight in or sidestep, circling) and at or above the minimums listed below.

6.16.4.1. For Precision Approach Radar (PAR) approaches, visibility will be no lower than RVR 2,400 (730 meters) or 1/2-mile visibility (800 meters) with no RVR readout available. **(T-1)**

6.16.4.2. If the ceiling is below the value depicted for published DoD or approved precision approach, but visibility is at or above authorized minimums, comply with fuel requirements of **Chapter 12** before initiating enroute descent, penetration, or approach.

6.16.5. Flight Instrumentation Requirements. Aircraft are limited to a Decision Height (DH)/MDA based on a height above touchdown (HAT) of 300 feet and RVR 4,000, or ¾-mile

visibility (1,220 meters) with no RVR if full flight instrumentation is not available and operational. Full flight instrumentation consists of the following:

6.16.5.1. PAR and Category I Instrument Landing System (ILS). Flight director, a HUD or PFD and NAV display at each station, and no shared ADC or IRU.

6.16.5.2. CAT II ILS. Flight director, a HUD or PFD and NAV display at each station, and no "No CAT II" warning message. A HUD will be used by the PF position, if available. **(T-2**)

6.16.6. Category I ILS Procedures. Decision Altitude for precision approaches will be as published, but no lower than 200 feet HAT. **(T-2)**

6.16.6.1. ILS Precision Runway Monitor (PRM) Approaches. Both pilots must be certified to conduct an ILS PRM approach. **(T-2)** Comply with the following operational procedures:

6.16.6.1.1. Two operational VHF-capable communication radios are required.

6.16.6.1.2. Brief the approach as an ILS/PRM approach referencing the PRM Attention All Users Page (PRM AAUP) corresponding to the approach.

6.16.6.1.3. All breakouts from the approach will be hand flown. **(T-1)** Autopilot must be disengaged when a breakout is directed. **(T-1)**

6.16.6.1.4. Should a TCAS Resolution Advisory (RA) be received, the pilot will immediately respond to the RA. **(T-1)** If following an RA requires deviating from an ATC clearance, the pilot must advise ATC as soon as practical. **(T-1)** While following an RA, comply with the turn portion of the ATC breakout instruction unless the pilot determines safety to be a factor.

6.16.6.2. IAW AIM 5-4-5, aircrews complying with the required equipment provisions listed on the approach may execute Category I Instrument Landing System (ILS) approaches with less than 2400 RVR, but 1800 RVR or greater, at locations without Touchdown Zone Lighting (TDZL) and Centerline Lighting (CL), or when such system is inoperative.

6.16.7. Category II ILS Procedures. DH is based on radar altitude. Minimum HAT is 100 feet. Minimum RVR is 1,200. Maximum crosswind component limitation is 10 knots. **(T-2)** Aircrews will not execute an actual Category II ILS to minimums unless both pilots are qualified and current in Category II ILS. **(T-3)** The AC must have logged at least 100 hours in command since AC certification. **(T-3)**

6.16.8. ILS Special Authorization (SA) CAT I Approach. Procedurally, the crew will fly a SA CAT I ILS approach the same as a CAT II ILS approach. **(T-1)** All CAT II limitations (crosswinds, autopilot status, required equipment, AFMAN 11-202V3, etc.) must be met. **(T-1)** If the crew receives a CAT II unsafe annunciation above 300 feet, they may elect to continue to the normal CAT I minimums to the same runway (no lower than 200 feet DH). If a CAT II unsafe annunciation is received below 300 feet, the crew will immediately commence a go-around, unless visual cues are sufficient to complete the approach to landing. **(T-1)** Use of the HUD to DH is mandatory. SA CAT II ILS approaches are not authorized.

6.16.9. Holding. An aircraft may hold at a destination that is below landing minimums, but forecast to be at or above minimums provided:

6.16.9.1. The aircraft has more fuel remaining than that required to fly to the alternate and hold for the appropriate holding time, and the weather at the alternate is forecast to remain at or above alternate filing minimums for the period, including the holding time.

6.16.9.2. Destination weather is forecast to be at or above minimums before excess fuel will be consumed.

6.17. NVG Approach and Landing.

6.17.1. NVG Approach Weather Minimums. Aircrews may fly NVG instrument approaches which transition to NVG landings with weather down to 600-2 (180m/3200m) (OG/CC or equivalent may approve down to 300-1 [90m/1600m]) or circling minimums (whichever is higher). **Note:** Noncurrent and unqualified aircrew are restricted to 1500-3.

6.17.2. NVG Crosswind Limits for Approach.

6.17.2.1. Runways 90 to 120 feet wide. Maximum crosswind component is 15 knots. **(T-3)**

6.17.2.2. Runways wider than 120 feet. Maximum crosswind component is 20 knots. **(T-3)**

6.17.3. NVG Failures During Approach and Landing. If the pilot or copilot experience NVG failure or other malfunctions at or below 300 feet above ground level (AGL), perform a go-around. The PIC will ensure NVG failure procedures are briefed. **(T-2)**

Section 6G—Miscellaneous

6.18. Hung Flare Procedures. Reference T.O. 1C-17A-1 Hung Flare Inspection procedure.

6.19. Enplane/Deplane Crewmember(s)/Maintenance Personnel with Engines Running. The Engine Running On/Offload (ERO) checklist may be run, but is not required, when enplaning/deplaning crewmembers and maintenance personnel actively working a maintenance issue through the crew entry door. If not run, the PIC will ensure the aircraft is depressurized and parking brakes are set prior to authorizing the door to be opened. The PIC will ensure the crew entry door is closed prior to aircraft movement.

Chapter 7

AIRCRAFT SECURITY

7.1. General. This chapter provides guidance on aircraft security and preventing and resisting aircraft piracy (hijacking) of the C-17 aircraft. AFI 13-207-O, *Preventing and Resisting Aircraft Piracy (Hijacking) (CUI)*, DAFI 31-101, *Integrated Defense (ID)*, and specific MAJCOM security publications contain additional guidance. Aircrews will not release information concerning hijacking attempts or identify armed aircrew members or missions to the public. **(T-2)**

7.2. Security. The C-17 is a "Protection Level-3" resource. Aircraft security at non-United States military installations is the responsibility of the controlling agency.

7.3. Integrated Defense. The following security procedures implement DAFI 31-101 requirements for C-17 aircraft:

7.3.1. The aircraft will be parked in an established restricted area and afforded protection IAW DAFI 31-101. **(T-2)**

7.3.2. When no permanent or established restricted area parking space is available, establish a temporary restricted area consisting of a raised rope barrier, and post with restricted area signs. Portable security lighting will be provided during the hours of darkness if sufficient permanent lighting is not available. **(T-2)** Post security forces IAW DAFI 31-101.

7.3.3. The security force must be made aware of all visits to the aircraft. **(T-3)** The security force point of contact must be identified to the PIC. **(T-3)**

7.3.4. Security support is a continual requirement and is not negated by the presence of aircrew or ground crewmembers. Security force support terminates only after the aircraft doors are closed and the aircraft taxis. **(T-2)**

7.4. Arming of Crewmembers. Crews will arm IAW mission directive (Mission Detail, SPINS, OPORD, etc.) **(T-2)** Arming is required anytime mission directives conflict. The following procedures apply when arming is directed:

7.4.1. Weapons Issue. Arming for anti-hijacking requires hollow-point ammunition while on the aircraft. If crews are also required to be armed for force protection away from the aircraft, use ball ammunition, unless approved/directed per regional guidance (SPINs, OPORD, FCG, etc.). **(T-2)** The PIC determines which crewmembers will be armed. For missions that require all crewmembers to be armed for force protection annotate the Flight Authorization with "Bulk Arming Required." (**T-2)**

7.4.2. When mission directive does not require crews to arm and transiting CONUS locations, arming is at the discretion of the PIC when enroute locations have an AMC or Logistics Readiness Squadron passenger terminal. Passenger movement will be accommodated to the maximum extent possible IAW DoDI 4515.13*, Air Transportation Eligibility.*

7.5. Force Protection. Remain alert to the possibility of terrorist activities at all times. Reference AFTTP 3-4, *Airman's Manual,* Chairman of the Joint Chiefs of Staff (CJCS) Guide 5260, *A Self-Help Guide to Antiterrorism,* and DoDIO-2000.16V1_DAFI31-145-O, *Antiterrorism (AT) Program Implementation*, for Force Protection measures.

Chapter 8

TRAINING AND OPERATING LIMITATIONS

8.1. Passengers on Training Missions. Passengers are defined as anyone listed on a passenger manifest. Mission Essential Personnel (MEP) are not considered passengers.

8.1.1. Do not carry passengers with an unqualified pilot occupying a primary crew position. **(T-2)** Initial qualification or re-qualification for pilots will not be conducted with passengers onboard. **(T-2)**

8.1.2. Mission certification training, upgrade training, evaluations, off station trainers, and Joint Airborne/Air Transportability Training (JA/ATTs) may carry passengers only if the aircrew in training is qualified. Tanker and receiver AAR is authorized if the pilot flying is qualified (AF Form 8 on file documenting successful completion of an aircraft flight evaluation including air refueling). **(T-2)**

8.1.3. Multiple practice approaches, touch-and-go landings, simulated emergency training, and airdrops are prohibited with passengers on board. **(T-2) Exception:** Personnel scheduled to jump following a heavy/CDS airdrop, safeties, MEP (defined in DAFMAN 11-401, *Aviation Management*), exercise participants that will be offloaded by "airland" procedures following the airdrop, or any personnel authorized by the JA/ATT tasking order may be transported on airdrop training missions.

8.2. Touch-and-go Landing Limitations.

8.2.1. Only accomplish touch-and-go landings under the direct supervision of an IP. **(T-2)**

8.2.2. Limitations.

8.2.2.1. Comply with all flight manual restrictions and procedures to include performance degradation with fuel, cargo limits, etc. **(T-2)**

8.2.2.2. Minimum runway length: 7,000 feet (if there is a published declared distance, the Landing Distance Available (LDA) must be 7,000 feet minimum). Minimum runway width: 120 feet. **(T-2)** Raised barriers reduce runway available for touch-and-go landings.

8.2.2.3. Minimum ceiling/visibility: 300 feet and RVR 40 (3/4 SM visibility). **(T-2)**

8.2.2.4. RCR will be 12 or higher. **(T-2)**

8.2.2.5. Do not accomplish touch-and-go landings on runways reported with standing water, slush, or snow. **(T-2)**

8.2.2.6. Maximum crosswind component: 25 knots. **(T-2)**

8.2.2.7. Touch-and-go landings may be performed with cargo onboard. After the first Approach Checklist is accomplished, cargo security is only required to be checked every hour. Touch-and-go landings with hazardous cargo onboard in quantities which require their identification on flight plans are prohibited. **(T-2)**

8.3. Training on Operational Missions. Crews may accomplish AAR training on operational missions provided the following requirements are met:

8.3.1. All mission-required fuel is onloaded prior to commencing any training. **(T-3)**

8.3.2. Passengers and MEPs are briefed on the activity.

8.3.3. AAR training may be accomplished with an unqualified receiver pilot at the controls if under IP supervision and no passengers are on board the aircraft. **(T-3)**

8.3.4. Unscheduled AAR is not authorized without coordination with appropriate C2 agencies. **(T-2)**

8.4. Simulated Emergency Flight Procedures.

8.4.1. Simulated emergency flight procedures will be conducted IAW AFMAN 11-202V3 and this manual. **(T-2)** Do not practice emergency procedures that degrade aircraft performance or flight control capabilities (in-flight). Emergency procedure training with degraded aircraft performance or non-standard configurations will only be accomplished in the simulator. **(T-2)**

8.4.2. The PIC or IP will alert all crewmembers prior to practicing emergency procedures. In an actual emergency, terminate all training and flight maneuvers practice. **(T-2)** Training should be resumed only when the PIC determines it is safe.

8.5. Prohibited Flight Maneuvers. Practice of the following maneuvers are prohibited in flight: **(T-1)**

8.5.1. Stall and approach to stalls including initial buffet.

8.5.2. Dutch roll.

8.5.3. Abnormal configuration approaches.

8.5.4. Unusual Attitudes.

8.5.5. Bank angles greater than 60-degrees.

8.5.6. Aerobraking during landing.

8.6. Flight Pilot Training.

8.6.1. A Mission Ready Mobility Pilot Development Phase I Pilot (FPC)/Mission Ready Mobility Pilot Development Phase II Pilot (FPQ) may practice AAR from either seat (to include the contact position) with the following restrictions:

8.6.1.1. Accomplished under direct IP supervision.

8.6.1.2. No passengers are authorized. (N/A for Senior Officer Qual FPs with a Form 8 documenting AR accomplishment.)

8.6.1.3. Contacts by non-AAR qualified pilots are only made after receiving acknowledgment from the tanker pilot and boom operator. An AAR qualified pilot will make a contact prior to any FP training to validate system functionality.

8.6.2. An FPC/FPQ may perform any tactical maneuver under the direct supervision of an IP.

8.7. Night Vision Goggle (NVG) Training.

8.7.1. Airland Training.

8.7.1.1. Ground Operations Training. NVG combat offloads and ground maneuvering are approved. Combat offloads may be conducted with cargo compartment lighting set to minimum red/ANVIS (overt) or IR (covert).

8.7.1.2. Takeoff and Landing Restrictions:

8.7.1.2.1. One RA and one GPS must be operational. **(T-3)**

8.7.1.2.2. Both HUDs must be operational. **(T-3)**

8.7.1.2.3. NVG touch-and-go landings are authorized.

8.7.1.3. Runway must be lit with an authorized covert/overt lighting pattern IAW DAFMAN 13-217. Expected runway lighting configuration should be briefed prior to executing the approach. Landings to unlit (AMP-4) runways/LZs are not authorized, unless specially certified.

8.7.2. Airdrop Training Restrictions. Drop zones will be lit IAW DAFMAN 13-217 lighting patterns (covert or overt) or in accordance with **paragraph 17.4**. **(T-3)**

8.8. Planned Go Around After Touchdown (GOAT). A GOAT is a planned training maneuver, accomplished in conjunction with a full flap approach to an Assault Landing Zone (ALZ) or simulated ALZ. The approach and landing for the GOAT are accomplished IAW the approach and assault landing sections of T.O. 1C-17A-1. No later than main gear touchdown, a go-around will be initiated IAW T.O. 1C-17A-1. **(T-3)**

8.8.1. An instructor pilot must occupy a primary pilot position during the maneuver. **(T-3)**

8.8.2. The following parameters must be met:

8.8.2.1. Landing data and brake temperatures must support a full stop. **(T-2)** For GOATs to a marked ALZ on a larger than 5,000-foot runway, crews may calculate brake temperature-corrected landing distance for the actual runway remaining, not the marked ALZ.

8.8.2.2. T.O. 1C-17A-1 requirements for landing gear cooling periods following consecutive landings apply.

8.8.2.3. Sink rate reduction and FPV movement are associated with the landing phase and accomplished in addition to the power advance for the go-around.

8.8.2.4. A "TAKEOFF" or "LANDING" cannot be logged for currency, but will be logged on AFTO Form 781H as touch-and-go landings. **(T-2)**

8.8.2.5. Will not be accomplished on Semi-Prepared Runways. **(T-2)**

8.9. Miscellaneous.

8.9.1. For unilateral training, center bundle targeting at an Airdrop Damage Estimation (ADE) Level 3 is authorized with actuals provided the minimum Drop Zone (DZ) size is met IAW DAFMAN 13-217 and the ADE Level 3 ellipse is within the confines of the surveyed drop zone. The mission specific point of impact (PI) will be validated by Weapons and Tactics Flight (OSK). **(T-3)**

8.9.2. Training without GPS updating is encouraged. When conducting GPS-off training, crews will be in VMC. **(T-2)** If IFR, do not exceed Required Navigation Performance (RNP) restrictions for the airspace. **(T-2)** Crews should use an additional crewmember with moving map and GPS puck for debrief and an added factor of safety.

8.9.3. **Table 8.1** summarizes additional limitations and restrictions.

Table 8.1. Training Flight Restrictions.

Maneuver	Altitude	Remarks
Instrument Missed/ Low Approaches	DDA/DH	Initiate practice instrument missed approaches no lower than the minimum altitude for the type of approach executed. **(T-3)**
Visual Low Approach/Planned Go Around	Initiate no lower than 100 feet HAT	**(T-2)**
Go Around due to Personnel and Equipment on Runway	Initiate above 500 feet HAT	**(T-2)**
CAT II ILS	As published	Minimum ceiling & visibility of 200-½. Crosswind components up to 15 knots are permitted. **(T-2)**
Tactical Descents	Complete above 2,000 feet AGL	Due to high descent rates, ensure ATC understands crew intentions. Minimum ceiling is 5000 feet AGL. NVG tactical descents for training must remain in VMC at all times. **(T-2)**
Assault Landing Zone Operations		Max brake temperature 150°C. **(T-2)**
Note: These restrictions do not apply to operational missions.		

Chapter 9

NAVIGATION PROCEDURES

9.1. Post Flight. For legs in which suspected or reported navigational error, procedural deviation, or other abnormality occurred, download and retain a copy of the mission plan, all pertinent forms, including, but not limited to DD Form 1801, *DOD International Flight Plan*, Mobility Air Forces Automated Flight Planning System (MAFPS) AMC Standard Report (ASR) or Computer Flight Plan (CFP), charts, and navigation log. Units will maintain all operational mission data on file (hard disk, paper copy, oceanic plotting chart, etc.) for 120 days. **(T-2)**

9.1.1. For suspected/reported airdrop malfunctions or off-DZ drops, record ADXX RECALL page information before leaving the aircraft. **(T-2)**

9.1.2. GPS and Global Navigation Satellite System (GNSS) are synonymous terms when referencing GPS equipment and/or procedures.

9.1.3. Sources for obtaining IRU Present Position (PPOS) coordinates will be in the following priority: **(T-2)**

9.1.3.1. GPS PPOS IAW flight manual.

9.1.3.2. FLIP/Jeppesen Airfield diagrams or MAJCOM-approved parking spot handouts.

9.1.3.3. DoD FLIP Enroute Supplement.

9.1.3.4. Best available chart.

9.2. Navigation Databases/Flight Plans/Data Verification.

9.2.1. Crosscheck WWNDB procedures Digital Aeronautical Flight Information File (DAFIF) against FLIP prior to flying any procedure IAW AFMAN 11-202V3. If a discrepancy is identified, submit the details to unit OGV or Stan/Eval office. Unclassified confirmed discrepancies will be emailed to National Geospatial-Intelligence Agency's (NGA) aero quality feedback section at **quality@nga.mil** and AMC/A3V. **(T-2)**

9.2.2. Mobility Air Forces Automated Flight Planning System (MAFPS) AMC Standard Report (ASR) Use. MAFPS ASR is the official source of performance, navigation, and climatic data, including enroute wind information. If stand-alone computer-based plans are used, each mission segment should utilize best wind data available. Use only MAJCOM validated ASRs. **(T-2)**

9.2.2.1. Use MAFPS ASRs to the maximum extent practical. A printed computer flight plan derived from other MAJCOM-approved planning software is an approved substitute if it includes all MAFPS navigational and waypoint data and enroute wind and temperature deviation data. The computer flight plan must also include the planned payload and ramp fuel. **(T-2)** The PIC has final responsibility for flight plan accuracy and diplomatic clearance compliance. **(T-2)**

9.2.2.2. Verify MAFPS ASR for route of flight and fuel computation accuracy before departure. **(T-2)** Pass any flight plan discrepancies to the 618 AOC flight planning office. **(T-2)** On flight-managed sorties, promptly notify the FM of any flight plan discrepancies. **(T-2)** Identify inaccurate MAFPS winds to the FM in-flight or 618 AOC flight planners

post flight when the wind at the MAFPS flight level exceeds either 30° error in direction or 25 knots in speed. **(T-2)**

9.2.2.3. Flight plan course values may differ between the MAFPS ASR and the mission computer flight plan. The MAFPS ASR course value (mag or true) is the average course direction for that leg. The mission computer flight plan course is the initial course outbound measured from the leg's beginning waypoint after any rollouts. Therefore, the mission computer flight plan course may differ by a few degrees from the MAFPS ASR flight plan course. The magnitude of this difference is more pronounced depending on leg length and direction.

9.2.3. Aeronautical Information Regulation and Control (AIRAC) cycle.

9.2.3.1. Effectivity of the aircraft's WWNDB and aircrew Aeronautical Information Publication (AIP)/FLIP documents are based on AIRAC cycles.

9.2.3.2. ICAO Document 8126 (Aeronautical Information Services Manual) specifies that 0001 UTC will be used to indicate the time when new AIP/FLIP documents become effective. **(T-2)** However, some countries have filed an exception. For example, the United States is 0901 UTC on the effective date while Australia is 1600 UTC the day prior to the effective date. AIRAC cycle changes require ATC computer systems to be updated. For that reason, some countries have modified their AIRAC cycle change day/time when air traffic is at the normal daily minimum for their airspace.

9.2.3.3. Pilot considerations during non-0001 UTC AIRAC cycle changes. Even though numerous NGA FLIP products have 0001 UTC effective times printed on them, plan not to receive a clearance for new or changed published procedures/routings until that ATC system's airspace computers have been updated.

9.2.4. WWNDB Procedure Omissions. Crews should be vigilant in their review/comparison of mission computer loaded procedures with published Flight Information Publications (FLIP). Numerous procedures that are published in DoD's FLIP may be omitted from the C-17 mission computer database.

9.2.5. Prior to takeoff, departure clearances may be available via CPDLC at some airports by logging onto KUSA in the Air Traffic Service Menu. The PIC is responsible for ensuring the uploaded flight plan is correct.

9.3. Navigation Capability / Airspace Requirements.

9.3.1. RNAV/RNP. C-17A aircraft are approved for unrestricted operations in the following enroute and terminal airspace: MNPS, Oceanic and remote airspace (RNP-10 & RNP-4), BRNAV (RNP-5), P-RNAV, RNAV 1, RNAV 2, and GPS Approach (RNP-0.3).

9.3.2. MNPS, RNP-x, P-RNAV, BRNAV, RNAV 2, and RNAV 1 are defined in FLIP and are considered special qualification airspace. Additional restrictions to include CPDLC/ADS equipment may be required in special qualification airspace.

9.3.2.1. Should any required equipment fail before entering such airspace, comply with FLIP restrictions or request a new clearance to avoid this airspace. **(T-1)**

9.3.2.2. Should any required equipment fail after entry into such airspace, immediately notify ATC and coordinate a plan of action. **(T-1)**

9.3.2.3. Document (using AFTO 781A) malfunctions or failures of required equipment, including the failure of this equipment to meet tolerances. **(T-1)**

9.3.3. RNAV airspace/procedure restrictions.

9.3.3.1. Manual entry of waypoints using Latitude/Longitude or Place/Bearing is prohibited on any RNAV procedure. **(T-1)** RNAV point-to-point navigation such as NAT track waypoints and remote oceanic navigation waypoints are the only times when Lat/Long entry is allowed for RNAV airspace operations. **(T-1)**

9.3.3.2. Pilots will not change any RNAV database waypoint overfly turn property from "no" to "yes" or vice versa. **(T-1)**

9.3.3.3. Waypoints may be deleted in order to comply with NOTAMs or ATC clearances (e.g., deleting the course reversal on an approach when cleared via a NO procedure turn routing).

9.3.3.4. Altitude and airspeed parameters may be adjusted to match current NOTAMs, ATC clearance, or cold weather adjustments.

9.3.3.5. Encrypted Precise Positioning Service (PPS) MGPS is authorized as primary navigation for civil navigation unless directed otherwise by SPINS, command authority, or host nation guidance for all phases of flight.

9.3.3.5.1. If MGPS cannot be properly verified with encrypted PPS via Technical Order guidance, then CGPS should be used as the primary navigation source.

9.3.3.5.2. When host nation directives are ambiguous, CGPS should be used as the primary navigation source.

9.3.3.6. RNAV procedures or routes requiring Satellite-Based Augmentation System or Ground-Based Augmentation System (GBAS) equipment are not currently authorized.

9.3.4. RNAV SID/STAR restrictions.

9.3.4.1. The use of non-United States Government developed Standard Terminal Arrival Routes (STARS) is authorized without Terminal Enroute Procedures (TERPS) review. The use of either the host nation sanctioned, or respective Jeppesen, STAR charts are authorized without further review. This does not alleviate pilots of the responsibility to check the Giant Report. **(T-2)** Additional guidance is found in the ASRR.

9.3.4.2. At least one GPS must be operational and updating the MC present position (MC PPOS) before departing on an RNAV departure. **(T-1)** Radio updating is not authorized as an update source on the ground prior to departing on an RNAV departure. **Exception:** If GPS updating fails during takeoff, the takeoff and planned RNAV departure may be continued. RNAV guidance remains valid for up to fifteen minutes following failure. When able, follow flight manual Section III procedures after safely airborne regarding failure of RNAV components. **Note:** Complete GPS failure is indicated by a NAV MCD/GPS MCD cue with accompanying MCD messages (MGPS INOP and CGPS INOP).

9.3.4.3. Aircrews are not authorized to fly SIDs that include Radius to Fix (RF) legs (Example: RNP1-RF required or RF required note). **(T-1)**

9.3.5. RNAV/GPS Approach restrictions.

9.3.5.1. Aircrews may fly any procedure with "RNAV (GPS) RWY" in the approach title or those that have ONLY "GPS RWY" in the approach title. These are stand-alone GPS-based approaches. These approaches can be identified on approach plates by the presence of GPS waypoint depictions (black diamond waypoints). Example: "RNAV (GPS) RWY 32" at Dover AFB (KDOV).

9.3.5.2. Instrument approach procedures titled "RNAV (RNP)", "RNAV (RNP AR)", or RNAV procedures that have "Special Aircraft & Aircrew Authorization Required (SAAR)" statements in the profile view are not authorized. **(T-1)**

9.3.5.2.1. ICAO instituted a new naming convention for Performance Based Navigation (PBN) approaches that some countries have adopted. PBN approaches titled RNAV (GNSS), RNAV (GPS), and RNAV (RNP) will transition to the title "RNP."

9.3.5.2.2. RNAV (RNP) Authorization Required (AR – formerly SAAR) will be titled "RNP AR" in counties that adopt the new naming convention. For example, RNP RWY 23 is authorized, but RNP RWY 23 (AR) is not authorized.

9.3.5.3. GPS approaches are flown to non-precision MDA (LNAV MDA or S-xx) or circling minima only.

9.3.5.4. C-17 aircrews may fly published PBN transition to conventional NAVAID final approach segment instrument approach procedures (IAPs) if available in the mission computer. That includes approaches with PBN initial, intermediate and/or missed approach segments. These IAPs must be extracted in their entirety (to include the missed approach segment) from the navigation database without alteration, and follow the same operational practices and procedures as RNAV (GPS) approaches. **(T-1)**

9.3.5.4.1. Each pilot must conduct a proper IAP review and brief how the approach will be flown to include transition to ground based NAVAIDs and, if necessary, back to PBN guidance. **(T-3)**

9.3.5.4.2. If the aircrew/aircraft is not capable of flying RNAV (GPS) approaches, the crew is authorized to fly the PBN segmented approach, provided they coordinate for vectors to final or alternate missed approach instructions, as authorized by the approach procedure.

9.3.5.4.3. Aircrews are not authorized to fly IAPs that include PBN transition-to-final with Radius to Fix (RF) legs. **(T-1)** Example: "RNP AR APCH, RF required" in the notes section of the IAP.

9.3.5.5. C-17 aircrews are not authorized to maintain overlay ground tracks based solely on GPS. **(T-1)** GPS overlay approach construction is subject to coding errors and may not provide course guidance over the proper ground track. These approaches can be identified on approach plates by the absence of GPS waypoint depictions (black diamond waypoints).

9.3.5.6. Accepting ATC vectors. Aircrews may accept ATC vectors to a published point on the approach or ATC vectors to intercept the final course.

9.3.5.7. VNAV is not authorized between the FAF and missed approach waypoint. **(T-2)** MSN VNAV (MSN descent) may be used between the Initial Approach Fix (IAF) and Final Approach Fix (FAF). However, crewmembers are reminded that the MSN VNAV algorithm remains a "dive and drive system" and may command descents greater than authorized during intermediate approach segments (TERPS criteria is based on a maximum descent of 500 feet/NM, approximately 1,500 VVI, between the IAF and FAF). Furthermore, MSN speeds may revert to tech order after vertical profile calculations. As a result, speed select mode (speed on thrust/pitch) and/or vertical speed mode (vertical speed command wheel) used in conjunction with MSN LNAV is the recommended mode for most RNAV approaches.

9.4. Enroute/Flight Progress.

9.4.1. Use all available FLIP enroute charts, and prior to departures and arrivals, terrain charts, to ensure navigational accuracy and terrain clearance.

9.4.2. For waypoints entered using Lat/Long, both pilots will use the ND CHART format in conjunction with the DEFINE/REVIEW WAYPOINT pages to ensure the proper coordinates were loaded for all points in the flight plan. **(T-2)**

9.4.3. Flying In Polar Regions.

9.4.3.1. Because the C-17 does not rely on magnetic field sensors for heading determination, navigation in Polar Regions poses no significant difficulties. Magnetic and grid headings are provided for pilot selection by converting true heading information from IRU sources.

9.4.3.2. Magnetic variation information is stored as a look-up table in the IRU and provides variation data from 72°N to 60°S. No variation data is supplied above/below these latitudes. Refer to the T.O. 1C-17A-1-2 for Polar Operations. Manual magnetic variation should only be selected when magnetic headings are flown.

9.4.3.3. Manual magnetic variation is not taken into consideration when displaying runways from the PERMANENT NAV DATABASE, as display logic assumes a zero magnetic variation. To get correct runway displays on the NAV display, create a new runway in the CUSTOM NAV DATABASE oriented to true or grid as applicable.

9.4.3.4. Whenever a magnetic heading is displayed, pilots see a jump in the heading scale as the magnetic variation is significantly changed. This occurs normally at 72°N/60°S or when manual magnetic variations are inserted into the MC. If the AP/FD is engaged in HDG or HDG HOLD modes while following this magnetic heading, the aircraft executes a turn in order to maintain the magnetic heading selected prior to the jump. These turns do not occur if the heading reference for the pilot ID switch was in TRUE or GRID when the heading was selected on the AP/FD. No turns occur if the AP/FD is engaged in the MSN lateral mode when crossing 72°N/60°S.

9.4.3.5. Grid heading reference is provided to pilots with a mathematical relationship defined in USAF air navigation manuals using a grid chart conversion factor of 1.0. While grid heading is selectable for display anytime via the MFC panel, it is only accurate at high latitudes where the grid convergence factor is actually 1.0. Pilots should consider this heading accurate at the latitudes near and above 72°N/60°S. The message GRID MODE

is displayed on the WAP when passing these latitudes for higher ones to indicate to the pilot that valid grid headings are available for navigation. If pilots have entered a manual magnetic variation as described above, the GRID MODE message is removed. Accurate grid headings are still available, but the MC interprets the pilot entered manual magnetic variation as a desire to fly magnetic headings, not grid headings.

9.4.3.6. The BDHI is oriented to magnetic headings. Grid reference is provided when GRID is selected on HDG REF SEL. As mentioned above, for latitudes above 72°N/60°S, the programmed magnetic variation is zero and the magnetic heading is incorrect; the BDHI essentially displays true headings. The VOR/TACAN/ADF needles still operate; the ADF continues to give relative bearing, and the VOR/TACAN needles are slaved to the correct radials. However, to determine correct magnetic headings for VOR/TACAN intercepts, input a manual magnetic variation if the stations are oriented to magnetic north. This is also true if trying to determine magnetic bearing from an NDB. If the VOR/TACAN/NDB is oriented to true north, then the magnetic variation needs to be zero for correct display on the BDHI.

9.4.3.7. For VOR and TACAN navigational aids, the course select knob on the CNC is consistent with the orientation of the NAVAID as defined in the NAV DATABASE. If the NAVAID is oriented to true, then a 300 course set in the CNC is a 300 true course; if the NAVAID is oriented to grid, then the course is a 300 grid course. Currently all NAVAIDs in the PERMANENT NAV DATABASE are defined as magnetic; pilots have to use the custom database to define a true or grid oriented NAVAID. Navigation position information is most easily interpreted when the displayed heading is consistent with the orientation of the NAVAID used (i.e., fly true headings when flying true TACAN radials, grid headings for grid courses, etc.).

9.4.3.8. For ILS approaches, the MC always assumes the course in the CNC to be magnetic. This has no effect on the ability of the aircraft to correctly fly the ILS, but is necessary for the MC to determine the ILS intercept angle and display NAV data. Pilots desiring to fly a magnetic ILS approach beyond 72°N/60°S should dial the magnetic course in the CNC and insert a manual magnetic variation in the MC. To fly a true ILS, pilots should dial the true course and ensure the MAG VAR is zero. Attempts to fly a magnetic ILS with incorrect magnetic variation should not cause problems once established on final; the intercept may be erratic and the NAV display shows a heading different from the ILS course equal to the error in magnetic variation as the aircraft tracks the ILS beam (no wind situation). Grid only ILS procedures are not supported; pilots convert the grid ILS approach course in the instrument approach procedure to a magnetic or true course, then ensure the correct MANUAL MAG VAR is inserted in the MC (local terminal magnetic variation for MAG ILS, zero magnetic variation for true ILS).

9.4.3.9. Special Aircrew and Aircraft Certification Required Approaches. All C-17As and aircrews are approved to fly “Special Aircrew and Aircraft Certification Required” approaches, except for CAT III ILS approaches and any RNAV approach with an RF leg.

9.5. Low-Level Navigation. Threat and emission control requirements permitting, use all available aids (i.e., mission computer data, navigational aid fixes, map reading) to remain position oriented.

9.5.1. While aircraft systems provide a self-contained adverse weather, day/night, worldwide navigation capability, pilots assume ultimate responsibility for enroute navigation, terrain avoidance, and time control. During low-level operations, attention should be focused outside the aircraft, emphasizing threat detection and situational awareness. Limit duties that distract attention from outside the aircraft to mission essential items only.

9.5.2. A Navigation Display (ND) Map format will be displayed on at least one MFD at all times. **(T-3)**

9.5.3. Time of Arrival control is primarily accomplished by airspeed adjustments.

9.5.4. Means of Navigation.

9.5.4.1. One radar altimeter will be operational for all low-level operations. **(T-2)** The mission computer navigation solution is the primary means of navigation backed-up with map reading, on VFR low-level routes. If conditions permit, ground-based navigational aids can be used as additional information sources.

9.5.4.2. The mission computer is the primary means of route navigation in IMC. Onboard radar, ground-based radar, and/or bearing/distance fixes should be used as backups.

9.5.5. Low-level modified contour flight may result in small deviations above and below the base altitude for smoothness of flight.

9.5.6. Altimeter Settings. In the absence of reported or forecasted barometric setting, crews may use the MC GPS barometric setting with the following restrictions:

9.5.6.1. Day VMC. Crews will use visual references to ensure terrain and obstacle clearance. **(T-3)**

9.5.6.2. Barometric altitudes (Minimum Safe Altitude (MSA), Emergency Route Abort Altitude (ERAA), etc.) will be increased by 1,000 feet. **(T-2)**

9.5.6.3. Night or IMC. Crews will add 1,000 feet to their planned minimum altitudes. **(T-2)**

9.6. Mission Computer Approaches.

9.6.1. The C-17 mission computer approach uses the MC APPROACH page of the mission computer. It only mathematically derives a final approach course and glide path from default or pilot input data. **Note:** Reference AFTTP 3-3.C-17 for techniques on programming Mission Computer approaches.

9.6.2. In a contingency, the crew will be provided an approved (TERPS'd) approach procedure leading to a mission computer approach final approach fix. **(T-2)** All course data and altitude restrictions required to program the approach in the mission computer will be included on the approach procedure. **(T-2)** This will include any high-precision waypoints, weather minimums, and missed approach instructions required to safely execute the approach. **(T-2)**

9.6.3. Restrictions on Mission Computer Approaches.

9.6.3.1. Mission computer approach procedures in IMC will only be used when no other published approach procedure is available and with specific authorization from MAJCOM/A3. **(T-2)** All crews are authorized to use these procedures for training in VFR conditions IAW AFMAN 11-202V3.

9.6.3.2. Mission Computer approaches require an RNP value of 0.3 NM from the FAF to the Runway (unless non-standard TERPs values used). PIC will ensure the RNP value is set to 0.3 prior to commencing a Mission Computer approach. Aircrews will input a RNP value of 0.3 on NAV SENSORS page 7, LSK 1L to ensure the aircraft will annunciate when the ANP is greater than 0.3 NM.

9.6.3.3. It is highly recommended that the approach be flown with the autopilot coupled.

9.6.3.4. Weather minima for approved MC approaches will be no lower than 600-2. **(T-2)**

9.6.3.5. Ensure Lateral Deviation Indicator (LDI) does not exceed ½ full scale deflection.

9.6.3.6. Execute the missed approach procedure if "UNABLE RNP" or "OVRFLY NOT POSSIBLE" appears on the MCD unless the runway is in sight. **(T-1)**

9.6.3.7. All of the waypoints required for flying the approach and missed approach must exist in the current navigation database. If the waypoints are not in the permanent navigation database, aircrews will be provided the required waypoints to be loaded electronically into the custom navigation database. **(T-2)** Aircrew will not manually enter or alter the latitude or longitude coordinates for any portion of the approach or missed approach. **(T-1)**

9.6.3.8. Enter all of the waypoints for the approach up to the FAF in the primary flight plan. **(T-2)**

9.6.3.9. Enter all of the waypoints for the missed approach, starting with the missed approach waypoint (MAWP), in the secondary flight plan. **(T-2)** Special attention should be paid to whether each waypoint is an "over fly" or "under fly." The MAWP is usually an "over fly" waypoint. If required to execute the missed approach, the PM will enable the secondary flight plan to provide lateral guidance. The secondary flight plan will be displayed on at least one ND Map display. **(T-2)**

9.6.3.10. Both pilots will review the entire procedure in the mission computer, verifying the coordinates and the MFD Map display for accuracy, prior to commencing the approach. **(T-2)** If any portion does not agree with the approved procedure, the approach will not be flown. **(T-2)**

9.7. Mission Computer Approach Planning Guidance. Planners will reference Federal Aviation Administration (FAA) Order 8260.3F, *United States Standard for Terminal Instrument Procedures (TERPS)* and/or AFMAN 11-2C-17, Volume 3, Addenda B, *C-17 Special Operations (CUI)*. **(T-1)**

Chapter 10

AIRCREW MAINTENANCE SUPPORT PROCEDURES

10.1. General. This chapter contains aircrew procedures not contained in the flight manual, other portions of this manual, or other publications.

10.2. Responsibilities. Aircrew may assist the normal maintenance function when critical contingency tasking dictates their use, provided this action does not impact crew duty and crew rest limits specified in **Chapter 3** of this manual.

10.3. Refueling/Defueling.

10.3.1. Aircraft Refueling. Aircrews normally only refuel in cases when maintenance support is not readily available and the mission would be delayed. If the Fuel Manifold Drain cannot be drained (lack of personal protective equipment /fuel container), make an AFTO 781A red diagonal entry stating why the step could not be completed and accomplishment causes delay.

10.3.2. Refer to T.O. 1C-17A-1, T.O. 1C-17A-2-12JG-28-1, *Servicing Fuel*, and T.O. 1C-17A-2-12JG-28-2, *Servicing Fuel* for refueling/defueling procedures.

10.3.3. Specialized Fueling Operations (SFO). Specialized Fueling Operations require additional certification IAW AMC Aircrew Tactics and Training Division (AMC/A3T) syllabus. For Specialized Fueling Procedures reference AFI 11-235, *Specialized Fueling Operations*, T.O. 00-25-172, *Ground Servicing of Aircraft and Static Grounding/Bonding*, and applicable AMC SFO Concept of Operations.

10.3.4. Concurrent Servicing. Concurrent Servicing Operations are defined as the simultaneous servicing of fuel or oxygen with any of the following: passengers on board, performance of minor maintenance, fleet servicing, or cargo loading/unloading. Concurrent Servicing Operations require a refueling supervisor (formerly Concurrent Servicing Supervisor) and other special precautions IAW T.O. 00-25-172. The C-17A is permitted to load/unload all classes of hazardous cargo during Concurrent Servicing Operations if only fuel is being serviced (not permitted during LOX servicing). Refer to T.O. 00-25-172 for further guidance.

10.4. Maintenance Monitor Panel. Aircrews will not erase any fault unless specifically directed by maintenance. **(T-2)**

10.5. Aircraft Servicing Requirements. When adequate maintenance support is not available, the aircrew may need to perform certain maintenance tasks. Use the Job Guides (JGs) in **Table 10.1**. If aircrew servicing is required at a base with AMC support, the PIC will notify C2 and describe the circumstances. **Note**: Aircrews should consult a qualified maintenance facility when questions/concerns arise as to performing any maintenance task. **(T-2)**

Table 10.1. Job Guides.

T.O. Number	Title
1C-17A-2-10JG-10-1	Ground Handling Parking/Mooring
1C-17A-2-10JG-30-1	Ground Handling Quick Turn-Around

1C-17A-2-10JG-50-1	Ground Handling Launch
1C-17A-2-10JG-60-1	Ground Handling Servicing Equipment Positions
1C-17A-2-10JG-70-1	Ground Handling Mission Reconfiguration Cargo Compartment
1C-17A-2-12JG-28-1	Servicing Fuel
1C-17A-2-12JG-28-2	Servicing Fuel
1C-17A-2-12JG-29-1	Servicing Hydraulic Power
1C-17A-2-12JG-79-1	Servicing Engine Oil
1C-17A-2-40JG-20-1	System Integration Displays
1C-17A-2-00GV-00-1	General Vehicle Manual
1C-17A-2-1	Aircraft Cross Servicing Guide
1C-17A-6	Inspection Requirements Manual

10.6. Aircraft Recovery Away from Main Operating Base (MOB). When an aircraft lands at a base other than the MOB, a crew chief should accompany the aircraft. The PIC is responsible for ensuring the aircraft is turned to meet subsequent mission taskings. If qualified maintenance specialists are unavailable, the aircrew is responsible for turning the aircraft to meet subsequent mission taskings.

10.6.1. Recovery items the aircrew may be responsible for include, but are not limited to, the following:

10.6.1.1. Parking and receiving.

10.6.1.2. Aircraft servicing.

10.6.1.3. Supervision of minor maintenance within local capability.

10.6.1.4. Minor configuration changes to meet mission tasking.

10.6.1.5. Securing the aircraft before entering crew rest.

10.6.1.6. Coordinating aircraft security requirements.

10.6.1.7. AFTO Form 781-series forms maintenance.

10.6.2. In all cases where aircrews service the aircraft without qualified maintenance specialist assistance, comply with procedures in this chapter. **(T-2)**

10.6.3. Aircrews are not qualified to accomplish the required ground inspections. In those instances where maintenance personnel are not available, the aircrew will enter a red dash symbol in the AFTO 781A, updating current status and enter a red dash symbol and a discrepancy that reflects that the applicable maintenance inspection (e.g., pre-flight, thru-flight, basic post-flight) is overdue. **(T-2)** Reference T.O. 00-20-1, *Aerospace Equipment Maintenance Inspection, Documentation, Policies, and Procedures*.

Chapter 11

CARGO AND PASSENGER PROCEDURES

11.1. Transportability Problem Items.

11.1.1. An air transportability problem item is any item in its proposed shipping configuration which may be denied transport aboard US Air Force cargo aircraft due to excessive size, weight, fragile or hazardous characteristics, lack of adequate means for handling, restraint, or a requirement for special support equipment.

11.1.2. Items defined as transportability problem items should be coordinated through AFLCMC/EZFC Air Transportability Test Loading Agency (ATTLA) prior to being offered for air shipment. Coordination for certification is the shipper's responsibility. Loadmasters are not responsible for determining if an item requires certification.

11.1.3. In addition to the procedures/limitations listed in T.O. 1C-17A-9, the following should be considered when determining the loadability of cargo:

11.1.3.1. Any vehicles, including any additional cargo they contain, must be fully restrained. Loose items such as those listed below will have provisions to be secured to the frame of the vehicle or they will be secured separately to the floor:

11.1.3.1.1. Spare wheels, tools and toolboxes, towing chains, pinch bars, etc.

11.1.3.1.2. Bulldozer blades and push arms.

11.1.3.1.3. Cranes or booms on wrecking trucks, etc.

11.1.3.1.4. Dump truck bodies and other hydraulic or mechanical lift mechanisms.

11.1.3.1.5. Machines and tools in shop trucks, shelters, and containers.

11.1.3.2. Commercial vehicle axle weights should be less than or equal to 80 percent of the manufacturer's gross weight rating. This information is normally found on the vehicle data plate. If sleeper shoring is used under the frame, the item may be airlifted at weights up to 100 percent of the manufacturer's gross weight rating.

11.1.3.3. Equipment must have a suitable number of tiedown provisions to allow for restraint to the aircraft using on-board tiedown devices. Examples of such tiedown provisions are vehicle frames, axles and crossmembers, pintle hooks, and cut-outs or other openings in structural members. Tiedown rings on non-military items (commercially manufactured items) should not be used unless the rating is provided. If tiedown rings are not present or rating available, restrain the vehicle using largest frame components, preferably at longitudinal main frames adjacent to lateral cross members.

11.1.4. Some non-standard cargo/equipment not contained in aircraft loading manuals requires certification for air shipment. The shipper will provide a copy of the certification to the designated aircrew member prior to loading. **(T-2)** If the certification letter with loading instructions/requirements is not provided to the designated aircrew member, the cargo should be carefully considered prior to accepting the cargo for airlift. Contact Air Transportability Test Loading Agency AFLCMC/EZFC (ATTLA) at Wright Patterson AFB, OH, (937) 255-

2330 or MAJCOM Stan/Eval for questions concerning cargo certification. ATTLA certification is required for:

11.1.4.1. Items that operate in flight or connect to the aircraft (e.g., vents, electrical outlets, antennas, etc.).

11.1.4.2. Non-palletized items with questionable structural integrity or items with significant damage to the frame or structural components (e.g., battle damaged equipment).

11.1.4.3. Items that exceed the allowable loading limits of the aircraft as described in the applicable aircraft TO 1C-XXX-9.

11.1.4.4. Items that require special equipment or loading procedures not listed in the applicable aircraft TOs.

11.1.4.5. Items designed to interface with the aircraft rail systems (e.g., LSA™ adapters), not contained in the applicable aircraft TO.

11.1.4.6. Any type of watercraft, fixed-wing and rotary-wing aircraft not identified in the applicable aircraft TO.

11.1.4.7. Enclosed items (airtight containers, on-board tanks, etc.) not designed with pressure relief devices or items that cannot be configured in a way to allow for aircraft cabin pressure changes.

11.1.4.8. Items that cannot be restrained using standard restraint procedures listed in the aircraft's TO or items requiring specific restraint procedures.

11.1.5. If the ability to restrain an item or its structural integrity is questioned, it will not be loaded. **(T-2)** If there is any question as to whether an item meets air transportability requirements contact ATTLA at: AFLCMC/EZFC (ATTLA), 2145 Monahan Way, Building 28, Wright-Patterson AFB, OH 45433-7101; (937) 255-2330; or visit the website at: **https://intelshare.intelink.gov/sites/attla/**.

11.2. Emergency Exits and Safety Aisles. A clear unobstructed path must be maintained in case of an emergency evacuation of personnel. **(T-1)** Loose equipment, crew or passenger bags will not be secured in a manner that will impede the use of emergency exits. **(T-2)** No part of the cargo load will extend over the edge of the walkway. **(T-2) Exception:** Equipment such as spare tires, mirrors, etc., protruding from rolling stock may extend over the walkway. Palletized cargo may extend beyond the vertical stacking line of the pallet only if specifically authorized in the loading manual or a load certification letter.

11.3. Pre-Mission Duties.

11.3.1. Duty Loadmaster Operations. Duty loadmasters are used as a means of flow control, for example at stations with limited aerial port personnel, when units are deployed as part of an Expeditionary Airlift Squadron or during a tactical/contingency operation. Duty loadmasters do not relieve the primary aircrew loadmaster(s) of their duties. Duty loadmasters ensure items loaded on aircraft do not exceed aircraft limitations and adequate restraint is applied to cargo to prevent movement. Duty loadmasters are not required to restrain cargo for flight limits. Duty Loadmasters will accomplish an Exterior and Interior Safety Inspection, Basic Aircraft Pre-flight and all appropriate loading preparation checklists prior to conducting loading operations. **(T-2)** After completion of aircraft loading, duty loadmasters will

accomplish the Before Leaving Aircraft Checklist unless maintenance personnel remain at the aircraft to monitor operating aircraft systems, or turnover to flight crew. **(T-2)** Primary aircrew loadmaster(s) will complete all pre-departure checklists to include the After Loading General Checklist. **(T-2)** Upon arrival at an aircraft where loading operations are being directed by a duty loadmaster, the primary loadmaster will coordinate with the duty loadmaster on assumption of direct loading supervision. **(T-2)**

11.3.2. Aerial Port Expediter (APEX) Loading Operations. APEX is an aerial port loading program directly managed and supervised by AMC Air Cargo Movement Policy Branch (AMC/A4TC). It provides aerial port management the flexibility to determine the best time to on/offload aircraft and the ability to evenly distribute port workloads. Loadmasters will not request APEX loading/unloading. It is not intended to function as an aircrew convenience or to alleviate the loadmaster's responsibility to on/offload aircraft.

11.3.2.1. APEX Load directors are qualified to on/offload all types of cargo (except airdrop equipment), operate the aircraft winch, cargo door, ramp and struts without the presence of an aircraft loadmaster. A maintenance representative must be present at the aircraft during APEX operations. Load directors are not authorized to apply/remove aircraft power or monitor aircraft operating systems. **(T-2)**

11.3.2.2. Load directors are required to ensure restraint of cargo meets flight criteria. Load directors will accomplish a basic cargo compartment inspection IAW current APEX guidance and appropriate loading preparation checklist prior to conducting loading operations. **(T-2)** APEX operations do not alleviate the loadmaster from accomplishing applicable checklists.

11.3.2.3. Load directors have overall control until relieved by the outbound loadmaster(s). A coordination briefing is required prior to the loadmaster assuming overall control.

11.3.2.4. Loadmasters who need on/offloads for training or evaluations should notify Air Terminal Operations Center (ATOC) prior to entering crew rest. ATOC should make every effort to accommodate the request.

11.3.2.5. Loadmasters treat APEX loaded aircraft like any duty-loaded or staged aircraft. Any cargo discrepancies that cannot be corrected by the loadmasters in a timely manner will be reported to ATOC. Forward negative APEX trend data to AMC/A3V through Stan/Eval channels.

11.3.3. Known tiedown equipment deficiencies. Tiedown devices found to be deficient must be removed from service and identified to Aerial Port for replacement either enroute or upon mission completion, as supplies allow.

11.3.3.1. Davis 08/08 CGU-3/E 25K tiedown device. Prior to use, ensure tiedown devices with manufacture date of "08/08" have a repair kit installed. Repair kits consist of a keeper plate on top side of release handle attached with three Philips head screws. Any devices that have not been repaired with these kits are defective and must be removed from service.

11.3.3.2. Peck and Hale CGU-4/E 10K tiedown device. The chain can be pulled out of clasp on these devices once locked. This can be accomplished by pulling on the excess chain (free end) while locked into the chain pocket of the device. Inspect by tensioning

the device, then pulling on the excess chain (free end). If chain comes out of the pocket, the device is defective and must be removed from service.

11.3.3.3. Davis CGU-4/E 10K device. The chain can be pulled out of the pocket when significant slack is present. This can be accomplished by pulling the loaded end of the chain away from the device, while locked into the chain pocket. Inspect by tensioning the device, then pulling on the loaded end of the chain. If chain comes out of the pocket, the device is defective and must be removed from service.

11.3.4. Missions with Passengers. Missions with 20 or more passengers should be planned with a pallet position available for baggage. Aerial Port will coordinate with the loadmaster to determine seat release/loose baggage capacity if a full position is not available. **(T-2)** Hand carried baggage will not obstruct egress pathways. **(T-2)**

11.4. Passenger Handling. The PIC will have final authority and may frustrate Hand-Carry-Only passengers that were selected if the baggage limitations cannot be met. **(T-2)** If a pallet is not available, the loadmaster may floor-load passenger baggage in the open pallet position if time and conditions permit.

11.4.1. When passengers are onboard, loadmasters occupying the forward loadmaster station will not position the seat or be distracted in a manner that would prevent them from immediately viewing or assisting passengers. **(T-2)**

11.4.2. Passengers will not be seated alongside pallets loaded in the logistics rails on the same side of the aircraft. These seats may be occupied by aircrew/MEPs at the PIC's discretion if non-netted cargo on a logistic pallet does not extend past X +/- 85 and a clear unobstructed path to an emergency exit is maintained. **(T-2)**

11.5. Passengers. Passengers are required to be seated with seatbelts fastened during critical phases of flight. Passengers will not be allowed to lay on top of ISUs, containers or palletized cargo. They will also not be allowed to lay on, inside of, or under rolling stock in flight. Laying or sleeping on the aircraft ramp is not allowed. Due to the possibility of impeding a safe egress, hammocks or slings are not to be utilized by attaching to the aircraft and/or cargo by passengers. PICs, with the Loadmaster's coordination, will maintain final authority on any requests for non-standard passenger accommodations. In all circumstances, passengers will have emergency provisions immediately available and will not impede safe egress of the aircraft in the event of an emergency.

11.6. Loaded Weapons. Weapons are considered loaded if a magazine is installed in the weapon. This applies even if the magazine is empty. Normally, personnel are not permitted to carry their basic combat load or individual issue of hazardous materials removed from its required packaging. See AFMAN 24-604, *Preparing Hazardous Materials for Military Air Shipments,* for procedural exceptions in support of the DoD, Federal agencies, and allies providing sustained, immediate, and responsive air movement, and delivery of personnel and hazardous material to, within, or from objective areas under tactical, contingency, or emergency conditions.

11.7. Weight and Balance. Accomplish weight and balance for this aircraft according to T.O. 1-1B-50, *Aircraft Weight and Balance*, T.O. 1C-17A-5-2, *Loading Data*, and Addenda A of this manual. **(T-1)** The unit possessing the aircraft maintains the primary weight and balance handbook containing the current aircraft status and provides a supplemental weight and balance handbook for each aircraft. **(T-2)**

11.7.1. The supplemental handbook will include T.O. 1C-17A-5-2, AFMAN 11-2C-17V3ADD-A, sufficient copies of DD Form 365-4, *Weight and Balance Clearance Form F—Transport/Tactical*, and a certified copy of the current DD Form 365-3, *Chart C - Basic Weight and Balance Record*. **(T-2)** Chart C will include the aircraft's basic weight, basic moment, and center of gravity. **(T-2) Exception:** T.O. 1C-17A-5-2 and AFMAN 11-2C-17V3ADD-A may be excluded from the supplemental handbook if carried in digital format.

11.7.2. The Weight and Balance section of the unit possessing the aircraft will provide the information required to maintain current and accurate documents to the appropriate agency. **(T-2)**

11.7.3. Crews will ensure a sufficient amount of printer paper is onboard to complete the mission. **(T-3)** Contact maintenance to restock paper supply if more is required.

11.8. Alcoholic Beverages. MAJCOM/A3 or NAF/CC may authorize the dispensing of alcoholic beverages.

11.9. Rucksacks. The following procedures apply to loading of rucksacks.

11.9.1. In all cases, rucksacks will be loaded on the same aircraft as the individual. **(T-2)**

11.9.2. Transported units must ensure that adequate space is provided on the load plan and aircraft to ensure all personnel have an unobstructed path to evacuate the aircraft during an emergency. **(T-1)**

11.9.3. During administrative/training deployments, rucksacks may be loaded on deploying vehicles, palletized, or floor loaded. Placing rucksacks on the aircraft floor may increase loading and offloading times. Also, this method may require more space and reduce the number of personnel or equipment airlifted.

11.9.3.1. During tactical deployments into a Forward Operating Base, rucksacks not loaded on vehicles will be carried by the individual onto the aircraft. **(T-2)** When this provision is used, the following procedures apply:

11.9.3.1.1. The troops may wear the rucksacks in the seat provided the seats are placed in the paratroop configuration.

11.9.3.1.2. All troops must have quick release straps on their rucksacks. **(T-2)**

11.9.3.1.3. Brief troops to leave their rucksacks on the seat if an emergency evacuation is necessary. **(T-2)**

11.9.3.2. The following procedures apply to transporting hazardous materials in rucksacks:

11.9.3.2.1. Personnel will only be permitted to carry their basic combat load or individual issue of hazardous material when they will engage an enemy force immediately upon arrival. **(T-2)** Personnel may retain small arms ammunition (cartridge for weapons, DOT 1.4) and nuclear, biological, and chemical equipment as long as it is retained in a carrier (e.g., bandoleers, pouches, bags). Weapons will remain clear until the aircraft has landed or as directed by the loadmaster. **(T-2)**

11.9.3.2.2. Protect munitions and other hazardous materials placed in rucksacks, field packs, or other authorized containers, removed from their shipping container from

accidental malfunction. For airland troops and airdrop troops who are not rigged prior to takeoff, all carriers will be consolidated in one central location on the aircraft (as directed by the loadmaster) and distributed to personnel after landing. **(T-2)** Paratroopers rigged prior to takeoff may retain individual carriers containing hazardous materials.

11.9.3.2.3. The troop commander or load team chief briefs the loadmaster concerning the individual issue of hazardous materials. The loadmaster will brief the PIC. **(T-2)**

11.9.3.2.4. Hazardous materials identified for sustainment must be prepared and certified according to AFMAN 24-604. **(T-1)**

11.10. Cargo and Material Handling Equipment (MHE) Issues.

11.10.1. ATOC will coordinate with the shipper or maintenance to connect/disconnect cargo to aircraft venting systems. **(T-2)**

11.10.2. Loadmasters will not connect/disconnect cargo to aircraft venting systems. **(T-1)**

11.11. Procedures for Loading Hazardous Cargo. Hazardous materials and cargo fall into many categories, handle and transport these items with caution. IAW AFMAN 24-604, the aircrew must have visual and physical access to all hazardous materials to mitigate any hazard posed by an in-flight incident. **(T-0)** Hazardous materials and cargo not properly packaged and documented in accordance with AFMAN 24-604 will be rejected for air shipment. **(T-1)**

11.11.1. Lithium Batteries and Pyrotechnic Material (Class/Division 1.3G). Aircraft halon fire extinguishers only help reduce the intensity of the fire until the lithium and pyrotechnic material expends itself. The prohibited transportation of damaged lithium batteries and strict adherence to packaging requirements has greatly reduced the inherent risks associated with shipment.

11.11.2. Load planners will position primary lithium battery and pyrotechnic material shipments on aircraft load plans in a manner that will permit easy access and jettisoning in-flight without moving other cargo. **(T-1)** Any agency providing hazardous cargo briefings will ensure aircrew personnel are aware of lithium battery and pyrotechnic material quantity and location aboard the aircraft. **(T-2)**

11.11.3. On/Offloading Procedures. Load teams should use extreme caution when handling or transporting these items and ensure hazard markings and warning labels are visible to aircrew and load team personnel.

11.11.4. Inflight/Emergency Procedures. Loadmasters must conduct frequent in-flight cargo checks to monitor primary lithium battery and pyrotechnic material shipments. **(T-2)** If a fire is suspected or indicated, loadmasters notify the PIC and complete the applicable aircraft emergency procedures. Use aircraft Halon fire extinguishers to slow down the thermal runaway and prevent the spread of fire to other cargo. Jettison if practical. **Note:** Do not use water as a fire suppression system for lithium batteries or pyrotechnic material. Water presents a hazard to electronic equipment and, in less than a large quantity, is ineffective.

11.12. Silver Bullet Command and Control Module (CCM).

11.12.1. When the Silver Bullet is assigned to a mission, the 618th Air Operations Center Special Assignment Airlift Missions (618 AOC/XOOO) planners will ensure the party is

briefed on the risk for in-flight occupancy. **(T-2) Note:** Due to lack of rigorous safety certification, AMC does not endorse occupancy of the Silver Bullet CCM during takeoff, landing, air refueling, or periods of moderate turbulence. If passengers occupy the Silver Bullet during these periods they do so at their own risk.

11.12.2. PICs are responsible for personally briefing the DV Contact and DV party (as applicable) on takeoff, landing, air refueling and turbulence occupancy restrictions. **(T-2)** The PIC retains overall authority to remove personnel from the Silver Bullet CCM when passenger safety may be jeopardized (in-flight emergency, combat threat environments, etc.).

11.12.3. PICs will ensure all crewmembers and passengers are briefed on:

11.12.3.1. Personnel Occupancy Limit (10 personnel total).

11.12.3.2. Aircraft Emergency Procedures.

11.12.3.3. Emergency Procedures specific to the Silver Bullet.

11.12.4. An Emergency Passenger Oxygen System (EPOS) and a life vest (when required) will be pre-positioned for every occupant.

11.12.5. In the event of a loss of cabin pressurization or smoke/fumes, Silver Bullet occupants will don EPOS and evacuate the Silver Bullet CCM. **(T-1)** After all occupants egress the Silver Bullet, electrical power should be removed.

11.12.6. The Silver Bullet/Steel Eagle wireless router and communications equipment must be turned off during air refueling operations.

11.13. Senior Leader In-Transit Conference Capsule (SLICC) and Senior Leader In-Transit Pallet (SLIP).

11.13.1. Only qualified SLICC maintenance personnel will install/remove the SLICC vestibule and connect/disconnect SLICC electrical power. **(T-2)**

11.13.2. ADS palletized seats or other cargo will not be loaded immediately forward of the SLICC Conference Capsule. Maintain a clear forward exit path from the Conference Capsule's forward exit door.

11.13.3. A maximum of four operational SLIPs may be airlifted on a single aircraft. When multiple occupied SLIPs are used on the same aircraft, all SLIPs should be loaded into the same side of the logistics rails to simplify the inter-SLIP electrical connections.

11.13.4. Due to availability of seat belts, if passenger restraint is required during turbulence, occupancy is limited to five in the Conference Capsule.

11.13.5. The SLICC Berthing Capsule will not be occupied during critical phases of flight (takeoff, landing, and aerial refueling).

11.13.6. Passengers may occupy all seats on the SLIP during all phases of flight.

11.13.6.1. Aircrew will ensure seats are facing either fully forward or fully aft for takeoff and landing. **(T-1)**

11.13.6.2. Each seat will be equipped with an EPOS and a life vest (when required). **(T-2)**

11.13.7. SLICC modules power can be shut off immediately in the event of an emergency by pressing the red emergency shutdown switch on either the external power distribution panel or

internal operator control panel of each module. Aircrews will familiarize themselves with the location of these emergency shutdown switches prior to transporting passengers in the SLICC modules. **(T-2)**

11.13.8. Confirm an emergency key is located in a pouch by each entry door. These are available for use by aircrew personnel in the event that an emergency situation would require immediate access to a locked module.

Chapter 12

FUEL PLANNING AND CONSERVATION

12.1. General. This chapter provides fuel planning procedures for all C-17 missions including flight managed and local missions. Missions should be planned at altitudes, routes, and airspeeds to minimize fuel usage.

12.2. Fuel Reserves.

12.2.1. **Table 12.1** shows Identified Extra Fuel additions for mission planning.

Table 12.1. Identified Extra Fuel.

THUNDERSTORMS Corrections are not cumulative Use only the highest applicable correction	Isolated - 1,300 lbs. Few - 2,500 lbs. Scattered or Numerous - 5,000 lbs.
ICING Corrections are cumulative Add fuel for forecast or actual conditions	Departure or Landing - 1,100 lbs. (min)/2,200 lbs. (max) Enroute - 1,100 lbs.
ATC Altitude Restrictions (1) (2) Applicable for airfields listed in the Flight Planning Fuel Policy Letter on the EFB.	Hold Down – 4,500 lbs. Early Descent – 4,500 lbs.
ENVIRONMENTAL HI FLOW	100 lbs./hour
GROUND OPERATIONS/ERO	100 lbs./minute
KNOWN HOLDING DELAYS	200 lbs./minute of expected delay
Notes: (1) Hold down fuel is added as a departure bias and is calculated as if it is burned upon reaching the top of climb. (2) Early descent fuel is added as an arrival bias and is calculated as if it is burned on the last leg of the flight plan.	

12.2.2. 2,200 lbs. of fuel is calculated by MAFPS and included in the "Unusable Fuel" block for Fuel Totalizer bias.

12.2.3. FMs will calculate the MAFPS ASR with a fuel burn bias, specific to aircraft tail number. **(T-2)** This bias will be annotated on the MAFPS ASR as "DGDP" (Degrade Percentage). **(T-2)** Enter the Tail Specific Fuel Bias (Fuel Flow Factor) value, as found on the C17/CL_Inserts fuel bias list, into the MC as a Fuel Factor (FF LSK 6L on the Performance Factors Page). **(T-2)** The number found on this checklist may not match the MAFPS "DGDP" number. **(T-2)** Crews may contact the flight manager for an updated Required Ramp Fuel Load (RRFL) following a tail swap.

12.2.4. Tankering fuel for convenience is prohibited. If FM planned tankered fuel is deemed operationally necessary, it will be included in the RRFL and included on the MAFPS ASR in the Identified Extra block. **(T-2)**

12.2.5. When filing an alternate located in Alaska or at latitudes greater than 59 degrees (north or south) use standard holding fuel calculations. **(T-2)**

12.2.6. Local supplements will not dictate IAF or Top of Descent fuel. **(T-2)**

12.3. Fuel Conservation. It is Air Force policy to conserve aviation fuel when it does not adversely affect training, flight safety, or operational readiness. Aircrew and mission planners manage aviation fuel as a limited commodity and precious resource. Fuel optimization will be considered throughout all phases of mission planning and execution. **(T-2)** Comply with the following whenever consistent with tech order guidance and safety:

12.3.1. Fuel Loads. Excessive ramp and recovery fuel adds to aircraft gross weight and increases fuel consumption. Do not ferry extra fuel beyond optimum requirements for safe mission accomplishment and training objectives.

12.3.2. Flight Planning. Aircrew and mission planners optimize flight plans and flight routing for fuel efficiency based upon wind, altitude, etc. Accepting a "direct to" ATC clearance is not necessarily fuel or time advantageous. Crews should analyze all factors prior to accepting a "direct to" clearance off the MAFPS route.

12.3.3. Takeoff. Consider a rolling takeoff and Derated Thrust (DRT) to save fuel and minimize engine wear. Use DRT for takeoffs unless dictated by performance, tactics, etc.

12.3.4. Climb/Descent. In-flight procedures such as climb/descent profiles and power settings should also be considered for efficient fuel usage.

12.3.5. Descent. The optimum descent is executed at idle power. Avoid early descents whenever possible.

12.3.6. Weather Deviations. Attempt to coordinate for off-course deviation early so gross maneuvering is not required.

12.3.7. Holding. If holding is required, hold clean at the most fuel efficient altitude and request a large holding pattern. Hold at endurance or performance manual recommended holding speeds, conditions permitting.

12.4. Fuel Planning Procedures.

12.4.1. Prior to departure, the minimum planned landing fuel at the primary destination is the greater of:

12.4.1.1. 18,000 lbs. or **(T-2)**

12.4.1.2. Required fuel reserve, fuel required to transit to alternate (if required), contingency fuel, and any authorized identified extra fuel. **(T-2)**

12.4.2. Fuel Calculations. PICs verify the RRFL in the MC by ensuring all required planning data is input into the MC. **(T-2)**

12.4.2.1. If the MC calculates a destination fuel less than 18,000 lbs., generates an INSUFFICIENT FUEL message, or in the PIC's judgment, the RRFL is insufficient to complete the mission, the PIC will:

12.4.2.1.1. Ensure the MC is properly programmed including any applicable drag indexes/factors into the performance factors page. **(T-2)**

12.4.2.1.2. Contact the FM to identify and resolve discrepancies in fuel planning calculations. The PIC is the ultimate authority for final fuel load.

12.4.3. When Actual Fuel Load exceeds the RRFL by more than 5,000 lbs., consider defueling the aircraft to the RRFL if time and mission requirements permit.

12.4.4. Once airborne: Change each destination and rendezvous (RZ) alternate holding fuel (LSK 4L) to 0+45 or 2+00 (if using holding in lieu of), and remove all ID extra except 2.2K (fuel totalizer bias) and authorized tankered fuel to reflect the aircrew's ability to use all, some or none of the contingency fuel. If the MC generates an INSUFFICIENT FUEL message, recheck flight plan/performance factors for accuracy. If the MC is accurate, consider divert options and contact a FM.

12.5. Computer Flight Plan. Reference MAFPS folder on EFB for current examples and descriptions.

12.6. Tactical Fuel Planning. Ensure low-level time and ground operations are accounted for during tactical missions. **(T-2)**

12.7. AAR Fuel Planning.

12.7.1. Preflight Fuel Planning. Enter MC information IAW T.O. 1C-17A-1-2. Ensure that UNID EXTRA and STORED fuel for any RZ is greater than or equal to 0.0.

12.7.2. In-flight Fuel Planning. For single and multiple AAR, PICs are responsible for computing recovery fuel requirements, and required on-load requirements. **(T-2)** Reference AFTTP 3-3.C-17 for AAR techniques.

Chapter 13

AIR-TO-AIR REFUELING

13.1. General. This chapter establishes air refueling procedures and policy applicable to C-17 aircraft and aircrews and is in addition to those prescribed by the flight manual and other applicable directives.

13.2. AAR Limitations.

13.2.1. Tanker Autopilot. Tanker pilots notify receiver pilots when any axis of the autopilot is not used. If a tanker pilot or receiver pilot requires the tanker aircraft to fly autopilot-off for training, the pilot flying the opposing aircraft will be qualified unless under the direct supervision of an instructor (N/A for Formal Training Unit or upgrade training). **(T-2)** Verbal notification and acknowledgement will take place between the tanker and receiver prior to conducting autopilot-off refueling. **(T-2)**

13.2.2. AAR Without Tanker Disconnect Capability. Without tanker disconnect capability means the boom operator cannot trigger an immediate disconnect. AAR will not be conducted after a loss of tanker disconnect capability. **(T-3) Exceptions:** Fuel emergency or contingency missions, Joint Chiefs of Staff alert (J-alert), and support missions under normal conditions when refueling is essential for home base recovery, or for any real-world mission when authorized. Do not accomplish any training, boom limit demonstrations, or practice emergency separations. **(T-2)**

13.2.3. Manual Boom Latching (MBL) (also referred to as Emergency Boom Latching (EBL), Override Boom Latching (OBL) and amplifier override). This is an emergency procedure. Normal tanker disconnect capability and automatic disconnect limits are inoperative. Use of this procedure must be authorized in the mission directive. **(T-2)** C-17s are not authorized to conduct MBL training. **Note:** The boom operator and receiver pilot will coordinate all actions as required by applicable directives and checklists when making AAR contacts using emergency boom latching procedures. **(T-2)**

13.2.4. Reverse AAR procedures will be accomplished for operational necessity IAW T.O. 1C-17A-1. **(T-2)**

13.2.5. Practice Emergency Separations:

13.2.5.1. Follow “Breakaways” guidance in T.O. 1C-17A-1 and ATP-3.3.4.2.

13.2.5.2. Prior to the actual accomplishment of a practice emergency separation, coordination between the tanker pilot, boom operator, and receiver pilot is mandatory. Coordination will include when the separation occurs and who will give the command of execution. **(T-2)** Pilots will verify the AAR system is not in override. **(T-2)** Tanker disconnect capability must be verified by a boom operator initiated disconnect prior to accomplishing the separation. **(T-3)** Tanker disconnect capability will not be verified on the same contact as the practice emergency separation. **(T-3) Note:** Practice emergency separations will terminate no lower than 500 feet below tanker altitude. **(T-2)**

13.2.6. Receiver AAR Training for Unqualified Receiver Pilots (includes first pilots, and aircraft commanders refueling from the right seat). In-flight training will be accomplished under direct IP supervision. **(T-3)** The following procedures apply:

13.2.6.1. The receiver pilot informs and receives acknowledgment from the tanker of crew qualifications and anticipated training. **(T-3)**

13.2.6.2. For receiver pilot initial qualification or requalification, the receiver instructor/examiner pilot will be in one of the pilot seats with immediate access to the controls through all phases of the refueling from the astern position until back to at least 100 feet in trail of the tanker. **(T-2)**

13.2.7. Planned changes of aircraft control should be accomplished from no closer than the astern position. **(T-2)**

13.2.8. If a tanker or receiver seat change takes place, move back to at least 100 feet in trail of the tanker and to a point where the receiver pilot can maintain visual contact with the tanker until the seat change is complete. **(T-2)**

13.2.9. Tanker disconnect capability will be demonstrated by a boom operator initiated disconnect before conducting a limit demonstration. **(T-2)** Tanker disconnect capability will not be verified on the same contact as the limit demonstration. **(T-2)**

13.2.10. Weather Limitations.

13.2.10.1. Turbulence. Do not plan AAR if severe turbulence is forecast on the refueling track. **(T-2)** Terminate refueling if moderate turbulence is encountered. **(T-2)**

13.2.10.2. Visibility. Do not close from 1 NM range (2 NM for receiver or tanker formations) unless you have visual contact with the tankers. **(T-2)** Discontinue refueling if in-flight visibility is insufficient to continue safe refueling operations. **(T-2)**

13.2.10.3. AAR alternate airfields must meet the criteria of AFMAN 11-202V3. **(T-2)**

13.2.11. NVG Use During AAR. Pilots may use NVGs during air refueling rendezvous to acquire the tanker but must remove goggles no later than (NLT) 1 NM from the tanker. **(T-3)**

13.3. Receiver PIC Responsibilities.

13.3.1. Always obtain a stable, astern position (no closure or drifting back) prior to closing to the contact position. **(T-3).** The astern position is approximately 50 feet aft and slightly below the tanker boom nozzle.

13.3.2. Will ensure planned air refueling altitude is within aircraft formatting capability. **(T-2)**

13.3.3. Must receive permission from controlling authority (i.e., OG/CC, 618 AOC) prior to accomplishing opportune AAR. **(T-3)**

13.3.4. Will not redistribute fuel to increase AAR onload rate. **(T-2)**

Chapter 14

COMBAT MISSION PLANNING

14.1. General. This chapter provides general combat mission planning guidance for planners and aircrews, standardizing procedures for planning, briefing, and reviewing all missions. Planners and aircrews should reference AFTTP 3-3.C-17 for additional mission planning techniques. Pilots will be given one full day of planning for missions employing low-level operations. **(T-3) Exception:** Aircrews may perform low-level operations using mission planning cell or equivalent developed charts and scenarios without a full day of mission planning when circumstances would prevent sortie execution. Aircrews operating under this exception will add 30 minutes to their sequence of events to accommodate a thorough low-level study on the day of the flight. **(T-3)**

14.2. Drop Zone (DZ) and ALZ Surveys. The AMC Zone Availability Report (ZAR) and AMC Guidance Packages (formerly Digital Tactics Binder) are the approved sources for DZ and ALZ surveys. **(T-2)** The ZAR is maintained by AMC Combat Operations Division-Tactics (AMC/A3TW), and AMC Guidance Packages are maintained by 618 AOC.

14.3. Route Planning. To the maximum extent possible, crews should reference techniques in AFTTP 3-3.C-17. Low-level flight will be planned using tactical corridors. **(T-3)** The standard width for a tactical corridor is 3 NM. Tactical corridor width can vary from 1 NM minimum either side of centerline, to as wide as desired. Corridors do not have to be symmetrical but must be annotated on the chart. **(T-3)** The following altitudes are the minimum established for C-17 operations. FLIP/ICAO procedures, training considerations, terrain, or operational directives may dictate higher altitudes.

14.3.1. Day VMC is 300 feet AGL, modified contour. **(T-1)** Fly modified contour by using radar altimeter, HUD, and visual references. **(T-2)**

14.3.2. Night Enroute Altitude is an indicated altitude of 500 feet above the highest obstruction to flight, within a tactical corridor. See example **Figure 14.2**. **Note:** If Digital Terrain Elevation Data (DTED) is not available, Night Enroute Altitude will be calculated using 400 feet plus one chart contour interval above the highest depicted terrain contour, if higher. See **Figure 14.1** for inherent chart errors. **(T-1)**

14.3.3. NVG Enroute Altitude.

14.3.3.1. Fly no lower than an indicated altitude of 500 feet above the highest spot terrain (charted or DTED spot elevation) within a tactical corridor. **(T-2) Note:** If DTED is not available, NVG Enroute Altitude will be calculated using 400 feet plus one chart contour interval above the highest depicted terrain contour, if higher. See **Figure 14.1** for inherent chart errors. **(T-1)**

14.3.3.2. NLT 3 NM prior to any charted man-made obstacle within the NVG tactical corridor, the aircrew must visually identify the obstacle. **(T-2)** If the obstacle is not identified by 3 NM or lateral deconfliction cannot be ensured with navigational aids (i.e., mission computer Ground Reference Points, chart study, etc.) climb to 500 feet above the obstacle until the aircrew confirms the aircraft is past the obstacle, or until the crew can identify the obstacle and maintain well clear. **(T-2)**

14.3.4. VFR Minimum Safe Altitude (VFR MSA). VFR MSA is an initial VFR altitude that provides additional terrain and obstacle clearance while the aircrew analyzes situations that require interruption of low-level operations (route disorientation and equipment malfunctions or when either pilot leaves the seat during low-level operations, etc.). Plan VFR MSA at an indicated altitude of 500 feet above the highest obstruction to flight, within 5 NM of route centerline. **Note:** If DTED is not available, MSA will be calculated using 400 feet plus one chart contour interval above the highest depicted terrain contour, if higher. An MSA will be computed for each leg, route segment, or entire low-level route. See **Figure 14.1** for inherent chart errors. **(T-2)**

14.3.5. Minimum IFR Enroute Altitude.

14.3.5.1. Fly no lower than 1,000 feet (2,000 feet in mountainous terrain) above the highest obstruction to flight within 4 NM of route centerline (10 NM outside the US unless 4 NM authorized by MAJCOM/A3). **(T-2)** Round this altitude to the next 100-foot increment. **(T-2)** **Note:** Mountainous terrain is defined in AFMAN 11-202V3.

14.3.5.2. Minimum altitudes for IFR operations in IMC within published Military Training Routes (MTRs)/Special Use Airspace (SUAS) in US sovereign airspace or operational Minimum Risk Routing (MRR) will be the computed leg minimum IFR Enroute Altitude unless a higher altitude is required by FLIP AP/1B or SPINS. **(T-2)**

14.3.6. Emergency Route Abort Altitude (ERAA). ERAA is 1,000 feet (2,000 feet in mountainous terrain) above the highest obstruction to flight within 22 NMs of planned route centerline.

14.3.6.1. Several ERAAs may be computed for route segments transiting significant terrain differentials or a single ERAA may be computed for the entire low-level route.

14.3.6.2. Climbing to ERAA may put the aircraft or formation in a controlled (i.e., IFR) altitude structure and require coordination with Air Traffic Control agencies.

14.3.7. ANP 1.0 or less is required to fly lower than MSA at night. **(T-2)** ANP 0.3 is required to fly tactical corridors less than 3 NM left and right of centerline at night. **(T-2)** In the event of suspected GPS degradation, crews may continue in a tactical corridor smaller than 3 NM either side of centerline so long as centerline can be identified via formation position and/or ground reference.

14.3.8. Obstruction to flight is defined as man-made obstacle, terrain feature, charted spot elevation, or DTED spot elevation.

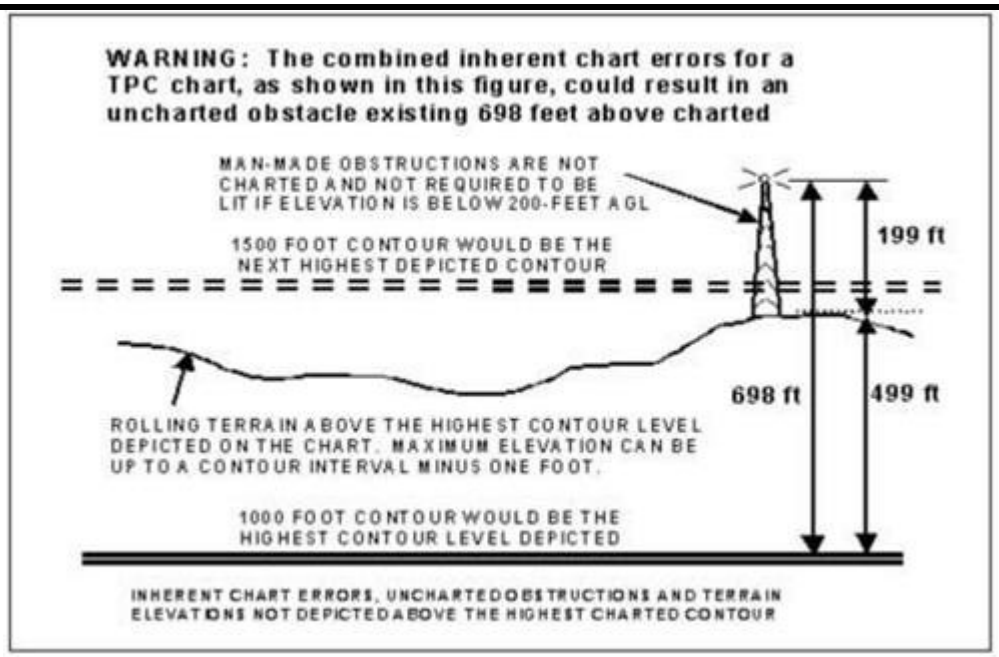
14.3.9. DTED information may be used but is not required in determining low-level altitudes.

14.3.10. Fly other than Day VMC (modified contour) altitudes or escape maneuvers with reference to the barometric altimeter, using the radar altimeter as backup. **(T-2)**

14.3.11. If the altitude for the next leg is higher than the current leg altitude, complete climbs before the turn point. If the altitude for the next leg is lower, do not initiate descent until after the turn point. **(T-2)**

14.3.12. Legs may be segmented to allow flight closer to the ground. Once the obstruction is visually identified and the aircraft can maintain well clear, the crew may descend to the next segmented altitude, if lower.

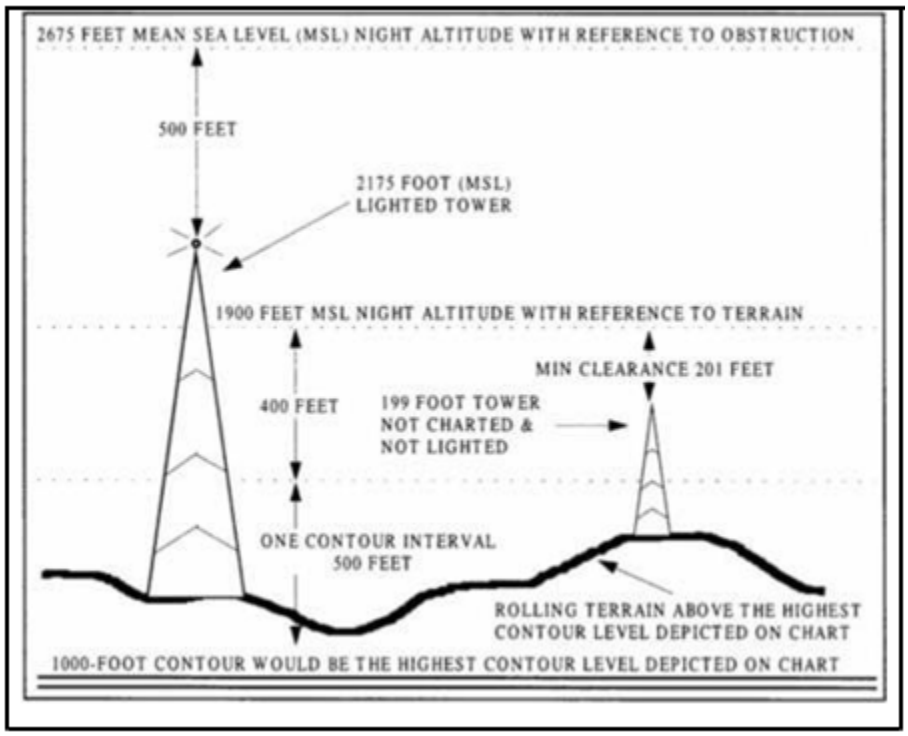
Figure 14.1. Inherent Chart Errors.



Warning: Aeronautical charts do not depict man-made obstacles less than 200 feet AGL or a change in terrain until it exceeds the chart contour interval. The worst situation would occur if a 199-foot tower sat on terrain with an elevation just below the next higher contour. For a TPC (1:500,000) with a contour interval of 500 feet, this results in an uncharted obstacle existing 698 feet above charted terrain. Additionally, the highest spot elevation on any given leg may not be the highest terrain as in the case of gradually rising elevations. Planners will ensure accurate terrain analysis by evaluating both spot elevations and the highest contour level. **(T-2)**

Caution: Some charts may depict terrain and obstacle altitudes in meters versus feet (e.g., JOG and TLM charts in some areas of the world).

Figure 14.2. Night VMC Enroute Altitude.



14.3.13. Temperature Correction. For all low-level operations, temperature corrections will be applied to the minimum IFR, VMC night and NVG altitudes. **(T-2)** Apply corrections IAW the temperature correction chart provided in the Flight Information Handbook (FIH) to ensure adequate obstacle clearance. **(T-2)** Add the values derived from the FIH temperature correction chart to the IMC and Night minimum enroute altitudes whenever the outside air temperature is 32°F/0°C or below. **(T-2)**

14.4. Peacetime Route Restrictions. In addition to restrictions in AFMAN 11-202V3, specific country or theater of operations publications, and FLIP area planning, routes should not be planned or flown:

14.4.1. With less than 1 NM of separation (3 NMs when in excess of 250 Knots Calibrated Airspeed (KCAS)) when below 2000 feet AGL from known sensitive environmental areas (i.e., hospitals, fish hatcheries, large poultry complexes, recreation areas, institutions, etc.).

14.4.2. With less than 3 NMs of separation from prohibited airspace.

14.4.3. With less than 3 NMs of separation from nuclear power plants.

14.4.4. Through Restricted airspace without clearance.

14.4.5. Below 1,000 feet AGL within a 2,000-foot radius over cities or towns shown as magenta shaded areas on 1:500,000 (TPC) scale charts, in addition to the restrictions in AFMAN 11-202V3.

14.4.6. Over or through active live fire or impact areas that may not be specifically designated as Prohibited or Restricted areas.

14.4.7. Below 500 feet AGL unless:

14.4.7.1. Host nation rules specifically allow such VFR operations.

14.4.7.2. Routes or training areas have been environmentally assessed and surveyed for operations at 300 feet AGL. This restriction does not apply to one-time-use routes. Consult FLIP AP/1B for published Military Training Route restrictions.

14.5. Navigation Chart Preparation. Low-level navigation charts will be updated and annotated with the most recent Vector Vertical Obstruction Data (VVOD) or supplement. **(T-2)** In no case will VVOD coverage be less than 22 NMs either side of the entire planned route centerline. **(T-2)** Crews may trim charts to no less than 10 NMs of the planned route centerline after establishing the emergency safe altitude (ESA). **Caution:** 1:50,000 and smaller scale maps do not depict aeronautical information, may not show man-made obstacles, and are rarely updated through the VVOD.

14.5.1. Chart Annotation. Individual chart annotations will have, as a minimum, turn points, initial point (IP), objective area, course line, navigation information, VVOD data and date, ERAA and chart series/date. **(T-2)** Reference AFTTP 3-3.C-17 for standard chart techniques.

14.5.2. Digital or paper low-level navigation charts are approved for use.

14.6. Route Study. Crew route study is mandatory before accomplishing flight in the low-level environment. The route study will include, at a minimum:

14.6.1. Route overview.

14.6.2. Obstacles/obstructions along the planned route.

14.6.3. Leg information (e.g., course, altitudes, and controlling obstacles).

14.6.4. Low-level hazards and mitigation.

14.6.5. Terrain features.

14.6.6. Individual crewmember responsibilities/duties.

14.6.7. Discussion of any airspace restrictions.

14.6.8. ERAA controlling obstacle and location.

14.7. Airlift Support Forces Coordination. Ensure coordination is complete with airlift and supporting forces. **(T-2)**

14.8. Briefings. The PIC will ensure all applicable briefings and de-briefings are completed for each mission. **(T-2)** Brief applicable items in sufficient detail to ensure a clear understanding of mission objectives and procedures. The PIC is responsible for ensuring all crewmembers are briefed on applicable mission items. **(T-2)**

Chapter 15

MISSION EMPLOYMENT/TACTICAL PROCEDURES

15.1. Tactical Checklists. Initiate the Combat Entry Checklist so it is completed prior to entering the combat area. Execute the Combat Exit Checklist after vacating the combat area. These checklists will be executed for all airdrop and tactical airland missions. **(T-2)**

15.1.1. As a minimum, the PF, PM and primary LM will remain on interphone from initiation of the Combat Entry Checklist until completion of the Combat Exit Checklist, unless crew duties require otherwise. **(T-2)**

15.1.2. Personnel performing duties requiring them to be mobile in the cargo compartment during airdrop, low-level operations, or threat environments will wear protective headgear. **(T-2)** All other personnel in the cargo compartment will be seated with seat belts fastened. **(T-2)**

15.1.3. During accomplishment of tactical checklists, do not change IFF or CALLSIGN X until appropriate FIR boundaries have been transited, or as directed by ATC. **(T-2)**

15.2. Tactical Descents, Arrivals and Departures.

15.2.1. Formation tactical descents are limited to VMC, 3-ship formations and a minimum of 6,000-foot spacing between aircraft. **(T-2)**

15.2.2. Unless operational necessity dictates, limit tactical descents to comply with all required airspace speed restrictions except as directed by T.O. 1C-17A-1 (i.e., 250 KIAS below 10,000 feet). **(T-2)**

15.3. Ground Operations.

15.3.1. NVG Taxi/Backing. Pilots may taxi using NVGs on airfields without lights (blacked out) or equipped with overt or covert lights. If taxiing or accomplishing ground operations on blacked out taxiways/runways/ramps, the PIC will ensure aircraft or environmental lighting provides for clear definition of taxiway/runway/ramp edge. **(T-2)** Comply with all taxi restrictions in **Chapter 5** and the ASRR. The LM may provide the pilot with directions to taxi the aircraft while using NVGs.

15.3.2. Engine Running Onload and Offload (ERO) Procedures. Do not use troop doors. **(T-2)** Do not onload or offload through the crew entrance door and cargo ramp/door at the same time unless both doors are directly supervised by a C-17A qualified pilot/loadmaster. **(T-2)** If a combat offload is to be accomplished before offloading vehicles, do not remove restraint until after the combat offload is completed. **(T-2)** When EROs are accomplished through the cargo ramp and door, place engines in reverse idle prior to the pilot giving clearance for offload/onload operations. **(T-2)** If a TR will not extend, consideration should be given to shutting down the affected engine. With exception of small arms ammunition (Division 1.4), do not ERO explosive cargo unless authorized in the JA/ATT, exercise, operation or contingency air tasking order, SPINS, etc. **(T-2)**

15.3.2.1. Do not release vehicle parking brakes until all restraint is removed and cleared by the loadmaster. **(T-2)**

15.3.2.2. Personnel to be offloaded will be briefed to secure their baggage aboard vehicles (when conditions allow). **(T-2)**

15.3.2.3. Vehicles and all personnel exiting via the ramp will proceed directly aft of the aircraft at least 25 feet before turning and/or at least 200 feet before stopping. **(T-2)**

15.3.2.4. The loadmaster directs all onload and offload operations using briefed signals. Other qualified personnel (Contingency Response Group (CRG) loadmaster, aerial port member, etc.) may perform these duties; however, the primary loadmaster retains overall responsibility for the operation. Passengers should be escorted (CRG, Aerial Port, airfield control personnel, etc.) when enplaning or deplaning. Deplane passengers before removing cargo and enplane after loading cargo unless cargo size and location dictates otherwise. **(T-2)**

15.3.2.5. Load Data. The loadmaster may use the load plan total weight and load center of balance for entry on the DD Form 365-4 provided these procedures are followed:

15.3.2.5.1. The load plan data will be checked by a qualified load plan validator (i.e., aircraft loadmaster, CRG loadmaster, aerial port specialist, or any individual who has completed the AMC Affiliation Program Airlift Planners Course). **(T-2)**

15.3.2.5.2. If downloading to an empty aircraft, a DD Form 365-4 is not required for the subsequent sortie.

15.3.2.6. Crew Entrance Door ERO Procedures. The PIC may approve the offload or onload of personnel and small cargo through the crew entrance door. In this instance, the throttles may be positioned to idle or reverse idle. Deplane the loadmaster to assure safety of deplaning/enplaning of personnel. **(T-2)**

15.3.2.7. Reduced Lighting EROs. Reduced lighting EROs must be accomplished with red/NVIS (overt) lighting in the cargo compartment sufficient to permit MHE drivers to see marshaller's signals and safely position MHE. **(T-2)** Dimming rheostat will be set to minimum amount of lighting to accomplish LM duties. **(T-2)** Marshallers will not use NVGs to position MHE within 25 feet of the aircraft for on/offloading operations. **(T-2)** MHE operators may utilize NVGs to the final parking position for situational awareness. LMs may use NVGs to maintain situational awareness behind the aircraft before and after actual loading.

15.3.3. Combat Offload Procedures. On operational missions, the controlling MAJCOM/A3 or Director of Mobility Forces (DIRMOBFOR) may authorize combat offloads when conditions warrant. OG/CC may approve combat offloads on training missions following a thorough risk assessment.

15.3.3.1. Many explosive items have specific "drop criteria" that, if exceeded, render the item useless or dangerous to the user. Small arms ammunition (Division 1.4) and all other Class 1 cargo rigged for airdrop (i.e., with cushioning material), may be combat offloaded. All other Class 1 cargo will not be combat offloaded, unless specifically directed by the approving authority. **(T-2)**

15.3.3.2. Excessively rough, sharply undulating or battle-damaged surfaces may cause damage to the aircraft during combat offload operations. Reducing forward taxi speed on these surfaces reduces aircraft oscillation. The PIC is responsible for ensuring the offload

area permits the offload operation to be conducted without damage to the aircraft. The PIC is ultimately responsible for ensuring the area in front and behind the aircraft is clear during the operation. The loadmaster ensures the area behind the aircraft is well clear of anything which may be damaged due to engine exhaust blast. To combat offload, a surface of at least 1,000 feet is required; however, 1,500 feet is desired to provide a margin of safety. **(T-2)**

15.3.3.3. Ensure the crew rest window to the cargo compartment is clear of obstructions for combat offload operations. **(T-2)**

15.3.3.4. If ground personnel are present, the crew will ensure the individuals are contacted to ensure no one disrupts the operation. **(T-2)**

15.3.3.5. If combat offloading to an empty aircraft, a DD Form 365-4 is not required for the subsequent sortie.

15.3.3.6. NVG Combat Offloads. Pilots may accomplish Combat Offloads on NVGs at airfield light levels down to and including blacked out, provided aircraft or airfield lighting permit clear definition of taxiway/runway/ramp edges. Loadmasters may accomplish Combat Offloads on NVGs with Red/NVIS (overt) or IR (covert) cargo compartment lighting set at minimum levels to perform loadmaster duties safely. Blacked out (no-light) operations in the cargo compartment are not authorized. **(T-2)**

15.4. Emergency Airlift of Personnel. Use these procedures for noncombatant evacuation and combat loading of personnel when directed by the controlling combatant commander, MAJCOM/A3, or DIRMOBFOR. Refer to AFTTP 3-3.C-17 for expanded guidance. Airlift of this nature is normally accomplished without the use of individual seats, seat belts, or litter stanchions. Emergency Oxygen sources are not available. For mission planning purposes, an estimated 300 personnel can be seated on the main floor and 50 on the ramp. The actual number of personnel may exceed this depending on their size, amount of baggage/equipment carried, and the number of tiedown straps onboard the aircraft. Mission planners should update planning factors for extended operations. Prior to departure for operations, ensure tiedown straps are fully stocked and consider carrying extra straps if available.

15.4.1. The following methods are recommended to maximize the number of personnel that can be carried:

15.4.1.1. Seat passengers in sidewall seats first and then facing forward on the cargo floor or ramp in lateral rows. Consider reserving the use of sidewall seats for passengers with special needs (elderly, pregnant, wheelchair bound, small children, etc.).

15.4.1.2. Attach the hook end of cargo straps to tiedown rings in rows A and G. After personnel are seated, route straps laterally across their legs and secure ratchet end of straps to the tiedown ring in row D. Use caution as overtightening the strap restricts blood circulation in the legs of passengers.

15.4.1.3. Combat troops may sit on rucksacks secured with tie-down straps or individual restraint device (last-resort belt) clipped into tiedown rings. If rucks are worn on backs and equipped with a quick-release strap, brief troops to leave their rucks on the aircraft in the event of an emergency.

15.4.1.4. If possible, secure baggage separately, either palletized or stacked/secured on the ramp.

15.4.2. The maximum altitude for emergency airlift will not exceed FL250. **(T-2)**

15.4.3. Outboard toes should be stowed after loading.

15.4.4. Consider adding additional emergency equipment, fleet service supplies and cargo straps.

15.4.5. For floor loading of litter patients, refer to AFMAN 11-2AEV3 Addenda-A, *Aeromedical Evacuation Operations Configuration/Mission Planning.*

Chapter 16

AIRCRAFT FORMATION

16.1. General. Formation procedures will be conducted IAW this chapter and T.O. 1C-17A-1. **(T-2)** Aircrews will follow formation standards described in AFTTP 3-3.C-17 unless otherwise briefed. **(T-2)** Consider safety, aircrew capability, proficiency, survivability and user requirements when planning any formation.

16.1.1. Vortices generated during departure, airdrop, and recovery can be significant in size, duration, and velocity. Due to the potential hazards, aircrews should be aware of their existence and attempt to avoid them.

16.1.2. Aircraft may alter position slightly to avoid vortices from preceding aircraft.

16.1.3. References to "SKE" in this publication apply to all forms of Station Keeping Equipment and Formation Flying System (SKE/FFS).

16.1.4. Buddy/Enroute Cell Departure Procedures are approved for C-17A crewmembers qualified in formation departures. Ensure a thorough pre-briefing is conducted with the tanker lead (e.g., speeds, vertical profile, rejoin coordination, etc.) prior to execution.

16.2. Specified Times. The mission commander determines the sequence of events and mission times based on Air Tasking Order, Fragmentary Order (FRAG), planning, staff input, fuel requirements, passenger/jumper comfort, taxi distances, briefing requirements, etc.

16.3. Weather Minimums.

16.3.1. Takeoff. Takeoff minimums are no lower than 200-foot ceiling and one-half mile visibility (RVR 2400). **(T-2)** If the departure ceiling or visibility is below published approach minimums, the formation may takeoff if the requirements for a departure alternate (IAW AFMAN 11-202V3_AMCSUP) are met. **(T-1)**

16.3.2. Landing. Landing minimums are the published minimums for the airport navigation aid used, but no lower than 200-foot ceiling and one half-mile visibility. **Note:** Formation NVG landing minimums are in accordance with **Chapter 6**, NVG Approach and Landing.

16.4. Ground Operations. The minimum taxi interval is one aircraft length. **(T-2)** Lead may increase the taxi interval if circumstances dictate.

16.5. Takeoff. For takeoff, aircraft taxi onto the runway individually. The takeoff EPR rating (i.e., DRT or MAX) will be briefed and all aircraft will take off with the same EPR rating. **(T-2)**

16.5.1. The minimum takeoff interval is 30 seconds. **(T-2)** It is possible at high power settings to cause a FOD hazard to following aircraft, so plan departure spacing accordingly. Begin timing for takeoff interval when the preceding aircraft starts its takeoff roll.

16.5.2. For aborts during takeoff, a briefed crewmember immediately transmits position number aborting three times on interplane frequency (e.g., "BASCO 70, number two aborting, number two aborting, number two aborting."). The PM transmits the same abort call on runway controlling frequency (after completing emergency procedures requiring immediate action). The aborting aircraft clears the runway as safety allows. Succeeding aircraft, if not

on takeoff roll, hold until the runway is clear, and a new takeoff clearance is obtained. Any aircraft on takeoff roll will abort their takeoff. **(T-1)**

16.6. Bank Angles, Airspeeds, and Rates of Ascent/Descent. Standard airspeeds and rates of ascent/descent are depicted in AFTTP 3-3.C-17. These standards will be followed unless otherwise briefed. **(T-2)** Lead should select MIN/20 degrees of bank when using SKE. Element leads should select NORM/20 for turn rate and bank angle on the AFCS panel when flying SKE.

16.7. Radio Discipline. Limit transmissions to those required for safety or control of the formation. **Note:** HAVE QUICK and secure communications should be used when available.

16.8. Airborne Aborts (Departure, Enroute and Element Lead Abort). Any aircraft that cannot maintain formation position will notify lead of the nature of the emergency and intentions. **(T-2)** If the emergency does not permit maintaining position until an individual clearance is obtained, establish a safe heading away from the formation and maintain visual or SKE contact. If unable to maintain visual or use SKE, the use of air to air (A/A) TACAN, TCAS, or radar skin paint may aid in maintaining separation from the formation. After departing the formation, the aborting aircraft will climb or descend out of the formation altitude prior to maneuvering across the flight path of the formation. **(T-2)**

16.8.1. Departure. Aircraft aborting during assembly will execute the briefed emergency procedures, hold clear of departing traffic, maintain VMC if possible, notify lead, and contact the appropriate controlling agency. **(T-2)** If possible, the aborting aircraft remains clear of the formation until a landing can be made without interfering with the remainder of the departing formation.

16.8.2. Enroute. Aircraft that abort after assembly turn away from the formation and, with mission commander's concurrence, may proceed to a suitable recovery airfield or rejoin at the end of the formation.

16.8.3. Element Lead. If an element lead aborts the formation, the second aircraft of that element moves to the element lead position. **(T-2)** If a formation is flying a three-ship element, aircraft within the element may move up to maintain a two-ship element. For SKE formations, the appropriate follower aircraft select the new lead, positively identify the new leader on the SKE MFD, reset cross-track and range as required. **(T-2)**

16.9. Visual Procedures. Visual formation geometry is driven by the tactical situation, which is determined by the threat environment, terrain, mission requirements and other factors. For a complete description of visual geometries reference AFTTP 3-3.C-17. Choose the geometry that gives the best tactical advantage. Different formation geometries may be required through the course of the mission. Flight leadership is critical to the success of these tactics.

16.9.1. Departure and Assembly. After crossing the field boundary, wingmen adhere to air traffic control requirements and close to enroute position. Lead maintains assembly airspeed, until briefed acceleration time/point.

16.9.2. Late Take Off. Aircraft joining a formation enroute contact lead and rejoin as briefed/directed. The rejoining aircraft will remain at least 500 feet above or below the formation until the formation is in sight and cleared to rejoin. **(T-2)**

16.9.3. Enroute Procedures:

16.9.3.1. Spacing. Plan a minimum of 2,000-foot spacing within each element and 12,000 feet between elements. **(T-2)**

16.9.3.2. Minimum Altitude.

16.9.3.2.1. Day VMC. Formation lead flies modified contour by using radar altimeter, HUD, and visual references. **(T-2)** Element leads/wingmen use a combination of the above and reference to other aircraft. **(T-2)**

16.9.3.2.2. Night VMC. Formation lead maintains altitude by reference to the barometric altimeter. **(T-2)** Element leads/wingmen use a combination of barometric altimeter and reference to other aircraft. **(T-2)**

16.9.3.3. Airspeed. Leads announce unplanned airspeed changes greater than 10 knots. **(T-2)**

16.9.3.4. Inadvertent weather penetration (IWP). Formation lead must take necessary actions to not operate a VFR formation into IMC conditions. **(T-2)** If clouds and/or areas of poor visibility are inadvertently entered by a formation operating under VFR, the primary concern of the formation lead is to provide safe aircraft separation and terrain clearance. Wingmen immediately notify lead of deteriorating visual conditions if they occur. **(T-2)** Terminate these procedures if the entire formation attains VMC, terrain clearance, and positive separation from other formation aircraft. **(T-2)**

16.9.3.4.1. It may be necessary to modify these procedures due to formation geometries other than visual in-trail, terrain, airspace restrictions, etc. Inadvertent weather penetration in mountainous terrain using these procedures may be hazardous. Mission planners should brief procedures that best suit the situation.

16.9.3.4.2. The following procedures are for emergency use and do not constitute authority to violate AFMAN 11-202V3 or Federal Aviation Regulations (FARs). Formation lead must take all practical measures to avoid entering controlled airspace without clearance. **(T-2)** Attempt to avoid leveling at IFR altitudes to minimize the possible conflict with IFR traffic.

16.9.3.4.3. IWP with SKE. Immediately upon penetrating the weather, formation lead announces, “XXXX flight, execute inadvertent weather penetration with SKE now, base altitude XXXX, base heading XXXX, base airspeed XXXX. Acknowledge.” **(T-2)** The formation then climbs to a base altitude at or above the ESA for the route. **(T-2)** Climb at cruise speed and 1,000 feet per minute. **(T-2)** After the element wingmen establish a 1,000 feet per minute climb, set AFCS mode select to SKE, set cross-track to 1,000 feet left or right as appropriate while maintaining current separation. **(T-2)** When level at the base altitude, lead announces “assume SKE interval now.” **(T-2)** At this command, followers select appropriate SKE in-trail interval in the along track (LTRK) and drift back to establish appropriate SKE in-trail interval. **(T-2)** Once established in appropriate SKE in-trail interval, select appropriate SKE cross track (XTRK). **(T-2)** If visual conditions cannot be promptly reestablished, lead contacts ATC for IFR clearance; declare an emergency if necessary. **(T-2) Note:** This procedure is only used if SKE was tested, all formation aircraft have SKE in Transmit (XMIT), SKE is operable, and no SKE cautions and warnings.

16.9.3.4.4. IWP without SKE. Immediately upon penetrating the weather, formation lead announces, "XXXX flight, execute inadvertent weather penetration without SKE now, base altitude XXXX, base heading XXXX, base airspeed XXXX. Acknowledge." **(T-2)** The formation climbs to a base altitude at or above the ESA for the route. **(T-2)** Climb at cruise speed and 1,000 feet per minute. **(T-2)** After the element wingmen establish a 1,000 feet per minute climb, #2 turns right using 30 degrees of bank to a heading 30 degrees from the base heading, #3 turns left using 30 degrees of bank to a heading 30 degrees from the base heading. **(T-2)** They maintain these divergent headings (once rolled out) for 30 seconds before resuming base heading. **(T-2)** The use of A/A TACAN, TCAS, or radar skin paint may aid in maintaining separation from other members of the formation. The last element in the formation occupies the base altitude. **(T-2)** All other elements stack 1000 feet higher than the following element, with the first element occupying the highest altitude. **(T-2)** Do not change base heading while in IMC unless required for terrain clearance or to avoid controlled airspace. **(T-2)** If visual conditions cannot be promptly reestablished, lead contacts ATC for IFR clearance; declare an emergency if necessary. **(T-2)**

16.9.4. Recovery. If aircraft weights differ significantly, the heaviest aircraft dictates the speeds flown. Aircraft will not descend below preceding aircraft during the recovery. **(T-2)**

16.9.5. Formation Landings. All aircraft land on runway centerline with the same flap detent setting. **(T-3) Exception:** Last aircraft in formation may use a higher flap setting. Continue to runway exit point without stopping in any position that would prevent succeeding aircraft from clearing the runway. **(T-2)** The minimum landing interval is 45 seconds (60 seconds desired). **(T-2)** Extend the interval as necessary for icy runway conditions, short/narrow runways, or other adverse conditions. **(T-2)**

16.9.5.1. Aircraft will not perform touch-and-go landings out of formation recoveries. **(T-3)**

16.9.5.2. Aircraft may perform GOATs for training provided it is pre-briefed, does not negatively affect the formation recovery, and all preceding formation aircraft are clear of the runway.

16.9.6. Lead Changes. To accomplish a visual lead change, the leader signals or commands the lead change (if it does not occur at a briefed point). **(T-3)** The aborting leader maneuvers in the safest direction to assume the new position.

16.9.7. SKE. Use during visual formation. Except for SKE out during visual training, SKE should be available to the max extent possible while flying visual procedures.

16.9.7.1. During visual procedures, aircrew will set "PROX WARN" to 1000 (1500 if in personnel spacing). **(T-2)**

16.9.7.2. Aircraft should select "wingman" and "AR MODE: Yes" on the SKE SETUP page.

16.10. SKE Procedures. Note: When in actual IMC, aircrews should be prepared to use an alternate method of identifying other aircraft in the formation (i.e., A/A TACAN, TCAS, and radar skin paint) in case of a SKE malfunction.

16.10.1. During SKE procedures, aircrew will set "PROX WARN" to 2000 (1500 if in personnel spacing) and AR MODE to "yes". **(T-2)**

16.10.2. Flight Command Indicator (FCI). Element leaders should transmit FCIs prior to turns. If FCIs are not received prior to turns, wingman SKE XTRK/LTRK positioning is not affected; FCIs are fundamentally used for situational awareness. Do not delay turns in order to send FCIs. FCIs may be sent after a lead has initiated a turn for additional situational awareness.

16.10.3. Departure and Assembly. After takeoff each aircraft flies an independent departure, (e.g., climb out instructions or SID), while maintaining formation integrity using the PPI until all aircraft attain formation position or the departure is accomplished, whichever occurs first.

16.10.4. Enroute Procedures. Wingmen maintain position with SKE selected for thrust and roll flight director guidance. Wingmen periodically crosscheck deviation indicators and the PPI SKE format. The primary altitude reference is the aircraft barometric altimeter.

16.10.4.1. Climb and Descent. Use the Vertical Deviation Indicator (VDI) to monitor the selected lead's altitude during climb or descent. **(T-3)** Wingmen report significant altitude deviations to lead. **(T-3)**

16.10.4.2. Airspeed Changes. Lead signals or announces all changes of 10 knots or greater from the established base airspeed. **(T-3)**

16.10.4.3. Domestic Reduced Vertical Separation Minimum (DRVSM). RVSM separation standards may be applied to formation flights consisting of all RVSM-compliant aircraft operating in DRVSM airspace. **Note**: DRVSM airspace includes the airspace of the lower 48 states of the United States, Alaska, Atlantic and Gulf of Mexico High Offshore Airspace, and the San Juan FIR between FL290-410 (inclusive).

16.10.4.3.1. RVSM formation flights may file for a single altitude if all formation aircraft fly the assigned altitude, either offset laterally from each other or in trail.

16.10.4.3.2. RVSM formation flights requiring multiple altitudes should request an altitude block. Air traffic control may then apply RVSM separation standards between this altitude block and other RVSM aircraft (e.g., an RVSM formation flight is assigned FL320-330; ATC assigns other RVSM aircraft at FL310 and FL340).

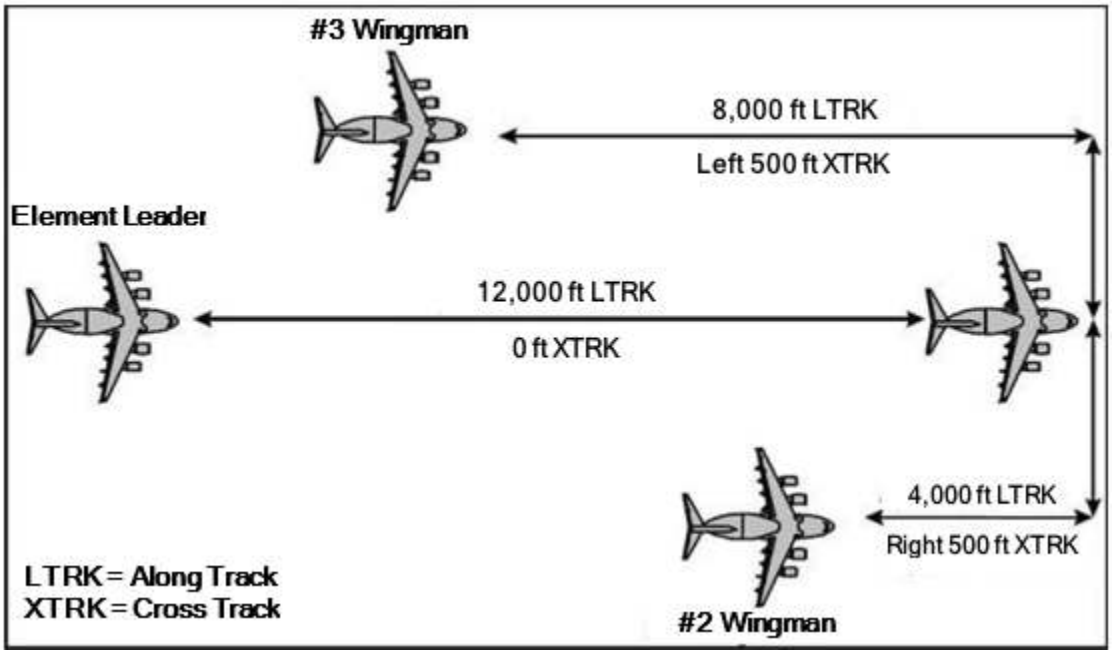
16.10.4.3.3. RVSM formation aircraft must use their automatic altitude control system to maintain the assigned altitude. **(T-2)** Aircraft maneuvering within an altitude block must ensure they do not exceed the vertical boundaries of the block by utilizing the aircraft altitude alerting system, altitude capture function (if installed) and automatic altitude control system. **(T-2)** If unable to maintain autopilot in vertical axis notify ATC of non-compliance with RVSM. **(T-2)**

16.10.4.3.4. Formation flights which do not consist of all RVSM aircraft will continue to be considered non-RVSM compliant and will have 2,000-foot vertical separation standards applied in DRVSM airspace. **(T-2)** In addition, aircraft formations conducting aerial refueling will continue to be considered non-RVSM compliant, regardless of the participating aircraft's single-ship status. **(T-2)**

16.10.5. Enroute Spacing.

16.10.5.1. Wingman. The second and third aircraft of each element maintain a minimum of 4,000 and 8,000-foot spacing, respectively, from their element lead. **(T-2)** Maintain spacing with reference to the element lead to reduce telescoping effects. **(T-2)** The minimum offset distance is 500 feet right for the #2 aircraft, and 500 feet left for the #3 aircraft. See example **Figure 16.1**. **(T-2)**

Figure 16.1. Standard SKE Enroute Formation.



16.10.5.2. Element lead. Each element lead maintains a minimum of 12,000-foot separation from the preceding element lead, stacks up 100 feet, and maintains "R0" cross-track separation. **(T-2)**

16.10.5.3. Long Missions. During long missions, mission commanders may extend enroute spacing and/or cross track to reduce fatigue, as required. **(T-3)**

16.10.5.4. Position errors. Formation aircraft detecting a significant position error immediately notify the offending aircraft to determine if the error is known. The offending aircraft immediately confirms or establishes position by other available means. **(T-2)**

16.10.6. Loss of SKE.

16.10.6.1. Single aircraft. Any aircraft that loses SKE notifies lead immediately. **(T-2)** If IMC, establish a safe heading away from the formation. **(T-2)** If last aircraft in the formation loses SKE, an airspeed reduction may be used to obtain immediate separation. Lead coordinates a separate IFR clearance for the malfunctioning aircraft as needed. **(T-2)**

16.10.6.2. Entire formation. If the entire formation loses SKE, attempt to correct the malfunction by changing the selected lead, provided an alternate means of maintaining formation position is being used (A/A TACAN, weather radar skin paint, radar beacon, TCAS). If the entire formation loses SKE and is unable to correct the malfunction or transition to a visual formation, execute inadvertent weather penetration without SKE procedures. **(T-2)**

16.10.7. Lead Change. The aborting lead commands the lead change and the new lead acknowledges receipt of this command. **(T-2)** If briefed, the change may be accomplished silently at a planned point or time. Do not accomplish lead changes in turns or descents. **(T-2)**

16.10.7.1. In IMC, the aborting lead turns 30 degrees away from base heading in the safest and most logical direction until at least 1 NM from the formation. Reset appropriate crosstrack, range, and lead number. Drift back to rejoin at the end of the formation. If VMC, the aborting lead may join at a coordinated position within the formation. **(T-2)**

16.10.7.2. Wingmen select the new lead and reset crosstrack and range as required. **(T-2)**

16.10.7.3. New lead completes an FCI check. **(T-2)**

16.10.8. Recoveries. If recovering a large SKE formation and the planned approach is not a straight-in, obtain a minimum of five minutes separation between sections of six or less aircraft prior to reaching the recovery base. **(T-2)** Ensure the airspace is available for each section of six or less to hold upon arrival at the recovery base if immediate landing is not possible. **(T-2)** If aircraft weights differ significantly, the heaviest aircraft dictates the speeds flown. **(T-2)** The formation may proceed visually to the field for a visual recovery, maintain SKE spacing and assume SKE in-line formation procedures, fly ATC locally approved SKE approach, or break up for individual approaches.

16.10.8.1. Instrument approaches.

16.10.8.1.1. Do not fly 45/180 procedure ground tracks. **(T-2)**

16.10.8.1.2. Holding Pattern, Holding Pattern in Lieu of Procedure Turn, or Procedure Turn pattern entry will be within 70 degrees of the published inbound course on the non-maneuvering side or within 20 degrees on the maneuvering side and a minimum of 1,000 feet above procedure turn or Ground Controller Approach (GCA) pickup altitude. **(T-1)** Do not enter from the quadrant requiring a turn to the non-maneuvering side. Request 2-minute holding legs with more than 3 aircraft in formation. **Caution:** In a non-radar environment or uncontrolled airspace, lead should consider increasing the formation's minimum altitude to ensure terrain/obstacle avoidance.

16.10.8.1.3. Lead signals the turn outbound over the IAF with the FCI at station passage. Wingmen delay the turn outbound based on SKE timing, maintain formation interval, and complete the approach in accordance with AFMAN 11-202V3. Formation lead ensures all aircraft stay in the "remain within" distance for the approach or coordinate with ATC for additional airspace. **(T-2)**

16.10.9. Landing. Minimum landing interval is 10,000 feet (12,000 feet desired). **(T-2)**

16.10.10. Missed Approach. Aircraft executing a missed approach fly the published or directed procedure. **(T-2)** If weather at the airfield is reported below minimums after the approach is started, the formation executes a missed approach, maintaining 160 KCAS (or minimum flap retract, whichever is higher) and approach separation. **(T-2)** Formation lead requests individual approaches, if possible. Coordinate for holding if required. In a radar environment, give the controlling agency the order in which aircraft are to depart the flight. **(T-2)** If individual approaches cannot be obtained or approach control is not available, the formation proceeds to an alternate airfield. **(T-2)**

16.11. Formation Air Refueling Procedures. The procedures contained in this section cover only the most common receiver/tanker formations and do not cover all possible situations. The procedures contained in this section do not relieve the mission commander and section lead of the responsibility to thoroughly plan and brief these procedures and cover all possible combinations. All AAR operations will be accomplished IAW T.O. 1C-17A-1, and ATP-3.3.4.2. **(T-2) Note:** The lead tanker is responsible for navigation of the entire formation (both tanker and receiver aircraft) from rendezvous through the end of air refueling operations. **(T-2)**

16.11.1. Briefing. The lead receiver PIC briefs all PICs within the receiver cell. This briefing will be in sufficient detail to cover all phases of cell operations. **(T-2)**

16.11.2. Aircraft will have operable SKE for formation air refueling flights. **(T-3)** If weather (WX) radar is inoperative, TCAS must be operational. **(T-3)**

16.11.3. During refueling, receivers address themselves as "Receiver 1" (R1), "Receiver 2" (R2), and "Receiver 3" (R3). **(T-2)** Address the individual tankers as "Tanker 1" (T1), "Tanker 2" (T2), etc. **(T-2)**

16.11.4. Rendezvous Procedures: R1 keeps heading and airspeed changes to a minimum during transitions to or from AR Echelon. **(T-2)** When 3 receivers are maneuvering to AR echelon, R1 has the option prior to the AR Echelon call to direct R3 to assume AR position. R3 selects R2 as their LEAD and change LTRK/XTRK to 4000/R500 or in-line as required for turns. R3 reports when established in AR position. **(T-2)** If R3 has not reported in "AR position" prior to the AR Echelon call, R2 maneuvers based on established procedures and does not select 3000/R5000 until approaching echelon position. **(T-2)** This option may be briefed to occur at a specific time and/or geographical location.

16.11.5. Emergency Actions: Upon losing all contact with the tanker lead or the respective tanker, or if unable to maintain formation due to disorientation, the wingman simultaneously executes the applicable lost wingman procedure while transitioning to instruments. **(T-2)**

16.11.5.1. Loss of VMC after rendezvous. In the event any aircraft momentarily and inadvertently enters into IMC, loses sight of other receivers or their respective tanker in the formation, the affected receiver informs T1 immediately. **(T-2)** If continuous radar skin paint off of the respective tanker and SKE position off of all other receivers can be maintained, all aircraft maintain their current position (awaiting AR, astern, post AR, etc.) until the formation reenters VMC. **(T-2)** If visual conditions do not return sufficiently to safely complete the formation AR procedures, T1, in coordination with the R1, takes action to ensure both altitude and lateral separation from all receivers and tankers. **(T-2)** Subsequent receiver rendezvous may be coordinated with ATC after obtaining proper separation from the tanker formation (i.e., minimum of 2 NM and 1000 feet between the lowest tanker and highest receiver).

16.11.5.2. Loss of SKE or WX radar in IMC. The loss of SKE or radar in IMC conditions after beginning the transition to AR echelon until completion of formation AAR procedures requires immediate action by R1 and T1. **(T-2)** R1, with the approval of T1, directs the appropriate action for the affected receiver to ensure both lateral and vertical separation from all other receivers and tankers. **(T-2)**

16.11.5.3. All other receiver aircraft with station keeping ability maintain their current position (awaiting AR, post AR, etc.) until reentering VMC and are subsequently cleared by R1 and T1. **(T-2)**

16.11.5.4. Lost Wingman Procedures During Receiver AR. Immediately contact T1 and establish visual, A/A TACAN, radar, SKE, or radio contact with any co-altitude aircraft. **(T-2)** If visual, A/A TACAN, radar, SKE, or radio contact cannot be established or maintained, descend to an altitude that provides positive separation from other aircraft and decrease airspeed to ensure separation. **(T-2)**

Chapter 17

AIRDROP

17.1. General. This chapter prescribes C-17 employment procedures for all airdrop operations. For additional guidance and information, refer to T.O. 1C-17A-1-4 and AFTTP 3-3.C-17. NAS training operations are required to comply with Title 14, Code of Federal Regulations, Part 105, *Parachute Operations*, current edition, restrictions. **(T-0)**

17.2. Radio Discipline. Unless used in conjunction with airdrop execution, avoid use of the words "GREEN" or "NO DROP" after the Slowdown Checklist and until completion of the Post Drop checklist. "GREEN LIGHT" can be seen or heard by the loadmaster for all drops.

17.3. NVG Airdrop. Airdrops may be accomplished on drop zones lit IAW DAFMAN 13-217, DZ marking patterns (covert and overt) while wearing NVGs.

17.3.1. Pilots are authorized to perform all airdrop methods while utilizing NVGs.

17.3.2. Loadmasters are authorized to perform all equipment type airdrops while utilizing NVGs. Loadmasters will not conduct personnel airdrop operations while utilizing NVGs. **(T-2)**

17.3.3. Red/NVIS (overt) or IR (covert) cargo compartment lighting is set to the lowest possible setting to accomplish the mission. Blacked out (no-light) operations in the cargo compartment are not authorized.

17.4. Unmarked DZ. Non-SOLL II crews may drop to an unmarked DZ in peacetime/training operations with actual cargo/personnel provided all of the following are met:

17.4.1. Positive communication method with the DZ Controller/Drop Zone Safety Officer (DZSO) and drop clearance relayed. **(T-2)**

17.4.2. No "UNABLE RNP" messages present. **(T-2)**

17.4.3. No "CGPS INOP or MGPS INOP" messages present. **(T-2)**

17.4.4. A "NO DROP" signal is established and coordinated with the ground party in the event of lost communications. **(T-2)**

17.5. Airdrop Equipment.

17.5.1. Airdrop Rigging Material. The loadmaster will obtain a sufficient amount of rigging material to satisfy load or mission requirements. **(T-1)**

17.5.2. Pole Knives. Loadmasters will ensure pole knives are carried on all heavy equipment airdrop sorties. **(T-1)**

17.6. Safety Equipment.

17.6.1. Personnel performing duties required to be mobile in the cargo compartment during airdrop, low-level operations, or threat environments will wear protective headgear (except personnel performing water jumps). **(T-1)** All other personnel in the cargo compartment will be seated with seat belts fastened. **(T-1)** Loadmasters will lower their helmet visor before opening the troop doors and keep them lowered until the troop doors are closed. **(T-1)** **Exception:** Helmet visor is not required while wearing NVGs.

17.6.2. All occupants of the cargo compartment will either wear a parachute or an attached restraint harness, or be seated with a seat belt fastened before a troop door and/or the cargo door and ramp is opened. **(T-1)** When a troop door(s) or the cargo door/ramp is open and the aircraft is below 800 feet AGL or above 25,000 feet MSL, occupant(s) will wear a restraint harness. **(T-1)** For operations over water, when parachute(s) are required, life preserver(s) will be worn. **(T-1) Exception:** For static line jumps, static lines are attached to anchor cables before troop door(s) are opened. Jumpers exiting on subsequent passes may stand and hook up with door(s) opened if they are forward of fuselage station (FS) 1027.

17.6.2.1. The restraint harness is adjusted to allow mobility to perform duties but not to a length that would allow the wearer to fall out of the aircraft. The restraint harness will be checked to ensure no frays or knots are present and fitted/adjusted prior to flight. **(T-2)** The lifeline is 18 feet 6 inches long. With the cargo ramp and door open the restraint harness may be connected to any tie down ring at or forward of FS 1188. With the paratroop doors open the restraint harness may be connected to any tie down ring. **Exception:** Loadmasters requiring mobility in the cargo compartment while using a restraint harness can be unattached as long as they remain forward of FS 1188 with the cargo door and ramp open or FS 1027 with the paratroop door(s) open. **Warning:** Except for a towed trooper, or emergency that threatens the survivability of the aircraft and crew, the restraint harness will not be disconnected or lengthened to a point that would allow the loadmaster to fall outside the aircraft. **(T-2)**

17.6.2.2. Aircrew Flight Equipment on Personnel Airdrops. EPOS and Life Preservers not properly secured in storage pouch will be stowed immediately prior to the airdrop. **(T-2)** Life preservers may be stowed earlier on non-overwater flights.

17.7. Airdrop Load Information.

17.7.1. The appropriate DD Form 1748 will be completed for all airdrop loads IAW AFMAN 13-210V2, *Joint Airdrop Inspection Records, Malfunction or Incident Investigations, and Activity Reporting Procedures*. **(T-2)**

17.7.2. Planners and aircrew will verify chute types planned with the user during mission planning. **(T-2)** Aircrews will be made aware of chute types on planning documents. **(T-2)** At a minimum, chute type and planned drop altitude will be annotated on the Mission Data Card. **(T-2)**

17.7.3. Joint Airdrop Inspector (JAI) will annotate actual chute type in the remarks section of the DD Form 1748. **(T-2)** For all heavy equipment platforms, the loadmaster will not arm the ADS locks without the JAI present to verify correct lock settings and arming sequence. **(T-2)**

17.7.4. During execution, aircrews will verify actual chute type loaded on the aircraft by referencing the remarks section on the DD Form 1748. **(T-2)** Crews will verify the correct drop altitude, chute type and number are entered in the MC. **(T-2)**

17.7.5. Load verification and marking. A pilot will verify with the loadmaster that the actual number and type of parachutes, load weights, sequence of extraction, and position of loads in the aircraft agree with entered mission computer data. **(T-2)** For training missions (e.g., unilateral, exercise, or JA/ATT), the PIC will ensure all equipment is marked with the aircraft call sign and date. **(T-2)** Markings are placed on the CDS bundle, platform, sling extraction

line bag, and drogue line. **(T-2)** If more than one CDS bundle is dropped on the same pass, mark only the first bundle out. **(T-2)**

17.7.6. If airdrop loads and airland cargo are carried at the same time, refer to the restrictions listed in **Table 17.1**. **(T-2)** These restrictions are designed to prevent airland loads from interfering with airdrop rigging equipment.

Table 17.1. Airdrop Configuration Restrictions.

Airdrop Configuration Restrictions	
ANCHOR CABLE HEIGHT FROM AIRCRAFT FLOOR	81 INCHES AT MIDSPAN WHEN INSTALLED
DISTANCE BETWEEN ANCHOR CABLES (a) CDS OR DUAL ROW (b) PERSONNEL (1) FORWARD BULKHEAD INTERMEDIATE SUPPORTS	182 INCHES 122 INCHES INBOARD, 182 INCHES OUTBOARD 155 INCHES INBOARD, 182 INCHES OUTBOARD
AIRLAND CARGO HEIGHT	CANNOT INTERFERE WITH INSTALLED ANCHOR CABLES, OR AIRDROP RIGGING EQUIPMENT
AIRLAND CARGO WIDTH ON HEAVY EQUIPMENT/DUAL ROW/CDS AIRDROPS	CANNOT OBSTRUCT VIEW OR HINDER ACCESS TO CARGO IN CASE OF EMERGENCY
CDS/DUAL ROW AIRDROPS (WHEN DROPPING ONE SIDE ONLY) (a) MAXIMUM WIDTH OF AIRLAND CARGO (b) POSITION OF AIRLAND CARGO	 110 INCHES AFT END OF AIRLAND CARGO FORWARD OF FS 1280
AIRLAND CARGO WIDTH ON PERSONNEL AIRDROPS	144 INCHES OR LESS WITH BOTH TROOP DOORS CONFIGURED MAY BE INCREASED TO 178 INCHES OR LESS IF ONLY ONE TROOP DOOR IS CONFIGURED

PERSONNEL DISTANCE FROM AIRDROP RIGGING EQUIPMENT (2)	30 INCHES MINIMUM

Notes:
1. Personnel airdrops may be performed with only one troop door configured for airdrop.
2. Floor-loaded rucksacks will be secured.

17.8. Required Navigation Performance (RNP).

17.8.1. IMC Airdrop. Lead may descend from minimum IFR Enroute altitudes to IMC drop altitude as long as UNABLE RNP messages are not displayed. IMC airdrop requires no UNABLE RNP message when using MC guidance. For SKE airdrops, wingmen may descend to IMC drop altitude and drop off the SKE timer regardless of MC ANP status, and drop off lead's SKE timer.

17.8.2. JPADS/Improved Container Delivery System (ICDS). Aircraft cannot drop with an UNABLE RNP message. **(T-2)**

17.9. Notice To Airmen (NOTAM) Requirements. For IFR airdrop in uncontrolled airspace, a Letter of Agreement between local ATC and the military is required. Also, provide a NOTAM to the FAA Flight Service Station nearest the objective area at least 6 hours in advance of the intended activity, regardless of actual or forecast weather. NOTAM information includes:

17.9.1. The name of the city or town nearest the route segment and the state. **(T-2)**

17.9.2. The date and time period of planned activity. **(T-2)**

17.9.3. The number and type of aircraft expected on the route. **(T-2)**

17.9.4. The ingress and egress points of the route segment expressed in fix/radial/distance from a VOR. **(T-2)**

17.9.5. The altitude at which the aircraft is flown. **(T-2)**

17.10. Minimum Drop Altitudes.

17.10.1. Day VMC Drop Altitude. Fly minimum airdrop altitude as specified in AFMAN 11-231, *Computed Air Release Point Procedures,* visually avoiding high terrain and obstacles in the vicinity of the drop zone. During training, flight below 500 feet AGL is only authorized on approved routes in accordance with **paragraph 14.4.7**.

17.10.2. Night Drop Altitude. If not on NVGs, fly no lower than an indicated altitude of 500 feet above the highest obstruction to flight (man-made obstacle, terrain feature, spot elevation, or DTED spot elevation) within 3 NM of run-in centerline from slowdown through escape, or as specified in AFMAN 11-231, whichever is higher. **(T-2) Note:** If DTED is not available, Night Drop Altitude will be calculated using 400 feet plus one chart contour interval above the highest depicted terrain contour, if higher. **(T-2)** After slowdown, when the drop zone is in sight and remains in sight, or when a positive position is identified and adequate terrain and obstacle clearance is assured, the aircraft may descend to the AFMAN 11-231 specified minimum drop altitude. **(T-1)**

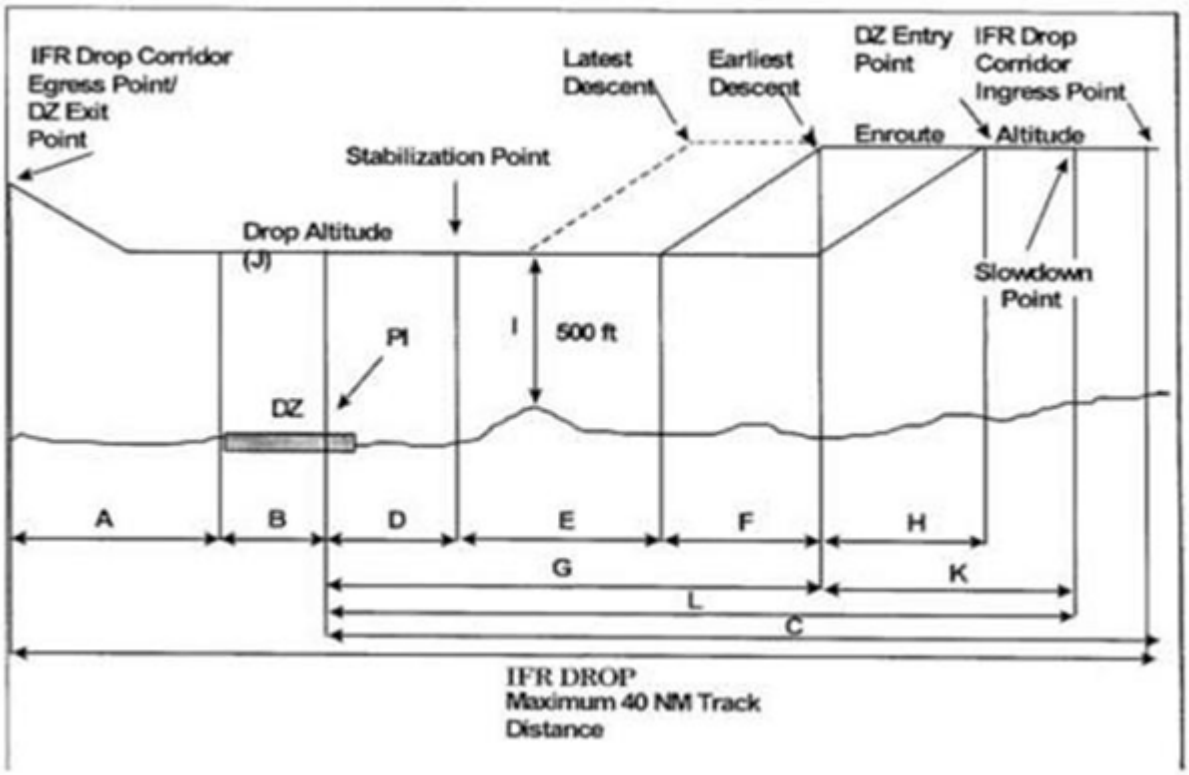
17.10.3. NVG Drop Altitude. Fly NVG Enroute Altitude through slowdown. Maintain the tactical corridor. After slowdown, when the DZ is in sight and remains in sight, or when a positive position is identified and adequate terrain and obstacle clearance is assured, the aircraft may descend to the AFMAN 11-231 specified minimum drop altitude. **(T-1)** NLT 3 NM prior to any charted man-made obstacle within the run-in corridor, the aircrew must visually identify the obstacle. **(T-1)** If the obstacle is not identified by 3 NM, climb to 500 feet above the obstacle until the aircrew confirms the aircraft is past the obstacle. **(T-1)**

17.10.4. IFR drop altitude is 500 feet above the highest obstruction to flight within 3 NM of run-in centerline from DZ entry point to DZ exit point. **(T-1)** If DTED is not available, IFR drop altitude will be calculated using 400 feet plus one chart contour interval above the highest depicted terrain contour, if higher. **(T-1) Warning:** Drop zone surveys do not assure terrain and obstruction clearance. The responsibility is incumbent upon planners and aircrew through thorough mission planning and chart updating. **Note:** AFMAN 11-231 minimum chute altitudes may require the aircraft to drop at a higher altitude than those listed above.

17.10.4.1. The provisions of FAA Exemption 4371 are not required to operate at IFR drop altitude in Special Use Airspace that supports operations at that altitude (e.g., an MTR/Minimum Crossing Altitude (MCA) /Restricted Area with a floor of 300 feet AGL).

17.11. IFR Drop Profile.

Figure 17.1. IFR Airdrop Profile.



17.11.1. IFR Drop Corridor (See **Figure 17.1** and **Table 17.2**.). The IFR Drop Corridor is the corridor where aircraft may operate below IFR Enroute altitude. The beginning of the corridor, the IFR Drop Corridor Ingress Point, is a maximum of 40 NM from the IFR Drop Corridor

Egress Point (co-located with the DZ Exit Point). Plan segmented corridor altitudes not lower than 500 feet above the highest obstruction to flight (man-made obstacle, terrain feature, or charted or DTED spot elevation), within 3 NMs of centerline. **Note:** If DTED is not available, IFR Drop Corridor altitudes will be calculated using 400 feet plus one contour interval above the highest depicted terrain contour, if higher. **(T-1)**

17.11.1.1. Drops conducted through or originating from IMC are only authorized from within or above an active restricted area or military operations in uncontrolled airspace.

17.11.1.2. In addition to required RNP, do not initiate descent from the minimum IFR Enroute altitude to IFR drop altitude unless all aircraft in the element are inside the DZ entry point, on course, with element lead's position positively known. **Note:** The 40 NM IFR Drop Corridor is an agreement with the FAA in designated FAA airspace. When conducting IFR airdrops outside of FAA airspace, reference these IFR drop corridor procedures for planning purposes, however, the 40 NM restriction does not apply.

17.11.2. IFR DZ Entry Point. A fixed point in the IFR Drop Corridor where an aircraft or formation may safely begin descent from IFR Enroute altitude or a segmented altitude to IFR drop altitude. Formation descent will not begin until the last aircraft is at or past the DZ entry point.

17.11.3. Earliest Descent Point (EDP). Earliest point in the IFR Drop Corridor where the formation lead may descend the entire formation to IFR drop altitude and be assured of terrain clearance for the entire formation. Computed by subtracting formation length (e.g., a 4-ship is 2 NMs long) from the computed DZ entry point. A minimum of 6 NM stabilization point is recommended in IMC.

17.11.4. IMC Stabilization Point. The point after the DZ entry point where the lead aircraft will plan to be stabilized at IFR drop altitude and airspeed (normally 6 NM from the PI).

17.11.5. Latest Descent Point (LDP). Latest possible point in the IFR Drop Corridor where formation lead may begin descent to IFR drop altitude and be assured of terrain clearance for the entire formation. This is the latest point that ensures all aircraft in the formation are stabilized on altitude and airspeed.

17.11.6. IFR DZ Exit Point. A fixed point on the DZ escape flight path centerline where each aircraft will be at minimum IFR Enroute altitude. Calculate the exit point based upon three-engine performance at airdrop gross weight. This point will be a minimum of 4 NMs track distance from the trailing edge of the DZ. Also referred to as the IFR Drop Corridor Egress point.

Table 17.2. IMC Drop Profile Calculation.

Step	Result
A DZ Exit Point. Compute this distance as 1 minute at 160 KCAS + climb at 1000 feet per minute (FPM) or as briefed (no greater than 3 engine climb performance). Climb at least 500 FPM or as required by terrain/obstructions.	
B DZ Length (Total DZ Length-Leading edge to PI Distance/2025 yd/NM) Example: (1688-550)/2025 = .6 NM	

C IFR Drop Corridor Entry Point. Computed by subtracting distances A and B above from 40 NM. Example: (40 NM - A - B) 40 - 4.1 - .6 = 35.3 NM	
D IMC Stabilization Point. Recommended at least 6 NM from the PI, the mission commander may extend this distance. Example: 6 NM	
E Slowdown from 160 KCAS to Drop Speed. Example 160 to 145 KCAS = .7 NM (See slowdown distance table)	
F Descent from IFR Enroute to IFR drop altitude. Example Descent from 2000 feet to 1000 feet MSL = 2.7 NM @ 160 ground speed and 1000 FPM (see attached distance)	
G Formation Length (3-ship elements, no ghosts) 2-ship .7 NM, 3-ship 1.3 NM, 4-ship 2.0 NM, 5-ship 2.6 NM, 6-ship 3.3 NM. Example: 3-ship 1.3 NM	
H DZ Entry Point. To extract this distance, compute the slowdown point for the last aircraft and subtract the initial deceleration distance, or add D + E + F + G above. Example (D + E + F + G) 6 + .7 +2.7 +1.3= 10.7 NM. Verify the result is not greater than the IFR Drop Corridor Entry Point.	
I Minimum IFR Drop Altitude. Example: 716 + 500 = 1216 feet	
J Planned Drop Altitude. Highest point on DZ plus AGL drop altitude. Will not be less than IFR Drop Altitude. Example: 289 + 800 = 1089 feet (use higher of Minimum IFR Drop Altitude and Min Chute Altitude)	
K Initial Slowdown Distance. Distance needed to slow from enroute airspeed to 160 KCAS* (see **Table 17.3**)	
L Slowdown Distance (See Table 17.3). Total distance from initial slowdown to PI. Item K + Item F + Item E + Item D Example (from 240 KCAS to drop airspeed, 2000 feet to 1216') 4.7 + 2.7 +.7 + 6	

Table 17.3. Slowdown Distance (part 1 of 3).

ENDING GROUNDSPEED									
		130	140	150	160	170	180	190	200	210
STARTING GROUNDSPEED	370	17.5	17.1	16.7	16.2	15.7	15.2	14.7	14.1	13.5
	360	16.4	16.0	15.6	15.2	14.7	14.2	13.6	13.1	12.5
	350	15.4	15.0	14.6	14.1	13.6	13.1	12.6	12.0	11.4
	340	14.4	14.0	13.6	13.1	12.6	12.1	11.6	11.0	10.4
	330	13.4	13.0	12.6	12.1	11.7	11.2	10.6	10.0	9.4
	320	12.5	12.1	11.6	11.2	10.7	10.2	9.7	9.1	8.5
	310	11.5	11.2	10.7	10.3	9.8	9.3	8.7	8.2	7.6
	300	10.7	10.3	9.8	9.4	8.9	8.4	7.9	7.3	6.7
	290	9.8	9.4	9.0	8.5	8.0	7.5	7.0	6.4	5.8
	280	9.0	8.6	8.1	7.7	7.2	6.7	6.2	5.6	5.0
	270	8.2	7.8	7.3	6.9	6.4	5.9	5.4	4.8	4.2
	260	7.4	7.0	6.6	6.1	5.6	5.1	4.6	4.0	3.4
	250	6.6	6.3	5.8	5.4	4.9	4.4	3.8	3.3	2.7
	240	5.9	5.5	5.1	4.7	4.2	3.7	3.1	2.6	2.0
	230	5.2	4.9	4.4	4.0	3.5	3.0	2.4	1.9	1.3
	220	4.6	4.2	3.8	3.3	2.8	2.3	1.8	1.2	0.6
	210	4.0	3.6	3.1	2.7	2.2	1.7	1.2	0.6	0.0
	200	3.4	3.0	2.6	2.1	1.6	1.1	0.6	0.0	

TABLE A (SLOWDOWN DISTANCES)

Table 17.4. Slowdown Distance (part 2 of 3).

GROUNDSPEED

DISTANCE	1000 FPM	200	190	180	170	160	150	140	130	120
	1000	3.3	3.2	3.0	2.8	2.7	2.5	2.3	2.2	2.0
	2000	6.7	6.3	6.0	5.7	5.3	5.0	4.7	4.3	4.0
	3000	10.0	9.5	9.0	8.5	8.0	7.5	7.0	6.5	6.0
	4000	13.3	12.7	12.0	11.3	10.7	10.0	9.3	8.7	8.0
	5000	16.7	15.8	15.0	14.2	13.3	12.5	11.7	10.8	10.0
	6000	20.0	19.0	18.0	17.0	16.0	15.0	14.0	13.0	12.0

DISTANCE	1500 FPM	200	190	180	170	160	150	140	130	120
	1000	2.2	2.1	2.0	1.9	1.8	1.7	1.6	1.4	1.3
	2000	4.4	4.2	4.0	3.8	3.6	3.3	3.1	2.9	2.7
	3000	6.7	6.3	6.0	5.7	5.3	5.0	4.7	4.3	4.0
	4000	8.9	8.4	8.0	7.6	7.1	6.7	6.2	5.8	5.3
	5000	11.1	10.6	10.0	9.4	8.9	8.3	7.8	7.2	6.7
	6000	13.3	12.7	12.0	11.3	10.7	10.0	9.3	8.7	8.0

DISTANCE	2000 FPM	200	190	180	170	160	150	140	130	120
	1000	1.7	1.6	1.5	1.4	1.3	1.3	1.2	1.1	1.0
	2000	3.3	3.2	3.0	2.8	2.7	2.5	2.3	2.2	2.0
	3000	5.0	4.8	4.5	4.3	4.0	3.8	3.5	3.3	3.0
	4000	6.7	6.3	6.0	5.7	5.3	5.0	4.7	4.3	4.0
	5000	8.3	7.9	7.5	7.1	6.7	6.3	5.8	5.4	5.0
	6000	10.0	9.5	9.0	8.5	8.0	7.5	7.0	6.5	6.0

TABLE B (SLOWDOWN DISTANCES)

Table 17.5. Slowdown Distance (part 3 of 3).

			ENDING GROUNDSPEED									
			100	110	120	130	140	145	150	160	170	180
	G	190	3.8	3.5	3.2	2.8	2.4	2.2	2.0	1.5	1.0	0.5
S	R	180	3.3	3.0	2.6	2.3	1.9	1.7	1.4	1.0	0.5	0.0
T	O	170	2.8	2.4	2.1	1.7	1.4	1.1	0.9	0.5	0.0	
A	U	160	2.3	2.0	1.6	1.3	0.9	0.7	0.5	0.0		
R	N	150	1.8	1.5	1.2	0.8	0.4	0.2	0.0			
T	D	140	1.4	1.1	0.8	0.4	0.0					
I	S	130	1.0	0.7	0.4	0.0						
N	P	120	0.6	0.3	0.0							
G	E	110	0.3	0.0								
	E	100	0.0									
	D		TABLE C (SLOWDOWN DISTANCES)									

17.12. VFR Drop Profile.

17.12.1. VFR DZ Entry Point. A fixed point in the Drop Corridor where the aircraft or formation transitions from VFR enroute procedures to VFR airdrop procedures (normally the planned slowdown point).

17.12.2. VFR DZ Exit Point. A fixed point on the DZ escape flight path centerline where each aircraft will be at minimum VFR enroute altitude. Calculate the exit point based upon three-engine performance at airdrop gross weight. **(T-2)**

17.13. Dual Row.

17.13.1. Reference dual row drop altitudes in accordance with Cargo Parachute Ballistic Data for C-17 Aircraft.

17.13.2. Formation Dual Row. Formation airdrop of dual row platforms is authorized from any position in the element. Increase minimum drop zone width by 400 yds (200 yds each side) in addition to DAFMAN 13-217 minimum drop zone calculations. **(T-2)**

17.14. High Altitude Airdrop Operations.

17.14.1. For communications and signals, interphone and hand signals are the primary methods of communications. Written messages may be necessary in some instances to communicate with individuals not connected to the aircraft interphone. Loadmasters will carry a suitable writing utensil and medium to write out messages that cannot be dealt with by using hand signals. **(T-3)** When dropping parachutists, the jumpmaster may monitor interphone. The loadmaster will coordinate all hand signals with the jumpmaster. **(T-3)**

17.14.2. Maintain interphone contact between the cockpit and the cargo compartment. The loadmaster must be on interphone from completion of applicable Pre-Slowdown Checklist until the completion of Post Drop Checklist and the cabin altitude is below 10,000 feet MSL. **(T-2)** The jumpmaster may also monitor interphone during high altitude personnel airdrops.

17.15. High Altitude Airdrop Oxygen Requirements.

17.15.1. See AFMAN 11-409, *High Altitude Airdrop Mission Support Capability Program*, for high altitude airdrop operations oxygen requirements (pre-breathe requirements and exposure limits, restrictions, Physiological Technician requirements, etc.).

17.15.2. For the purposes of high-altitude oxygen requirements, supplemental oxygen with a fitted helmet and mask will be used when cabin altitude exceeds 10,000 feet MSL. **(T-1)**

17.16. High Altitude Personnel Airdrop Procedures.

17.16.1. Air deflectors must be operational if paratroop doors are used. **(T-2)** If an air deflector does not extend, do not open the affected troop door. **(T-2)**

17.16.2. The cargo ramp and door or paratroop door may remain open during racetracks if required, provided racetrack altitude is at or above a safe drop altitude and paratroopers are rigged for high altitude airdrops.

17.16.3. For High Altitude Low Opening (HALO)/High Altitude High Opening (HAHO) drops, the allowable methods determining the release point are Launch Acceptability Region (LAR) using a MAJCOM-approved source (e.g., Consolidated Airdrop Tool), AFMAN 11-231, Jumpmaster directed, or user defined (jointly agreed upon release point/region).

17.16.3.1. For LAR drop, the green light will be turned on when entering the LAR, and the red light will be turned on when exiting the LAR. **(T-2)** The pilot will provide a standard "Green Light" call at the jointly agreed upon release point. User assumes responsibility for drop accuracy once a release point/LAR has been jointly agreed upon. **(T-2)**

17.16.3.2. For High Altitude Release Point (HARP) Jumpmaster directed HAHO/HALO drops, the green light may be turned on one minute prior to the release point. The pilot will provide a standard "Green Light" call at the jointly agreed upon release point. **(T-2)** User assumes responsibility for drop accuracy.

17.16.3.3. SOLL II crews are authorized to drop on a special operations user specified release point. User assumes responsibility for drop accuracy.

17.16.4. Normally, the jumpers will exit the aircraft at their own discretion; however, their exit must occur during the green light time. **(T-2)**

17.16.5. Ensure any paratroopers remaining on-board de-arm their parachutes before cabin altitude descends below set parachute activation altitude. **(T-2)**

17.17. Personnel Airdrop.

17.17.1. Drop parameters. Aircraft will no-drop if not within the following tolerances at green light. **(T-2)**

17.17.1.1. Element leads must maintain a minimum of 32,000 feet to preceding element lead. **(T-2)**

17.17.1.2. Wingman Spacing Within Elements (For up to 12 Degrees of Drift). The second aircraft in each element will be positioned 3,000 feet aft and 650 feet right for a right echelon, 650 feet left for a left echelon. **(T-2)** The third aircraft in each element will be positioned 6,000 feet aft and 1,500 feet right for a right echelon, 1,500 feet left for a left echelon. **(T-2)** For drifts greater than 12 degrees, use appropriate spacing in **Table 17.6**. **(T-2)** Lead signals the direction of echelon and expected drift prior to the pre-IP. **(T-2)** Wingmen reset along track LTRK on lead's command and automatically echelon in the descent to drop altitude (or on lead's command in case of level slowdown or pop-up). **(T-2)** Compression to 3,000 or 6,000-foot spacing should be initiated not later than the slowdown point. All elements fly the same drop altitude. **(T-2)**

Table 17.6. Personnel Formation Drift.

PERSONNEL FORMATION DRIFT		
Drift	#2 Wing	#3 Wing
L/R	L/R	L/R
1 to 12	650	1500
13	700	1500
14	750	1500
15	800	1600
16	850	1750
17	950	1850
18	1000	1950
19	1050	2050
20	1100	2200

17.17.1.3. Wingmen restrictions are ±500 feet fore/aft, ±200 feet left/right of position. **Warning:** If the second aircraft is not within these tolerances, a no drop will be called for both Number Two and Number Three aircraft by Number Two. **(T-2) Note:** To prevent course correction at mission computer red light, consider using split axis or heading hold.

17.18. Slowdown Planning. For equipment and CDS drops, the aircraft should be at drop altitude and drop airspeed by 10 seconds prior to the computed air release point. During personnel drops, the aircraft should be at or above drop altitude and at airspeed not later than 1 minute to go (2 minutes for jumpmaster-directed drops) to allow the jumpmasters access to the paratroop doors.

17.19. Navigation to the Computed Air Release Point (CARP). The primary method for navigating to the CARP is using a MC-calculated release solution updated with GPS. Alternate approved methods include SKE, Ground Radar Aerial Delivery System (GRADS), Ground Marked Release System (GMRS), Verbally Initiated Release System (VIRS), and Pilot-Directed Airdrop (PDA).

17.19.1. The PF is responsible for ensuring the aircraft is positioned at the release point at green light and for maintaining drift-corrected track through red light. The PM is responsible for ensuring a countdown to green light, the green light call, and the red light calls are given.

17.19.2. Element Lead. After slowdown, each element lead flies an independent run-in to the CARP. **(T-2)**

17.19.3. Pilot-Directed Airdrops. SOLL II, 57th Weapons Squadron cadre, and pilots who successfully accomplished the AMC-approved Pilot-Directed Airdrop (PDA) training syllabus are authorized to conduct PDAs. All others require MAJCOM/A3 approval before accomplishing actual PDAs. **Exception:** PDAs may be accomplished by any airdrop pilot for training on dry passes.

17.20. GPS Airdrop. A 300-yard preflight drop box will be computed and updated prior to initiation of the release point checklist. **(T-2).** If the aircrew determines that the aircraft will be outside the drop box at green light, a no drop will be called. **(T-2)**

17.20.1. During the Slowdown Checklist, the following items must be confirmed if planning to drop in IMC:

17.20.1.1. At least one GPS receiver will be updating with no "CGPS INOP and/or MGPS INOP" WAP messages present. **(T-2)**

17.20.1.2. No “UNABLE RNP” messages will be present. **(T-2)**

17.20.2. Crews will load GPS PPS codes and update PPOS with MGPS for all airdrop operations to enhance GPS accuracy. **(T-2)**

17.21. JPADS/ICDS Airdrop.

17.21.1. JPADS certified PADS Operators (PO) are authorized to use the JPADS mission planner and software to calculate release points for JPADS/ICDS airdrop operations.

17.21.2. The PO or mission planner is required to provide Joint Precision Airdrop System-Mission Planner (JPADS-MP) derived CARP(s) for each airdrop pass and a completed Airdrop Damage Estimate (ADE) prior to an airdrop mission. Both pilots will review preflight CARP(s)/ADE for each respective airdrop. **(T-2)**

17.21.3. Airdrop Damage Estimate (ADE). Units must perform a full ADE prior to JPADS/ICDS airdrops. **(T-2)** The ADE must be coordinated and approved by the area controlling agency. **(T-2)** Coordinate with the owning agency of the restricted airspace or controlled airspace and landowners with property surrounding the DZ for all JPADS/ICDS operations. Examine the area in the vicinity of the DZ for potential damage or hazards in the course of normal operations or during extraordinary system failure events. If the ADE demonstrates potential damage or hazards restrict airdrop release LAR, lower the drop altitude, change the run-in, change parachute type or cancel operations. Inform the controlling unit of the risk to their operations; the controlling unit, and the Joint Force Controller (JFC) designated agency are approving authorities for risk to the area surrounding the DZ. Intelligence personnel are responsible for providing the JFC-designated agency close-up and overview imagery to facilitate ADE. For actual JPADS training drops, AMC units are required to contact their Wing Tactics two weeks prior in order to ensure all planning, coordination and reviews/assessments have been accomplished. **(T-3)** Operations conducted at Yuma Proving Ground under JPADS-related test plans do not need Wing Tactics review. See AFTTP 3-3.C-17 and DAFMAN 13-217 for further information. The ADE must include, at a minimum, a review of the airspace and ground space with respect to:

17.21.3.1. CARP and LAR location.

17.21.3.2. ICDS success ellipse.

17.21.3.3. Chute failure footprint.

17.21.3.4. Guidance failure footprint.

17.21.4. Weather restrictions. CONUS training drops in/through IMC will be flown IAW **para 17.11.1**. **(T-2)**

17.21.4.1. JPADS parachutes will not be dropped through severe turbulence or severe icing.

17.21.4.2. Wind Limits. Surface wind limitations are unrestricted for dropsonde operations, 17 knots for JPADS Ultra-Light Weight 2K/2K-M/10K, and as published in DAFMAN 13-217 for all other parachutes.

17.21.5. When dropping JPADS, dropsondes are not required. When dropping near the edge of the LAR or in strong/variable wind conditions, dropping a dropsonde is recommended to improve the drop solution and reduce risk.

17.21.6. Altitude and Weight Limits.

17.21.6.1. JPADS 2K/2K-M operations conducted from 5,000 feet AGL to 24,500 feet MSL have a weight range of 850 to 2,280 lbs. rigged weight. **(T-2)** Training payloads may be dropped as low as 3,500 feet AGL. When dropping 3,500 to 5,000 feet AGL, payload weights will be within 1,380 to 1,780 lbs. rigged weight.

17.21.6.2. JPADS 10K operations conducted from 5,000 feet AGL to 24,500 feet MSL have a weight range of 5,000 to 10,000 lbs. rigged weight. **(T-2)** Training payloads may be dropped as low as 3,500 feet AGL.

17.21.7. Drop Zone Size. Drop zone size criteria for JPADS and ICDS drops during contingency operations is at the discretion of the user. DAFMAN 13-217 drop zone size restrictions apply during training.

17.21.8. JPADS Guided and Dropsonde Footprint Locations. During normal training operations, a JPADS and Dropsonde DZ, CARP, chute failure footprint and guidance failure footprint will be located within restricted airspace. If winds force the CARP outside of restricted airspace additional coordination with ATC is required prior to airdrop operations. This includes coordination with the ATC agency, filing a NOTAM, and ensuring airspace is clear for the entire guided system's flight profile from the drop altitude to the ground. **(T-2)**

17.21.9. During training operations, the entire ICDS success footprint will be located within the surveyed DZ boundaries. The chute failure footprint must fall within restricted airspace. If outside of restricted airspace, the chute failure must fall on the surveyed DZ. If operating in a restricted area and winds force the CARP outside of restricted airspace, coordination with ATC is required prior to airdrop operations. This includes coordination with the ATC agency, filing a NOTAM and ensuring airspace is clear from the drop altitude to the ground. **(T-2)**

17.21.10. JPADS Modular Autonomous Guidance Unit (MAGU) Military Global Positioning System (MILGPS) Procedures. Follow MILGPS keying procedures. The JPADS contains a Selective Availability Anti-Spoofing Module (SAASM) GPS within the guidance unit (referred to as "JPADS MILGPS"). The JPADS MILGPS is approved to receive crypto-variable GPS keys and is an UNCLASSIFIED but controlled item and must be handled to

preclude unauthorized access, tampering, theft, or loss. Due to the general application and associated security protocols, black GPS keys will be used. **(T-2)**

17.21.10.1. Jettison of JPADS MAGU with Military GPS (MILGPS). Instances of jettison of the JPADS with MILGPS must be reported to the GPS controlling authority. Each such report must include the Avionics Module/enclosure serial number and must state whether the system was keyed or unkeyed. This serial number will be recorded on the DD Form 1748, *Joint Airdrop Inspection Record* during the JAI Inspection.

17.21.10.2. Reporting is submitted to US Army Soldier Systems Center, Assistant Product Manager Cargo Aerial Delivery Systems Team, Product Manager Force Sustainment Systems (PM FSS) SFAE-CSS-FP-F, 10 General Greene Avenue, Natick, MA 01760-5057. **(T-2)** Time permitting, the LM, with concurrence from the PIC/PO, will remove the Avionics Module from the JPADS prior to load jettison. **(T-2)**

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Attachment 1

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Abbreviations and Acronym

A/A—Air to Air

AAR—Air to Air Refueling

AC—Aircraft Commander

ACF—Acceptance Check Flight

AD—Airdrop

ADC—Air Data Computer

ADE—Airdrop Damage Estimate

ADIZ—Air Defense Identification Zone

ADS-B—Automatic Dependent Surveillance – Broadcast

ADS-C—Automatic Dependent Surveillance – Contract

ADS—Aerial Delivery System

ADSC—Aerial Delivery System Controller

ADTD—Aircrew Data Transfer Device

AE—Aeromedical Evacuation

AETC—Air Education and Training Command

AFCS—Automatic Flight Control System

AFH—Air Force Handbook

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFPD—Air Force Policy Directive

AFRC—Air Force Reserve Command

AFTO—Air Force Technical Order

AGE—Aircraft Ground Equipment

AGL—Above Ground Level

MAGU—Modular Autonomous Guidance Unit

AIP—Aeronautical Information Publication

AIRAC—Aeronautical Information Regulation and Control

ALZ—Assault Landing Zone

AMC—Air Mobility Command

AMP—Airfield Marking Pattern

ANP—Actual Navigation Performance

AOA—Angle of Attack

AOC—Air Operations Center

AOR—Aera of Responsibility

AP—Autopilot

A/PDMC—Aircraft/Propulsion Data Management Computer

APEX—Aerial Port Expeditor

APU—Auxiliary Power Unit

AQP—Advanced Qualification Program

AR—Authorization Required

A/R—Air Refueling

ARC—Air Reserve Component

ARCT—Air Refueling Control Time

ARMS—Aviation Resource Management System

ARTCC—Air Route Traffic Control Center

ASR—AMC Standard Report

ASRR—Airfield Suitability and Restriction Report

AT—Autothrottle

ATC—Air Traffic Control

ATGL—Air Transportable Galley Lavatory

ATIS—Automatic Terminal Information Service

ATOC—Air Terminal Operations Center

ATT—Attitude

ATTLA—Air Transportability Test Loading Agency

BDHI—Bearing Distance Heading Indicator

BRNAV—Basic Area Navigation Airspace

C2—Command and Control

CARP—Computed Air Release Point

CBP—Customs and Border Protection

CCM—Command and Control Module

CCU—Communication Control Unit

CDFA—Continuous Descent Final Approach

CDI—Course Deviation Indicator

CDS—Container Delivery System

CDT—Crew Duty Time

CFP—Computer Flight Plan

CGPS—Civilian Global Positioning System

CIP—Core Integrated Processor

CJCS—Chairman of the Joint Chiefs of Staff

CL—Centerline Lighting

CNC—Communication Navigation Control

CONUS—Contiguous United States

CPDLC—Controller-Pilot Direct Link Communication

CRE—Contingency Response Element

CRG—Contingency Response Group

CRM—Crew Resource Management

CVAM—Assistant Vice Chief of Staff of the Air Force, Special Air Missions

CVR—Cockpit Voice Recorder

DAF—Department of the Air Force

DAFI—Department of the Air Force Instruction

DAFIF—Digital Aeronautical Flight Information File

DAFMAN—Department of the Air Force Manual

DCP—Dynamic Cone Penetrometer

DDA—Derived Decision Altitude

DGDP—Degrade Percentage

DH—Decision Height

DIRMOBFOR—Director of Mobility Forces

DLC—Direct Lift Control

DME—Distance Measuring Equipment

DoD—Department of Defense

DoDD—Department of Defense Directive

DoDI—Department of Defense Instruction

DRT—Derated Thrust

DRVSM—Domestic Reduced Vertical Separation Minimum

DTED—Digital Terrain Elevation Data

DV—Distinguished Visitor

DZ—Drop Zone

DZSO—Drop Zone Safety Officer

EBL—Emergency Boom Latching

ED—Engineering Disposition

EDP—Earliest Decent Point

EFB—Electronic Flight Bag

EFCS—Electronic Flight Control System

EGT—Exhaust Gas Temperature

ELT—Emergency Locator Transmitter

EPOS—Emergency Passenger Oxygen System

EPR—Engine Pressure Ratio

ER—Extended Range

ERAA—Emergency Route Abort Altitude

ERO—Engine Running Onload/Offload

ESA—Emergency Safe Altitude

ETP—Equal Time Point

FAA—Federal Aviation Administration

FAF—Final Approach Fix

FAR—Federal Aviation Regulation

FCC—Flight Control Computer

FCF—Functional Check Flight

FCG—Foreign Clearance Guide

FCI—Flight Command Indicator

FD—Flight Director

FDP—Flight Duty Period

FEDS—Flotation Equipment Deployment System

FF—Fuel Flow

FFS—Formation Flight System

FIH—Flight Information Handbook

FIR—Flight Information Region

FL—Flight Level

FLIP—Flight Information Publication

FM—Flight Manager

FMA—Flight Mode Annunciation

FOD—Foreign Object Damage

FPC—Mission Ready Mobility Pilot Development Phase I Pilot

FPM—Feet Per Minute

FPQ—Mission Ready Mobility Pilot Development Phase II Pilot

FPV—Flight Path Vector

FRAG—Fragmentary Order

FS—Fuselage Station

FSAF—First Suitable Airfield

GBAS—Ground-Based Augmentation System

GCA—Ground Controlled Approach

GDSS—Global Decision Support System

GMRS—Ground Marked Release System

GNSS—Global Navigation Satellite System

GOAT—Go-Around After Touchdown

GP—General Planning

GPS—Global Positioning System

GPWS—Ground Proximity Warning System

GRADS—Ground Radar Aerial Delivery System

HAHO—High Altitude High Opening

HALO—High Altitude Low Opening

HARP—High-Altitude Release Point

HAT—Height Above Touchdown

HATR—Hazardous Air Traffic Report

HAZMAT—Hazardous Cargo or Materials

HF—High Frequency

HQ—Headquarters

HUD—Heads Up Display

IAF—Initial Approach Fix

IAP—Instrument Approach Procedure

IAW—In Accordance With

ICAO—International Civil Aviation Organization

ICDS—Improved Container Delivery System

ICS—Intercom Control Set

IDG—Integrated Drive Generator

IFF—Identification Friend or Foe

IFCM—Integrated Flight Control Module

IFM—Integrated Flight Management

IFR—Instrument Flight Rules

ILS—Instrument Landing System

IMC—Instrument Meteorological Condition

INS—Inertial Navigation System

IP—Initial Point

IP—Instructor Pilot

IR—Infrared

IRU—Inertial Reference Unit

IWP—Inadvertent Weather Penetration

JA/ATT—Joint Airborne/Air Transportability Training

JAI—Joint Airdrop Inspector

JFC—Joint Force Controller

JG—Job Guide

JOG—Joint Operation Graphics

JPADS—Joint Precision Airdrop System

JPADS-MP—Joint Precision Airdrop System-Mission Planner

KCAS—Knots Calibrated Airspeed

KIAS—Knots Indicated Airspeed

LAIRCM—Large Aircraft Infrared Counter-Measure

LACP—Loadmaster's Aft Control Panel

LAR—Launch Acceptability Region

LFCP—Loadmaster's Forward Control Panel

LHUD—Legacy Heads Up Display

LDA—Landing Distance Available

LDI—Lateral Deviation Indicator

LDP—Latest Descent Point

LM—Loadmaster

LNAV—Lateral Navigation

LOX—Liquid Oxygen

LSAF—Last Suitable Airfield

LSK—Line Select Key

LTRK—Along Track

LZ—Landing Zone

MAF—Mobility Air Forces

MAFPS—Mobility Air Forces Automated Flight Planning System

MAJCOM—Major Command

MAWP—Missed Approach Waypoint

MBL—Manual Boom Latching

MC—Mission Computer

MC—Mission Contributing

MCA—Minimum Crossing Altitude

MCD—Mission Computer Display

MCK—Mission Computer Communications Keyboard

MDA—Minimum Descent Altitude

MDS—Mission Design Series

ME—Mission Essential

MEL—Minimum Equipment List

MEP—Mission Essential Personnel

MFC—Multi-function Control

MFD—Multi-function Display

MHE—Material Handling Equipment

MGPS—Military Global Positioning System

MILGPS—Military Global Positioning System

MNPS—Minimum Navigation Performance Specifications

MOB—Main Operating Base

MRR—Minimum Risk Routing

MSA—Minimum Safe Altitude

MSL—Mean Sea Level

MTR—Military Training Route

NAS—National Airspace System

NAT HLA—North Atlantic Track High Level Airspace

NATO—North Atlantic Treaty Organization

NAVAID—Navigation Aid

ND—Navigation Display

NDB—Non-Directional Beacon

NGA—National Geospatial-Intelligence Agency

NGB—National Guard Bureau

NLT—No Later Than

NM—Nautical Mile

NOTAM—Notice to Airmen

NPA—Non-Precision Approach

NVG—Night Vision Goggle

NVIS—Night Vision Imaging System

OB—Operating Base

OBL—Override Boom Latching

OCF—Operational Check Flight

OCONUS—Outside Contiguous United States

OG—Operations Group

OGV—Operations Group Standardization and Evaluation

OI—Open Item

OPORD—Operations Order

OPR—Office of Primary Responsibility

OPT—On-board Performance Tool

PA—Public Address

PACAF—Pacific Air Forces

PADS—Precision Air Drop System

PAPI—Precision Approach Path Indicator

PAR—Precision Approach Radar

PBN—Performance-Based Navigation

PDA—Pilot-Directed Airdrop

PF—Pilot Flying

PFBIT—Pre-Flight Built-In Test

PFD—Primary Flight Display

PFR—Primary Flight Reference

PI—Point of Impact

PIC—Pilot in Command

PLSR—Precision Landing System Receiver

PM—Pilot Monitoring

PM FSS—Product Manager Force Sustainment System

PO—PADS Operator

PPI—Plan Position Indicator

PPOS—Present Position

PPS—Precise Positioning Service

PRM—Precision Runway Monitor

PRM AAUP—Precision Runway Monitor Attention All Users Page

P-RNAV—Precision Area Navigation

QAR—Quick Access Recorder

QRC—Quick Reaction Card

RA—Resolution Advisory

RAIM—Receiver Autonomous Integrity Monitoring

RAT—Ram Air Turbine

RCAM—Runway Condition Assessment Matrix

RCR—Runway Condition Reading

RF—Radius-to-Fix

RHUD—Replacement Heads Up Display

RNAV—Area Navigation

RNP—Required Navigation Performance

RRFL—Required Ramp Fuel Load

RSC—Runway Surface Condition

RVR—Runway Visual Range

RVSM—Reduced Vertical Separation Minimum

RZ—Rendezvous

SA—Special Authorization

SAAM—Special Assignment Airlift Mission

SAAR—Special Aircraft & Aircrew Requirement

SAASM—Selective Availability Anti-Spoofing Module

SC/EFC—Spoiler Control/Electronic Flap Computer

SDP—Special Departure Procedure

SDRR—Signal Data Recorder-Reproducer

SED—Standby Engine Display

SF—Standard Form

SFO—Specialized Fueling Operations

SID—Standard Instrument Departure

SKE—Station Keeping Equipment

SLICC—Senior Leader In-Transit Conference Capsule

SLIP—Senior Leader In-Transit Pallet

SM—Statue Mile

SOLL II—Special Operations Low Level II

SOV—Shut-Off Valve

SPINS—Special Instructions

SPIC—Special Pilot In Command

SPRO—Semi-Prepared Runway Operations

SQ—Squadron

STAN/EVAL—Standardization/Evaluation

STAR—Standard Terminal Arrival Route

SUAS—Special Use Airspace

TA—Traffic Advisory

TACAN—Tactical Aircraft Control and Navigation

TAWS—Terrain Awareness Warning System

TCA—Turbine Cooling Air Valve

TCAS—Traffic Alerting and Collision Avoidance System

TDY—Temporary Duty

TDZL—Touchdown Zone Lighting

TEM—Threat and Error Management

TERPS—Terminal Enroute Procedures

TLM—Topographic Line Map

T.O.—Technical Order

TOGA—Takeoff Go-Around

TOLD—Takeoff and Landing Data

TPC—Tactical Pilotage Chart

TR—Thrust Reverser

TRANSCOM—Transportation Command

TTF—Tanker Task Force

UDM—User Data Module

UHF—Ultra High Frequency

UIR—Upper Information Region

UTC—Coordinated Universal Time

WAP—Warning Annunciator Panel

WCC—Warning and Caution Computer

WG—Wing

WPS—Weapons Squadron

WWNDB—Worldwide Navigation Database

VASI—Visual Approach Slope Indicator

VDI—Vertical Deviation Indicator

VFR—Visual Flight Rules

VHF—Very High Frequency

VIRS—Verbally Initiated Release System

VMC—Visual Meteorological Conditions

VNAV—Vertical Navigation

VOR—Very High Frequency Omnidirectional Range

VPROF—Vertical Profile

VVI—Vertical Velocity Indicator

VVOD—Vector Vertical Obstruction Data

WX—Weather

XMIT—Transmit

XTRK—Cross Track

ZAR—Zone Availability Report

Office Symbols

AMC/A3A—Airspace Ranges/Airfield Operations Division

AMC/A3AS—Airfield Suitability Branch

AMC/A3T—Aircrew Tactics and Training Division

AMC/A3TW—Combat Operations Division-Tactics

AMC/A3V—Aircrew Standardization/Evaluation and Readiness

AMC/A4TC—Air Cargo Movement Policy Branch

AFLCMC/EZFC—Air Force Life Cycle Management Center Crew Systems Branch

OSK—Weapons and Tactics Flight

19 AF/A3—19th AF Director of Operations

618 AOC—Air Operations Center

618 AOC/XOOO—Special Assignment Airlift Missions

Terms

618 AOC (Air Operations Center)—Operations center that controls tanker and airlift forces worldwide through a network of computer systems. The 618 AOC is organized into geographic cells consisting of East, West, and Emergency Action Cells. The 618 AOC contains the following functions: Mobility Management, Global Channel Operations, Operations Management, Current Operations, Global Readiness, Weather, Logistics Readiness Center, Aerial Port Control Center, International Clearances, and Flight Plans.

Airdrop Damage Estimation (ADE)—A balance of science and art that produces the best estimate of potential damage concerns.

Aeromedical Evacuation (AE)—Movement of patients under medical supervision between medical treatment facilities (MTFs) by air transportation.

Air Force Component Commander (AFCC)—In a unified, sub-unified, or joint task force command, the Air Force commander charged with the overall conduct of Air Force air operations.

Airlift—Aircraft is considered to be performing airlift when manifested passengers or cargo are carried.

Air Reserve Component (ARC)—Refers to National Guard Bureau (NGB) and Air Force Reserve Command (AFRC) forces, both Associate and Unit-Equipped.

Air Route Traffic Control Center (ARTCC)—A facility that provides Air Traffic Control (ATC) services to aircraft operating on IFR flight plans within controlled airspace, principally during the enroute phase of flight.

Air Traffic Control (ATC)—A service provided by an appropriate authority to promote the safe, orderly and expeditious use of the air transportation system and to maximize airspace utility.

Augmented Crew—Basic aircrew supplemented by additional qualified aircrew members to permit in-flight rest periods.

BLUE BARK—US military personnel, US citizen civilian employees of the Department of Defense (DoD), and the dependents of both categories who travel in connection with the death of an immediate family member. It also applies to escorts for dependents of military members traveling under competent orders.

COIN ASSIST—Nickname used to designate dependent spouses accompanying dependent children and dependent parents of military personnel reported missing or captured who may travel space available on military aircraft for humanitarian purposes on approval of the Chief of Staff, United States Army; Chief of Staff, United States Air Force; Chief of Space Operations; Chief of Naval Operations; or the Commandant of the Marine Corps.

Command and Control (C2)—Exercise of direction and authority over assigned forces by a properly designated command echelon in the accomplishment of the mission.

Command and Control Center—Each C2 center provides supervision, guidance, and control within its assigned area of responsibility. For the purpose of this manual, C2 centers include operations centers, local AMC C2s, air mobility elements, contingency response elements/groups (CRE/CRGs), air mobility control centers, unit command posts, and tanker task forces.

Contingency Fuel—An additional 15 minutes of fuel to compensate for unforeseen circumstances during any phase of flight (e.g., unforecasted weather, launch delay, etc.). Contingency fuel will not be considered reserve fuel since crews may burn some or all of the contingency fuel at any time during the mission. Identified extra to compensate for unforeseen circumstances during any phase of flight (e.g., unforecasted weather, launch delay, etc.).

Contingency Mission—Mission operated in direct support of an OPORD, OPLAN, disaster, or emergency.

Contingency Response Elements/Groups (CRE/CRGs)—Team of qualified Air Force personnel established to control, coordinate, and function as an Air Force tanker and airlift C2 facility at a base where normal AMC C2 facilities are not established or require augmentation. CRE/CRGs support and control contingency operations on both a planned and no-notice basis.

Crew Resource Management (CRM)—The effective use of all available resources (people, weapon systems, facilities, equipment, and environment) by crews to safely and efficiently accomplish an assigned mission or task.

Critical Phase of Flight—Terminal area operations including taxi, takeoff and landing, low-level flight, air refueling, airdrop, weapons employment, flight using NVGs, tactical or air combat and formation operations (other than cruise), all portions of any test or functional check flight or any aerial demonstration IAW AFMAN 11-202V3.

Depressurization Fuel—Depressurization fuel will be calculated in MAFPS at FL250/10,000 foot altitude. If additional fuel is required from the Depressurized Engine Out ETP, then MAFPS automatically adds the additional fuel into the Ramp Fuel.

Deviation—A deviation occurs when takeoff time is not within -20/+14 minutes of scheduled takeoff time.

Director, Mobility Forces (DIRMOBFOR)—Individual responsible for theater mobility force management. The Air Force component commander exercises operational control of assigned or attached mobility forces through the DIRMOBFOR. The DIRMOBFOR monitors and manages assigned mobility forces operating in theater. The DIRMOBFOR provides direction to the Air Mobility Division in the AOR to execute the air mobility mission and will normally be a senior officer familiar with the AOR.

Distinguished Visitor (DV)—Passengers, including those of friendly nations, of star or flag rank or equivalent status, to include diplomats, cabinet members, members of Congress, and other individuals designated by the DoD due to their mission or position (includes BLUE BARK and COIN ASSIST).

Due Regard—Operational situations that do not lend themselves to International Civil Aviation Organization (ICAO) flight procedures, such as military contingencies, classified missions, politically sensitive missions, or training activities. Flight under "Due Regard" obligates the PIC to be his or her own air traffic control (ATC) agency and to separate his or her aircraft from all other air traffic. See FLIP General Planning, Chapter 8.

Equal Time Point (ETP)—Point along a route at which an aircraft may either proceed to First Suitable Airfield (FSAF) or return to Last Suitable Airfield (LSAF) in the same amount of time based on all engines operating. FSAF/LSAF are airports closest to the coast out and coast in route of flight that meet applicable destination alternate requirements.

Execution—Command-level approval for initiation of a mission or portion thereof after due consideration of all pertinent factors. Execution authority is restricted to designated command authority.

Fuel Reserve—Amount of usable fuel that must be carried beyond that required to complete the flight as planned.

Global Decision Support System (GDSS)—AMC’s primary execution command and control system. GDSS is used to manage the execution of AMC airlift and tanker missions.

Ground Time—Interval between engine shut down (or arrival in the blocks if engine shutdown is not scheduled) and next takeoff time.

Hazardous Cargo or Materials (HAZMAT)—Articles or substances that are capable of posing significant risk to health, safety, or property when transported by air and classified as explosive (class 1), compressed gas (class 2), flammable liquid (class 3), flammable solid (class 4), oxidizer and organic peroxide (class 5), poison and infectious substances (class 6), radioactive material (class 7), corrosive material (class 8), or miscellaneous dangerous goods (class 9). Classes may be subdivided into divisions to further identify hazard (e.g., 1.1, 2.3, 6.1, etc.).

Joint Airborne/Air Transportability Training (JA/ATT)—Continuation and proficiency combat airlift training conducted in support of DoD agencies. Includes aircraft load training and service school support. JA/ATT information can be found in AMCI 10-2101, *Joint Airborne/Air Transportability Training*.

Loading Time—Specific time established jointly by the commanders concerned when aircraft loading will begin.

Local Training Mission—A mission scheduled to originate and terminate at home station (or an off-station training mission), generated for training or evaluation, and executed at the local level.

Mission—Movement of aircraft from a designated point of origin to a designated destination as defined by assigned mission identifier, mission nickname, or both in the schedule, mission directive, OPORD, OPLAN, or FRAG order.

Mobility Air Force (MAF)—Forces assigned to mobility aircraft or MAJCOMs with operational or tactical control of mobility aircraft.

Mobility Air Forces Automated Flight Planning System (MAFPS)—The software used for AOC-level mission planning for Mobility Air Forces global strategic airlift, aerial refueling, and tactical airlift missions. Replaces Legacy AMC Advanced Computer Flight Plan mission planning system. MAFPS is further designed to provide a means for mission planners to optimize route planning with respect to flight mission time and fuel considerations.

Operational Control (OPCON)—Functions of command and control involving composition of subordinate forces, authority to approve allocation of assets to specific missions, assignment of tasks, designation of objectives, and authoritative direction necessary to accomplish the mission. This is a higher authority than the command that performs specific mission functions.

Operational Missions—Missions executed at or above 618 AOC level. Operational missions termed "CLOSE WATCH" include CORONET missions and AFI 11-221, *Air Refueling Management (KC-10, KC-46, and KC-135)*, priority 1, 2, and 3 missions tasked by the 618 AOC. Other operational missions such as deployment, re-deployment, reconnaissance operations, operational readiness inspections (ORI), AMC channel or SAAM, and JA/ATT missions may be designated "CLOSE WATCH" as necessary.

Operational Necessity—A mission associated with war or peacetime operations in which the consequences of an action justify the risk of loss of aircraft and crews.

Operations Plan (OPLAN)—A plan for a single or a series of connected operations to be carried out simultaneously or in succession, based on stated assumptions; a directive to permit subordinate commanders to prepare supporting plans and orders.

Overwater Flight—Any flight that exceeds power off gliding distance from land.

Passenger—Personnel listed on a passenger manifest (does not include aircrew member or MEP).

Permit to Proceed—Aircraft not cleared at the first US port of entry may move to another US airport on a permit to proceed issued by customs officials at the first port of entry. This permit lists the requirements to be met at the next point of landing (i.e., number of crew and passengers, cargo not yet cleared). PIC is responsible to deliver the permit to proceed to the customs inspector at the base where final clearance is performed. **Note**: Heavy monetary fines can be imposed on the PIC for not complying with permit to proceed procedures.

Positioning and De-positioning Missions—Positioning missions are performed to relocate aircraft for the purpose of conducting a mission. De-positioning missions are made to return aircraft from bases at which missions have terminated.

Repair Capable Facility—Stations with the necessary skilled maintenance personnel, support equipment, and facilities to accomplish required repairs.

Required Ramp Fuel Load (RRFL)—Minimum fuel required at engine start to complete tasked mission. Required ramp fuel load will consist of all fuel required for engine start, taxi, APU operation, takeoff, enroute, enroute reserve, contingency, air refueling, decompression (depressurization), descent, approach and landing, alternate, transition, holding/minimum landing.

Scheduled Takeoff Time—Takeoff time is established in the schedule or OPORD. For air aborts and diversions, this will be engine shut down time (or arrival in the blocks if engine shutdown is not scheduled) plus authorized ground time. Early deviation does not apply to aborts or diversions unless the mission is formally rescheduled by current operations. Scheduled takeoff time may be adjusted to make good an ARCT. Notify controlling agency prior to takeoff to adjust the scheduled takeoff time.

Section—Subdivision of a formation. A section normally consists of 6 aircraft (2 elements).

Special Assignment Airlift Mission (SAAM)—Funded airlift that cannot be supported by channel missions due to unusual nature, sensitivity/urgency of the cargo, or destinations outside established channel structure.

Tactical Event—A flight event that, due to its complexity and/or increased risk due to fatigue, limits the period within a crew's duty day the event may be performed. Not all tactical maneuvers are considered a tactical event. C-17 tactical events are Airdrop, Low Level, Non- SKE Formation, SKE Formation Greater Than 2-Ship, Assault Landing/Takeoff, Tactical Approaches/Departures, and Landings at AMP-3 or AMP-4 configured airfields at night. Events not listed, such as penetration descents and standard traffic pattern altitude downwind and overhead approaches, are not tactical events.

Tanker Task Force (TTF)—Force of tanker aircraft assembled and tasked to perform a specific function.

Tankered Fuel—Additional fuel carried through a primary destination for use on a subsequent leg.

Training Mission—Mission executed at the unit level for the sole purpose of aircrew training for upgrade or proficiency. Does not include operational missions as defined in this manual.

Unilateral—Operations confined to a single service.